Distributional Learning of Multiple Context-Free Languages under Fixed Finite-Semigroup Typing: Local Groups and the Length of Characteristic Data

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Abstract

We study the positive data needed to learn multiple context-free languages when constituents are compared only if they agree under a typing fixed in advance, a morphism onto a finite semigroup. Identification in the limit follows from Yoshinaka's framework; we bound the total length of characteristic samples by the size and rule thickness of a target grammar, and ask for which typings this length is polynomial. If the typing is onto a local group, a finite semigroup S in which $e\mathsf{S}e$ is a group for every idempotent e, then for every fixed fan-out and rank, length quadratic in grammar size and linear in rule thickness suffices, and these classes are identifiable in polynomial time and thick data. The proof shortens interleaved tuple components simultaneously while preserving ordered products in noncommutative groups. Non-context-free targets of fan-out two separate the typing regimes: one lies in the class of a local-group typing but of no typing combining a bounded window with a group morphism, another in the class of such a typing only for noncommutative groups. Length linear in grammar size times rule thickness is necessary under every typing over large enough alphabets. At fan-out one and rank at least two, local groups are exactly the boundary: a pseudovariety yields polynomial data for all its typings if and only if it consists of local groups; otherwise some targets need samples exponentially longer than their grammars, for every set-driven learner with characteristic samples throughout the class. For typings onto semilattices three letters are needed for this: over two letters such typings yield polynomial data at every fan-out. At fan-out one, a typing onto the two-element semilattice has polynomial data if and only if exactly one letter is sent to its identity. The boundary at fan-out two reduces to typings recording the set of letters of a word and is open from three letters on; the fan-out-one lower-bound family provably does not lift.

1 Introduction

Positive-data grammatical inference asks when a language can be identified in the limit from positive examples alone [1, 2]. Distributional learning treats words, or tuples of words, that occur in common contexts as candidate constituents. It began with substitutable CFGs [3, 12] and was extended to multiple context-free grammars (MCFGs), whose tuple-valued nonterminals express discontinuous and cross-serial dependencies [13, 14, 28, 29]. Yoshinaka's finite-signature framework [15] gives positive-data identification with characteristic samples of polynomially many words. Polynomially many words may still be exponentially long, so we measure a finite sample K by its total encoded length $\|K\|_+ = \sum_{w\in K}(|w|+1)$, following the thick-data perspective of [10, Section 2.5.2]. Qualitative identification follows from Yoshinaka's framework; our subject is the length of characteristic samples relative to an MCFG presentation of the target, and the question is: *for which comparison biases is this length polynomial?*

The setting. Before any data are seen, we fix a fan-out bound f and a surjective finite-semigroup morphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$, a *typing*. The reconstruction compares observed nonempty tuple components only when their s-values agree, and the target class $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ consists of the f-MCFLs that are tuple-substitutable under this restriction (Definition 2.8). The typing is a class-level bias in the sense of [5], not chosen per target; other biases restrict derivations by control sets [6, 7] or restrict the evidence used to compare factors [9, 11]. Upper bounds concern Yoshinaka's reconstruction operator in which two observed tuples are linked only when their types agree, $\mathcal{B}_{f,s}$, and its variant $\mathcal{B}^{\leq r_{\max}}_{f,s}$ with composition rules of rank at most $r_{\max}$ (Definition 3.4). A target is given by a reduced good MCFG G (Definition 2.6) of size $\|G\|$ and rule thickness $\bar{\theta}_G$, the largest length of a shortest sentence using a given rule; it is polynomially equivalent to the nonterminal thickness t_G of [10] and of the companion manuscript [23], written τ_G there, since $t_G \leq \theta_G \leq 2\|G\| \max\{1, t_G\}$ (Lemma 4.1).

Why a typing. A typing enlarges the class that the reconstruction identifies. Already at fan-out two, a $\mathbb{Z}_2$-typing admits a non-context-free target outside Yoshinaka's untyped class, on which the untyped operator over-generalizes from a sample of three words (Proposition 3.18). The price is paid in data: a characteristic sample must contain a word of every type that the target realizes, and some types may be realized only by long words (Proposition 4.6). Our results say for which typings this price stays polynomial.

Which typings are local groups. A finite semigroup S is a *local group* if $e\mathsf{S}e$ is a group for every idempotent e. They form a *pseudovariety* **LG** (a class closed under subsemigroups, homomorphic images and finite products), and a finite semigroup is a local group if and only if it has no subsemigroup isomorphic to the two-element semilattice $U_1 = \{1, \zeta\}$ (Theorem B.4). Windows (locally trivial semigroups, **LI**), finite groups (**Grp**) and the divisors of their products (**LI** $\vee$ **Grp**) are local groups, strictly: over $\{\mathtt{a}, \mathtt{b}\}$, recording the parity of the number of occurrences of $\mathtt{ab}$ is a local-group typing, but no prefix–suffix window combined with a group morphism determines it: for each window size and group morphism there are m, n such that $\mathtt{b}^m\mathtt{a}^n\mathtt{b}^m$ and $\mathtt{b}^{2m}$ agree on both but differ in that parity (Proposition B.6). A DFA transition morphism need not be a local group: a letter acting as the identity and another acting as a constant map on at least two states form a copy of U_1.

Main results. In Theorems A and B the alphabet Σ and the fan-out f are fixed.

Theorem A (Local groups)**.** *Let $r_{\max} \geq 1$ and suppose $s(\Sigma^+) \in$ **LG**. For every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and every reduced good presentation G of $L \cap \Sigma^+$ of fan-out $\leq f$ and rank $\leq r_{\max}$, the operators $\mathcal{B}^{\leq r_{\max}}_{f,s}$ and $\mathcal{B}_{f,s}$ have a common characteristic sample for L of total length $O(\|G\|^2\bar{\theta}_G)$; the constant depends only on f, $r_{\max}$, s and Σ (Theorem 5.7).*

The typing regimes of Theorem A differ already on non-context-free targets. A non-context-free language of fan-out two lies in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ for a local-group typing s ($f \geq 2$) but in the class of no window–group typing, of any window size and group (Proposition 5.9); another lies in the class of a window–group typing over S_3 but in the class of no window–abelian typing (Proposition 5.10). Thus, even when restricted to non-context-free targets of fan-out two, window–group typings cannot replace local-group typings, and window–abelian typings cannot replace window–group typings, in these class-coverage results. The separations concern the scope of the typing regimes, not the necessity of a particular shortening proof or the optimality of its bound.

Theorem B (The linear factor)**.** (i) *Suppose that Σ has at least $4f + 1$ letters, or that $f = 1$ and Σ has at least two letters. Then for every typing s there are finite targets in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, with*

rank-two presentations G within a factor $7f + 17$ of the size of the smallest reduced good ones, whose characteristic samples all have length $\Omega(\|G\|\bar{\theta}_G)$ for $\mathcal{B}_{f,s}^{\leq r}$ $(r \geq 1)$ and for every set-driven operator with characteristic samples throughout the class (Proposition 6.1).

(ii) *For the trivial typing and reduced good presentations of rank $\leq r_{\max}$, the operator $\mathcal{B}_{f,s_{\mathrm{triv}}}$ satisfies Yoshinaka's bound $O(\|G\|\bar{\theta}_G)$ [14]; if $r_{\max} \geq 2$ and the alphabet condition of* (i) *holds, the bound is attained up to a constant factor by the targets of* (i) *(Corollary 6.2).*

Theorem C (The boundary at fan-out one)**.** *Let $f = 1$ and $r_{\max} \geq 2$.*

(i) *A pseudovariety $\mathbf{V}$ of finite semigroups is contained in $\mathbf{LG}$ if and only if it has* uniform polynomial characteristic-data length *(Definition 7.2): for every nonempty alphabet Σ and every typing s with $s(\Sigma^+) \in \mathbf{V}$ there is a polynomial p, which may depend on s and Σ but not on the target, such that every $L \in \mathcal{C}_{1,s}^{\mathrm{mcf}}$ has a characteristic sample for $\mathcal{B}_{1,s}^{\leq r_{\max}}$ of length at most $p(\|G\|, \bar{\theta}_G)$ for each reduced good presentation G of $L \cap \Sigma^+$ of rank $\leq r_{\max}$ (Theorem 7.3).*

(ii) *Outside $\mathbf{LG}$ the failure is exponential: for a typing s_U of three letters onto U_1 there are targets $L_n \in \mathcal{C}_{1,s_U}^{\mathrm{mcf}}$ with reduced good rank-two presentations of size and rule thickness $O(n)$ whose characteristic samples all have length at least $3 \cdot 2^n - 1 \geq 2^{(\|G_n\| + \bar{\theta}_{G_n})/17}$ for $\mathcal{B}_{1,s_U}^{\leq r_{\max}}$ and for every set-driven operator with characteristic samples throughout the class (Theorem 7.3). So the data length is either $O(\|G\|^2\bar{\theta}_G)$ for all typings into $\mathbf{V}$, or exponential in $\|G\| + \bar{\theta}_G$ for some of them; no intermediate growth occurs.*

Hence, for $s(\Sigma^+) \in \mathbf{LG}$, the classes $\mathcal{C}_{f,s}^{\mathrm{mcf}} \cap \mathsf{MCFL}(f, r_{\max})$ (languages of MCFGs of fan-out $\leq f$ and rule rank $\leq r_{\max}$) are identifiable in polynomial time and thick data [10, 12] at every fan-out, and at fan-out one $\mathbf{LG}$ is exactly the pseudovariety boundary of this property (Corollary 7.6). An MCFG of fan-out one is a context-free grammar, so Theorem C is a statement about context-free languages; it turns the sufficient condition $s(\Sigma^+) \in \mathbf{LI} \vee \mathbf{Grp}$ of the companion manuscript [23] into a characterization. We include it as the benchmark against which Theorem A is measured: at fan-out one the local-group sufficient condition cannot be weakened at the level of pseudovarieties.

Two letters and isolated letters. The three letters in Theorem C(ii) cannot be reduced to two for typings onto semilattices, and two of them must have type 1. Call a letter *isolated* if its type is realized only by its own powers and all other words containing it share one type (Definition 7.9). At fan-out one, the type of an isolated letter has representatives of length polynomial in $\|G\|$ and $\bar{\theta}_G$, for reduced good presentations of every rank and over every alphabet (Theorem 7.10). Hence over $\{\mathtt{a}, \mathtt{b}\}$ every typing whose positive image is a semilattice, and over every alphabet every typing onto U_1 with exactly one letter of type 1, has characteristic samples of length $O(r_{\max}(\|G\| + 2)^{23}\bar{\theta}_G^8)$ at fan-out one, with an absolute constant (Corollary 7.11); over two letters the same holds at every fan-out $f \geq 2$ (Corollary 8.7). Since two letters of type 1 already force exponential data (Corollary 7.7), this classifies the typings onto U_1 at fan-out one (Corollary 7.12). These data are not of the local-group form: they can be quadratic in $\bar{\theta}_G$ for grammars of logarithmic size (Proposition 7.8). The bound is on the representative length τ, not on the typed thickness, which is exponential already for Σ^+ under such a typing (Proposition 4.7).

The remaining question. Theorem A gives the *if* direction of Theorem C(i) at every fan-out. Whether $\mathbf{LG}$ is still the boundary at fan-out $f \geq 2$ is open (Open Problem 9.1). The family of Theorem C(ii) cannot settle it: under every fixed typing, all but finitely many of its members leave the class at fan-out two (Theorem 8.2). The question reduces to content typings, the typings

recording the set of letters of a word (Proposition 8.5); over two letters these have polynomial data at every fan-out (Theorem 8.6), so it remains open from three letters on.

Approach. Refining the grammar by types can make it exponentially thicker even for a two-element typing (Proposition 4.7), so we bound characteristic data in terms of the original presentation. Short tuples of prescribed type are obtained by replacing long subderivations by short ones whose group values may differ but whose combined effect does not. At fan-out one a replaced subtree occupies one contiguous block, and a pigeonhole argument on prefix products suffices. At fan-out $f \geq 2$ a subtree sends up to f blocks to different components, interleaved with the blocks of other subtrees, and in a noncommutative group the effect of simultaneous replacements depends on this order. For instance, if two subtrees write (x_1, x_2) and (y_1, y_2) and the root components are $\cdots x_1 \cdots y_1 \cdots$ and $\cdots y_2 \cdots x_2 \cdots$, then the two subtrees appear in opposite orders in the two products, and a choice of replacements that keeps one product fixed may change the other; with many subtrees, every pattern of crossings occurs. The key step is a simultaneous choice of replacements that preserves the ordered product of every component at once (Proposition 5.2). For local groups, whose values also depend on factors straddling block boundaries, choosing blocks far apart and padding them with unchanged boundary letters reduces to the same step (Lemma B.7). The lower bounds use finite families all of whose subsets stay in the class (Proposition 6.1) and types realized only by long words (Proposition 4.6).

Relation to prior work. Yoshinaka bounded characteristic data of (k, ℓ)-substitutable CFGs with a thickness parameter [12], learned multidimensionally substitutable MCFGs from positive data [13, 14], and gave the finite-signature learning principle with finite witness cardinality [15]; the qualitative results and the reconstruction operator used here are his. The finite-semigroup structure theory and local-group representations are classical [33–35]. Our contribution is the encoded-length analysis and the boundary results above. Other mildly context-sensitive formalisms are learned from positive data in [19], and MCFGs with queries or dual context sets in [16, 18, 20]; the MCFL substitution lemma of [21] is a membership condition, not a learning criterion.

Relation to the companion CFG manuscript. The companion manuscript [23] treats fan-out one. In short, it proves polynomial data for typings in $\mathbf{LI} \vee \mathbf{Grp}$ as a sufficient condition and exponential data for one typing onto U_1; here the fan-out-one boundary is determined exactly (**LG**), and the upper bound for **LG** and the linear-factor lower bound hold at every fan-out. Table 1 separates its results from ours (numbers in the second column refer to the companion). No result of the companion is used as a premise: instead of its bracketed doubling family [23, Proposition 6.4] we use a three-letter variant, proved in Appendix G, and its expressive results (counter and Dyck obstructions, comparison with congruential grammars) are not used. The results without a counterpart in the companion are not listed in the table: the separating targets (Propositions 5.9 and 5.10), the necessity of $\|G\|\bar{\theta}_G$ (Proposition 6.1), isolated letters and two letters (Theorem 7.10, Corollaries 7.11 and 8.7, Proposition 7.8), and, at fan-out two, power transfer and the reduction to content typings (Theorem 8.2, Proposition 8.5).

Scope of the claims. All bounds concern the total encoded length of characteristic samples for the stated operators, relative to the stated presentation; they bound neither an intrinsic grammar size nor a stage of Gold convergence. The typing is fixed before the target is seen; refining it enlarges the class but changes the reconstruction operator, so we neither infer nor optimize it. Table 2 at

Table 1: Results of the companion CFG manuscript [23] (fan-out one) and of the present paper.

Topic	Companion [23]	Here
Typed reconstruction, Gold learner, typed-thickness witness bound	Theorem 4.2, Theorems 5.2–5.3, Corollary 5.4; Lemma 6.2, Corollary 6.3	Tuple versions (Section 3, Theorem 4.3); qualitative results credited to Yoshinaka; not new
Window and window–group bounds	Proposition 3.5, Lemma 7.1, Theorem 7.2, Lemma 7.4, Corollary 7.5	Not new at $f = 1$; new for $f \geq 2$: owner-wise replacement across interleaved components (Proposition 5.2)
Local-group typings	Not treated (only $\mathbf{LI} \vee \mathbf{Grp}$; Lemma 7.6)	New for data length, at every f (Theorems B.4 and 5.7); the characterization of $\mathbf{LG}$ by U_1 and the semigroup algebra are classical [33]
Doubling family under a typing onto U_1	Proposition 6.4, Corollary 6.6, Appendix C (five letters, brackets)	Three-letter variant with the same bounds, proved here (Proposition 7.1)
Rare-type lower bound	Proposition 6.4(iii)–(iv), for a single word	Every realized type and every f (Proposition 4.6)
Fan-out-one pseudovariety boundary	Sufficiency for $\mathbf{LI} \vee \mathbf{Grp}$ (Corollary 7.5); necessity not claimed (Remark 7.7)	New (Theorem 7.3)
Individual typings onto U_1	One typing (Proposition 6.4)	New: at fan-out one, polynomial data iff exactly one letter has type 1 (Corollary 7.12); exponential data whenever two letter-disjoint words of type 1 exist (Corollary 7.7)
Typed versus untyped class	Separations at $f = 1$ (Propositions 6.4 and 8.4)	At $f \geq 2$, a non-context-free member outside Yoshinaka's class (Proposition 3.18)

the end of Section 7 lists, for each main claim, the fixed parameters, the operators and the side conditions, and Section 9 discusses the exponents.

Organization. Section 2 fixes the setting. Section 3 recalls the reconstruction operator and separates the typed from the untyped class. Section 4 reduces characteristic data to typed thickness and to representative length. Section 5 proves Theorem A and gives the separating targets, Section 6 proves Theorem B, and Section 7 proves Theorem C and classifies the typings onto U_1 and treats isolated letters and two letters at fan-out one. Section 8 treats fan-out two: non-lifting, the reduction to content typings, and two letters. Section 9 discusses the open problem and the exponents. The appendices contain the technical proofs.

2 Preliminaries

Throughout, Σ is a finite alphabet, ε the empty word and $\Sigma^+ := \Sigma^* \setminus \{\varepsilon\}$. If $\Sigma = \emptyset$, the only languages are $\emptyset$ and $\{\varepsilon\}$ and are handled directly. For a finite $K \subseteq \Sigma^*$ put $\|K\|_+ := \sum_{w \in K}(|w| + 1)$, and for a tuple $\vec{u} = (u_1, \ldots, u_d)$ put $\|\vec{u}\|_1 := \sum_i |u_i|$. Every polynomial bound below states whether it concerns cardinality or $\|\cdot\|_+$. The table fixes symbols used across sections; all others (sliding-factor labellings, auxiliary morphisms, padding lengths) are defined where they occur.

Symbol	Meaning
G, G_{in}	a (reduced good) MCFG; an arbitrary standard input MCFG
B, C, X, Y	nonterminals; X, Y also denote typed nonterminals where specified
$\|G\|$, θ_G, t_G	encoding size (Definition 2.5), rule thickness, nonterminal thickness
s, $\widehat{s}$, S	positive typing, its fresh-identity completion, its positive image
$\preceq$	kernel refinement (Definition 2.3), also between monoid morphisms on Σ^*
$\mathbb{G}$, $\mathbb{H}$	finite groups ($\mathbb{G}$ the group component of a typing, $\mathbb{H}$ a Rees-matrix group or an auxiliary group)
g	a monoid morphism $\Sigma^* \to \mathbb{G}$ into a finite group
T, N	a finite semigroup; the finite monoid receiving an auxiliary morphism λ
λ, $\lambda_{k,\nu}$	an auxiliary finite-monoid morphism refining $\widehat{s}$ (Section 4); the sliding-factor morphism (Definition 5.6)
T	a derivation tree
n	the index of a grammar family
t_{typ}, τ, μ_m	shortest typed yield, representative length, shortest word of type m (Section 4); none of them is a thickness of G
S_L	the set $s(L \cap \Sigma^+)$ of realized positive types
$\mathcal{B}_{f,s}$, $\mathcal{B}_{f,s}^{\leq r}$	uncapped and rank-capped reconstruction operators (Definition 3.4)

2.1 Positive-Data Learning

Definition 2.1 (Text, TxtEx identification, conservativeness)**.** Fix a pause symbol $\perp \notin \Sigma$. A *text* for $L \subseteq \Sigma^*$ is an infinite sequence over $L \cup \{\perp\}$ in which every element of L occurs; the *content* of a finite prefix is the set of elements of Σ^* occurring in it. A *learner* is a computable map from finite prefixes to grammars. It *TxtEx-identifies* L [1] if on every text for L its outputs are, from some stage on, one fixed grammar H with $L(H) = L$; it TxtEx-identifies a class if it does so for each member. It is *conservative* if it changes its hypothesis only on a datum $w \in \Sigma^*$ not generated by the current hypothesis. We fix a hypothesis H_0 with $L(H_0) = \emptyset$.

Definition 2.2 (Set-driven operator and characteristic sample)**.** A *set-driven reconstruction operator* $\mathcal{B}$ maps each finite $K \subseteq \Sigma^*$ to a grammar $\mathcal{B}(K)$ depending only on the set K; it is *sample-consistent* if $K \subseteq L(\mathcal{B}(K))$ for all K. A finite $\mathcal{F} \subseteq L$ is a *characteristic sample for $\mathcal{B}$ and L* if $L(\mathcal{B}(K)) = L$ for every finite K with $\mathcal{F} \subseteq K \subseteq L$ (cf. [4]).

The definition constrains generated languages only: $\mathcal{B}(K)$ need not stabilize syntactically as K grows.

2.2 Fixed Finite-Semigroup Typing

Definition 2.3 (Typing, fresh completion, refinement)**.** A *typing* is a surjective semigroup morphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ onto a finite semigroup, given by the multiplication table of S and the images of the letters, so that $s(w)$ is computable in time $O(|w|)$; S is its *positive image*. (A morphism into a larger finite semigroup is replaced by its corestriction to the image.) Let $\mathsf{S}^\sharp := \mathsf{S} \uplus \{\mathbf{1}_\sharp\}$ be obtained by adjoining a *fresh* identity, even if S already has one, and let $\widehat{s} : \Sigma^* \twoheadrightarrow \mathsf{S}^\sharp$ be the monoid morphism with $\widehat{s}(\varepsilon) = \mathbf{1}_\sharp$ and $\widehat{s}|_{\Sigma^+} = s$. For $\vec{w} \in (\Sigma^*)^d$ put $\widehat{s}^{(d)}(\vec{w}) := (\widehat{s}(w_1), \ldots, \widehat{s}(w_d))$; on $(\Sigma^+)^d$ this is $s^{(d)}(\vec{w})$. For typings s, s' write $s \preceq s'$ if $\ker s' \subseteq \ker s$ on Σ^+, and let (s, s') be the typing $w \mapsto (s(w), s'(w))$ onto its image in $\mathsf{S} \times \mathsf{T}$, where $\mathsf{T} := s'(\Sigma^+)$; thus $s, s' \preceq (s, s')$. Write s_{triv} for the one-element typing.

The fresh completion evaluates grammar templates (including empty terminal gaps) and types tuple yields in the refinement of Section 3.2; its new identity is not a positive-word type. Algebraic conditions such as **LG** concern S, not $\mathsf{S}^\sharp$; indeed $\mathsf{S}^\sharp$ is never a local group, since $\{\mathbf{1}_\sharp, e\} \cong U_1$ for any idempotent e of the minimal ideal of S. A finite-monoid morphism $q : \Sigma^* \to \mathsf{N}$ determines the

positive typing $q|_{\Sigma^+}$: by Lemma E.1(i), it gives the same target class and set-driven reconstruction rules as the fresh completion of that typing. Lemma E.1(ii) additionally identifies their trimmed yield-type refinements for *good* grammars. This last identification is not asserted for arbitrary standard grammars with empty-component yields.

Definition 2.4 (Prefix–suffix typing)**.** For $k, \ell \geq 0$ put $m_0 := \max\{1, k+\ell\}$, and for $|w| \geq m_0$ let $\mathrm{pref}_k(w)$, $\mathrm{suf}_\ell(w)$ be the prefix and suffix of the indicated lengths (length-zero windows are ε). Define

$$\widehat{s}_{k,\ell}(w) = \begin{cases} w, & |w| < m_0, \\ \langle \mathrm{pref}_k(w), \mathrm{suf}_\ell(w) \rangle, & |w| \geq m_0, \end{cases}$$

with disjoint tags for short words and pairs, and put $M_{k,\ell} := \widehat{s}_{k,\ell}(\Sigma^*)$. For short x, y and pairs $\langle u, v\rangle, \langle u', v'\rangle$ define

$$\begin{aligned} x \cdot y &:= \widehat{s}_{k,\ell}(xy), & x \cdot \langle u, v\rangle &:= \langle \mathrm{pref}_k(xu), v\rangle, \\ \langle u, v\rangle \cdot y &:= \langle u, \mathrm{suf}_\ell(vy)\rangle, & \langle u, v\rangle \cdot \langle u', v'\rangle &:= \langle u, v'\rangle. \end{aligned}$$

Put $s_{k,\ell} := \widehat{s}_{k,\ell}|_{\Sigma^+}$; a typing s with $s \preceq s_{k,\ell}$ for some $k, \ell \geq 0$ is a *window typing*.

A case check on the lengths of $x, y \in \Sigma^*$ gives $\widehat{s}_{k,\ell}(xy) = \widehat{s}_{k,\ell}(x) \cdot \widehat{s}_{k,\ell}(y)$. Since $\widehat{s}_{k,\ell}$ is onto $M_{k,\ell}$, the product is associative and $\widehat{s}_{k,\ell}(\varepsilon)$ is an identity, so $\widehat{s}_{k,\ell}$ is a finite-monoid morphism. The short-word tag separates ε from $\widehat{s}_{k,\ell}(\Sigma^+)$, so $\widehat{s}_{k,\ell}$ is isomorphic to the completion $\widehat{s_{k,\ell}}$.

2.3 Multiple Context-Free Grammars

Definition 2.5 (Standard MCFG presentation and size)**.** An MCFG [28, 29] is $G = (V, \Sigma, P, S, \mathrm{fan})$ with finite V, $S \in V$, $\mathrm{fan}(S) = 1$ and $\mathrm{fan}(A) \geq 1$. A rule of rank $r \geq 0$ is $A \to \rho(B_1, \ldots, B_r)$ with template $\rho = (\alpha_1, \ldots, \alpha_{d_0})$, $d_0 = \mathrm{fan}(A)$, $d_j = \mathrm{fan}(B_j)$, where each α_i is a word over $\Gamma_\rho := \Sigma \uplus \Xi_1 \uplus \cdots \uplus \Xi_r$, $\Xi_j := \{\xi_j^1, \ldots, \xi_j^{d_j}\}$, and every variable occurs at most once in $\alpha_1 \cdots \alpha_{d_0}$ (*linearity*). For $\vec{u}_j \in (\Sigma^*)^{d_j}$, $\rho(\vec{u}_1, \ldots, \vec{u}_r)$ substitutes $u_{j,\kappa}$ for ξ_j^κ; for $r = 0$ the rule is a *constant rule*. The tuple languages $L_A(G) \subseteq (\Sigma^*)^{\mathrm{fan}(A)}$ form the least family closed under the rules, and $L(G) := \{w : (w) \in L_S(G)\}$. We write $A \Rightarrow_G^* \vec{u}$ for a *terminal derivation* (a finite derivation tree rooted at A, evaluated bottom-up); one from S is *successful*. G is an f-MCFG if $\mathrm{fan}(A) \leq f$ for all A, and has rule-rank cap $r_{\max}$ if all ranks are at most $r_{\max}$. Such a presentation, with no further restriction, is called *standard*. Its *size* is

$$\|G\| := |V| + \sum_{(A \to \rho(B_1, \ldots, B_r)) \in P} \Big(1 + r + \textstyle\sum_i |\alpha_i|\Big),$$

where $|\alpha_i|$ counts terminal and variable occurrences. All bounds use this symbol-level size; a bit-level encoding with binary nonterminal names changes it by at most a factor $O(\log \|G\|)$.

$\mathsf{MCFL}(f, r)$ is the class of languages of MCFGs of fan-out $\leq f$ and rule rank $\leq r$; f-MCFL $= \bigcup_{r<\infty} \mathsf{MCFL}(f, r)$, and the minimum rule rank of a language is taken over all its fan-out-$\leq f$ presentations. For nonterminals of equal fan-out d, $A \to B$ abbreviates $A \to (\xi_1^1, \ldots, \xi_1^d)(B)$, and $A \to \vec{c}$ a constant rule. Rule variables are always ξ_j^κ; terminals in examples are in `typewriter` font. Fan-out and rank are distinct resources. At fixed fan-out at least two the rule-rank hierarchy does not collapse in general, although adjacent levels can coincide; in particular $\mathsf{MCFL}(2, 2) = \mathsf{MCFL}(2, 3)$ [22, 31]. By contrast, fan-out-one CFGs can be binarized. The reconstruction below therefore assumes neither a binary normal form nor a known rank cap.

Definition 2.6 (Grammar properties)**.** A rule is *nondeleting* if every variable occurs exactly once in its template; a nondeleting rule is *nonpermuting* if for each j the variables $\xi_j^1, \dots, \xi_j^{d_j}$ occur in this order in $\alpha_1 \cdots \alpha_{d_0}$, and *nonmerging* if moreover no α_i contains a factor $\xi_j^\kappa \xi_j^{\kappa+1}$. Following [14, Section 2.3], a rule is *ε-free* ("λ-free" there) if no α_i is empty. A rule, template or grammar is *good* if it is ε-free, nondeleting, nonpermuting and nonmerging. A nonterminal A is *reachable* if there is a chain $S = A_0, \dots, A_m = A$ in which each A_{i+1} is a child in a rule for A_i, and *productive* if $L_A(G) \neq \emptyset$; G is *reduced* if every nonterminal is both.

In a ε-free grammar every component of every derived tuple is nonempty (induction on derivations). ε-freeness concerns template components and is distinct from $\varepsilon \notin L(G)$.

2.4 Tuple Contexts and Substitutability

A *sentence context of arity d* is a word $E = v_0 \square_1 v_1 \cdots v_{d-1} \square_d v_d$ with each hole occurring once, $v_0, v_d \in \Sigma^*$ and $v_1, \dots, v_{d-1} \in \Sigma^+$; $\mathcal{E}_d$ is the set of these *admissible* contexts [14, Section 3.1]. $E[\vec{w}]$ fills $\square_i$ by w_i, $|E|_{\mathrm{fix}} := \sum_{i=0}^{d} |v_i|$, and E *accepts* $\vec{w}$ in L if $E[\vec{w}] \in L$. The *d-tuple distribution* of $\vec{x} \in (\Sigma^*)^d$ is $\mathcal{D}_L^{(d)}(\vec{x}) := \{E \in \mathcal{E}_d : E[\vec{x}] \in L\}$; at fan-out one it is the two-sided distribution $\{(u, v) : uxv \in L\}$.

Definition 2.7 (Tuple occurrence)**.** For $w = a_1 \cdots a_n$ number the cut positions $\textcircled{0} a_1 \textcircled{1} \cdots a_n \textcircled{n}$ and write $w[i : i'] := a_{i+1} \cdots a_{i'}$. An *arity-$d$ tuple occurrence* in w is a sequence $0 \le i_1 < i_1' < i_2 < i_2' < \cdots < i_d < i_d' \le n$; it determines $\vec{x} = (w[i_1 : i_1'], \dots, w[i_d : i_d']) \in (\Sigma^+)^d$ and the context $E \in \mathcal{E}_d$ obtained by replacing these intervals by $\square_1, \dots, \square_d$, so that $E[\vec{x}] = w$. For finite K, $\mathrm{Obs}_d(K) := \{\vec{x} \in (\Sigma^+)^d : E[\vec{x}] \in K \text{ for some } E \in \mathcal{E}_d\}$ is the set of tuples arising from such occurrences, and for a fixed fan-out bound $f \ge 1$, put $\mathrm{Obs}(K) := \bigcup_{1 \le d \le f} \mathrm{Obs}_d(K)$.

Definition 2.8 ((f, s)-tuple substitutability and the target class)**.** Let $f \ge 1$ and let s be a typing. A language $L \subseteq \Sigma^*$ is *(f, s)-tuple-substitutable* if for every $1 \le d \le f$ and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$,

$$s^{(d)}(\vec{x}) = s^{(d)}(\vec{y}) \ \text{ and } \ \mathcal{D}_L^{(d)}(\vec{x}) \cap \mathcal{D}_L^{(d)}(\vec{y}) \neq \emptyset \implies \mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y}).$$

$\mathcal{C}_{f,s}^{\mathrm{mcf}}$ is the class of (f, s)-tuple-substitutable f-MCFLs. For $\vec{x}, \vec{y} \in (\Sigma^+)^d$ write $\vec{x} \equiv_{L,s}^d \vec{y}$ (or $\vec{x} \equiv_{L,s} \vec{y}$) if $s^{(d)}(\vec{x}) = s^{(d)}(\vec{y})$ and $\mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y})$; this is an equivalence relation.

Empty components are never compared, following [14, Section 3.1 and footnote 3] (see [8] for CFGs), and $\varepsilon \in L$ is handled at the start symbol; otherwise already $\mathtt{a}^+$ would fail, since $\mathtt{a}$ and ε share a context but not their distributions. The bias consists of both s and f:

(a) If $s \preceq s'$, equal s'-types are equal s-types, so $\mathcal{C}_{f,s}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,s'}^{\mathrm{mcf}}$; in particular $\mathcal{C}_{f,s_{\mathrm{triv}}}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$. This compares classes only: the operators $\mathcal{B}_{f,s}$ and $\mathcal{B}_{f,s'}$ of Section 3 differ in general (they coincide when $\ker s = \ker s'$).

(b) $\mathcal{C}_{f+1,s}^{\mathrm{mcf}} \cap f\text{-MCFL} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$, but the class is not monotone in f: for the trivial typing $\{w \# w^R : w \in \{\mathtt{a}, \mathtt{b}\}^*\}$ is 1- but not 2-dimensionally substitutable [14, Section 3.1, Example 5].

(c) For every $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$, $\widehat{s}^{-1}(\mathsf{Acc}) \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$: it is regular, and componentwise equal types give equal total types in every context.

Proposition 2.9 (Intersection and abelian filtering)**.** *Fix f and s.*

(i) If L_1, L_2 are (f, s)-tuple-substitutable, so is $L_1 \cap L_2$.

(ii) If L is (f,s)-tuple-substitutable, A is an abelian group, $g:\Sigma^*\to\mathsf{A}$ a monoid morphism, $\mathsf{H}\le\mathsf{A}$ and $\gamma\in\mathsf{A}$, then $L\cap g^{-1}(\gamma+\mathsf{H})$ is (f,s)-tuple-substitutable.

(iii) If $L\in\mathcal{C}^{\mathrm{mcf}}_{f,s}$, $\mathsf{Acc}\subseteq\mathsf{S}^\sharp$ and, in (ii), A is finite, then $L\cap\widehat{s}^{-1}(\mathsf{Acc})\cap g^{-1}(\gamma+\mathsf{H})\in\mathcal{C}^{\mathrm{mcf}}_{f,s}$.

For infinite A the intersection in (ii) belongs to $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ whenever it is an f-MCFL.

Proof. (i) A context accepting equal-type $\vec{x},\vec{y}$ in $L_1\cap L_2$ accepts both in each L_i, so their distributions agree in each L_i, hence in the intersection. (ii) If some context E accepts $\vec{x},\vec{y}$ in $g^{-1}(\gamma+\mathsf{H})$, then $\delta:=\sum_i(g(y_i)-g(x_i))\in\mathsf{H}$, and by commutativity $g(E'[\vec{y}])-g(E'[\vec{x}])=\delta$ for every $E'\in\mathcal{E}_d$, so coset membership transfers in both directions. Thus $g^{-1}(\gamma+\mathsf{H})$ is (f,s_{triv})-, hence (f,s)-tuple-substitutable; apply (i). (iii) The language $\widehat{s}^{-1}(\mathsf{Acc})$ is (f,s)-tuple-substitutable by item (c) above, so the intersection is, by (i) and (ii); it is the intersection of an f-MCFL with a regular language, hence an f-MCFL [28]. □

The factorizations in (ii) and (iii) are a tool for constructing members of the class; they are not supplied to the reconstruction.

3 Reconstruction and Qualitative Learning

This section fixes the reconstruction operator analysed later and records its qualitative guarantees, none of which is claimed as new. For fixed f, rank cap and typing, our operator follows the distributional reconstruction pattern for substitutable languages over finite signatures in [15]. We use the MCFG signature of [15, Example 3] with tuple sorts refined by s-types (cf. [15, Section 8]). The related qualitative results are

Here	Related prior result	Treatment here
Prop. 3.10 (soundness)	[15, Lemma 2]	—
Prop. 3.11 (completeness)	[15, Lemma 3]	arbitrary same-type representatives
Thm. 3.13 (exact reconstruction)	[15, Lemmas 2–3]	as for Prop. 3.11
Cor. 3.16 (identification)	[15, Theorem 1]	unbounded rank over the class

The table records analogues, not literal identities: all statements of this section are proved directly for the concrete grammar in Appendices D and E, without embedding into the signature framework. Identifying all of $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, whose members have unbounded minimum rank, uses the sample-dependent cap $\|K\|_+$, outside the bounded-arity assumption of [15]. For a fixed finite signature, that paper bounds the *number* of words of a characteristic sample, polynomially in the grammar; it does not bound their total *length*, which is the subject of this paper.

3.1 Set-Driven Reconstruction

All interaction between templates and contexts rests on one observation.

Lemma 3.1 (Induced child contexts)**.** *Let $\rho=(\alpha_1,\dots,\alpha_{d_0})$ be a good template of rank r with child arities $d_1,\dots,d_r$, let $E\in\mathcal{E}_{d_0}$, let $1\le j\le r$, and fix $\vec{u}_i\in(\Sigma^+)^{d_i}$ for $i\ne j$. Let E_j be the word obtained from $E[(\alpha_1,\dots,\alpha_{d_0})]$ by replacing ξ^κ_j by $\square_\kappa$ and every other variable $\xi^{\kappa'}_i$ by $u_{i,\kappa'}$. Then $E_j\in\mathcal{E}_{d_j}$ and $E_j[\vec{x}]=E[\rho(\vec{u}_1,\dots,\vec{u}_{j-1},\vec{x},\vec{u}_{j+1},\dots,\vec{u}_r)]$ for every $\vec{x}\in(\Sigma^*)^{d_j}$.*

Lemma 3.2 (Composition preserves equivalence)**.** *Let $L\subseteq\Sigma^*$, let ρ be a good template of rank r, and let $\vec{x}_j,\vec{u}_j\in(\Sigma^+)^{d_j}$ satisfy $\vec{u}_j\equiv_{L,s}\vec{x}_j$ for every j. Then $\rho(\vec{u}_1,\dots,\vec{u}_r)\equiv_{L,s}\rho(\vec{x}_1,\dots,\vec{x}_r)$.*

Definition 3.3 (Composition witness)**.** Let K be finite and $r \geq 1$. A *composition witness of rank r* in K is a triple $(\vec{z}, \rho, (\vec{u}_1, \ldots, \vec{u}_r))$ in which ρ is a good template of rank r with output arity $d_0 \leq f$ and child arities $d_1, \ldots, d_r \leq f$, $\vec{z} \in \mathrm{Obs}_{d_0}(K)$, $\vec{u}_j \in (\Sigma^+)^{d_j}$, and $\vec{z} = \rho(\vec{u}_1, \ldots, \vec{u}_r)$.

Equivalently, a composition witness is a segmentation of the components of an observed $\vec{z}$ into terminal gaps and nonempty intervals labelled by variables, subject to the good-template conditions. If $E[\vec{z}] \in K$, Lemma 3.1 gives $E_j[\vec{u}_j] = E[\vec{z}] \in K$ with E_j admissible, so every $\vec{u}_j \in \mathrm{Obs}(K)$. The children occupy disjoint, nonempty parts of $\vec{z}$, so

$$r \leq \textstyle\sum_j \|\vec{u}_j\|_1 \leq \|\vec{z}\|_1 \leq \max_{w \in K} |w| \leq \|K\|_+. \tag{3.1}$$

Definition 3.4 (Capped and uncapped reconstruction operators)**.** Fix f and a typing s. For $r_{\max} \geq 1$ let $\mathcal{B}_{f,s}^{\leq r_{\max}}(K) = \widehat{G}_{\leq r_{\max}}(K)$ be the grammar with start symbol $\widehat{S}$, a nonterminal $[\vec{x}]$ of arity d for each $\vec{x} \in \mathrm{Obs}_d(K)$, $1 \leq d \leq f$, and the rules

(Start) $\widehat{S} \to [(w)]$ for $w \in K \cap \Sigma^+$, and $\widehat{S} \to \varepsilon$ if $\varepsilon \in K$;

(Const) $[\vec{x}] \to \vec{x}$ for every $\vec{x} \in \mathrm{Obs}(K)$;

(Comp) $[\vec{z}] \to \rho([\vec{u}_1], \ldots, [\vec{u}_r])$ for every composition witness of rank $1 \leq r \leq r_{\max}$;

(Link) $[\vec{x}] \to [\vec{y}]$ for $\vec{x}, \vec{y} \in \mathrm{Obs}_d(K)$ with $s^{(d)}(\vec{x}) = s^{(d)}(\vec{y})$ and $E[\vec{x}], E[\vec{y}] \in K$ for some $E \in \mathcal{E}_d$.

For $K \neq \emptyset$ put $\mathcal{B}_{f,s}(K) := \widehat{G}(K) := \widehat{G}_{\leq \|K\|_+}(K)$; $\mathcal{B}_{f,s}(\emptyset)$ has no rules. Fixed encoding orders make both operators deterministic and set-driven.

By (3.1), $\mathcal{B}_{f,s}(K)$ contains a Comp rule for every composition witness in K. Both operators are sample-consistent, via $\widehat{S} \to [(w)] \to (w)$ and $\widehat{S} \to \varepsilon$. All non-start templates are ε-free, so $\widehat{S} \Rightarrow^* \varepsilon$ only by the rule $\widehat{S} \to \varepsilon$. For the trivial typing and $\varepsilon \notin K$, Comp/Const, Link and Start correspond to the rules of Types I–III of [14, Section 4.1], and $\mathcal{B}_{f,s}$ corresponds to the operator of the unbounded-rank learner $\mathcal{A}(p, *)$ of [14, Corollary 2].

3.2 Yield-Typed Refinement and Witnesses

Definition 3.5 (Yield-type refinement)**.** Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a finite nondeleting MCFG and s a typing. For a template ρ and $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{d_j}$, let $\mathrm{out}_\rho(\mathbf{t}_1, \ldots, \mathbf{t}_r) \in (\mathsf{S}^\sharp)^{d_0}$ have ith component the image of α_i under the monoid morphism $\Gamma_\rho^* \to \mathsf{S}^\sharp$ extending $a \mapsto \widehat{s}(a)$ and $\xi_j^\kappa \mapsto (\mathbf{t}_j)_\kappa$. The *full yield-type refinement* $G^{\widehat{s}}$ has a fresh start symbol $\widetilde{S}$, nonterminals $A_{\mathbf{t}}$ for $A \in V$ and $\mathbf{t} \in (\mathsf{S}^\sharp)^{\mathrm{fan}(A)}$, start rules $\widetilde{S} \to S_{(m)}$ for $m \in \mathsf{S}^\sharp$, and for every rule $A \to \rho(B_1, \ldots, B_r)$ (including $r = 0$) and all $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{\mathrm{fan}(B_j)}$ the rule $A_{\mathrm{out}_\rho(\mathbf{t}_1, \ldots, \mathbf{t}_r)} \to \rho(B_{1,\mathbf{t}_1}, \ldots, B_{r,\mathbf{t}_r})$. The *trimmed refinement* $\widetilde{G}_0$ keeps exactly the nonterminals and rules occurring in some successful derivation from $\widetilde{S}$. Unless $G^{\widehat{s}}$ is named, *typed nonterminals* and *typed rules* are the non-start nonterminals and rules of $\widetilde{G}_0$, and $L_X := \{\vec{u} : X \Rightarrow^*_{\widetilde{G}_0} \vec{u}\}$.

Typed rules have the templates of their sources, so $\widetilde{G}_0$ is good whenever G is; each L_X is nonempty, and its tuples have nonempty components if G is ε-free.

Proposition 3.6 (Yield-type invariants)**.** *(i) If $A_{\mathbf{t}} \Rightarrow^* \vec{u}$ in $G^{\widehat{s}}$, then $\widehat{s}^{(\mathrm{fan}(A))}(\vec{u}) = \mathbf{t}$, and erasing indices gives a G-derivation of $\vec{u}$.*

(ii) Every terminal derivation $A \Rightarrow^*_G \vec{u}$ lifts uniquely to $A_{\widehat{s}^{(\mathrm{fan}(A))}(\vec{u})} \Rightarrow^*_{G^{\widehat{s}}} \vec{u}$; a successful derivation of w lifts uniquely through $\widetilde{S} \to S_{(\widehat{s}(w))}$.

(iii) $L(\widetilde{G}_0) = L(G^{\widehat{s}}) = L(G)$.

(iv) *For every typed nonterminal X, every terminal derivation $X \Rightarrow^*_{\widehat{G^s}} \vec{u}$ uses only rules of $\widetilde{G}_0$; hence* $L_X = \{\vec{u} : X \Rightarrow^*_{\widehat{G^s}} \vec{u}\}$.

Lemma 3.7 (Canonical contexts for typed nonterminals)**.** *Let G be good. Every typed nonterminal X of arity d has a context $\chi(X) \in \mathcal{E}_d$ with $\chi(X)[\vec{u}] \in L(G)$ for all $\vec{u} \in L_X$; if $\widetilde{S} \to X$ is in $\widetilde{G}_0$, one may take $\chi(X) = \square_1$.*

Lemma 3.8 (Typed nonterminals are distributionally homogeneous)**.** *Let G be good of fan-out $\le f$ with $L := L(G) \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. For every typed nonterminal X of arity d, L_X lies in a single $\equiv^d_{L,s}$-class, denoted* $\mathrm{Cl}_{L,s}(X)$.

The semantic class $\mathrm{Cl}_{L,s}(X)$ is distinct from the syntactic reconstruction nonterminal $[\vec{x}]$ of Definition 3.4.

The proofs of these three statements are in Appendix E. The homogeneity of Lemma 3.8 is weaker than that of congruential MCFGs [17], whose contexts may have empty separators.

Definition 3.9 (Canonical tuples, contexts and the witness set)**.** Let G be good. For every typed nonterminal X fix a *canonical tuple* $\omega(X) \in L_X$ and a *canonical context* $\chi(X)$ as in Lemma 3.7, with $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Let $\mathsf{W}(\widetilde{G}_0)$ consist of the words

$$\chi(X)[\omega(X)] \ \ (X \text{ typed}), \qquad \chi(X)[\rho(\omega(Y_1), \ldots, \omega(Y_q))] \ \ (X \to \rho(Y_1, \ldots, Y_q) \text{ typed}, \ q \ge 0).$$

Since $\omega(X)$ and $\rho(\omega(Y_1), \ldots, \omega(Y_q))$ belong to L_X, $\mathsf{W}(\widetilde{G}_0)$ is a finite subset of $L(G)$, nonempty if $L(G) \neq \emptyset$, and $|\mathsf{W}(\widetilde{G}_0)|$ is at most the number of typed nonterminals and rules. Canonical objects are properties of the target, not inputs of the operator: qualitatively any choice works, while Section 4 chooses short ones.

3.3 Qualitative Guarantees

Proposition 3.10 (Soundness of every hypothesis)**.** *Let L be (f,s)-tuple-substitutable and $K \subseteq L$ finite. For every $r_{\max} \ge 1$, if $[\vec{x}] \Rightarrow^* \vec{u}$ in $\widehat{G}_{\le r_{\max}}(K)$, then $\vec{u} \equiv_{L,s} \vec{x}$. Consequently $L(\mathcal{B}^{\le r_{\max}}_{f,s}(K)) \subseteq L$ and $L(\mathcal{B}_{f,s}(K)) \subseteq L$.*

Proposition 3.11 (Completeness from a witness assignment)**.** *Let G be a finite good MCFG of fan-out $\le f$, let r_G be the maximum rank of a typed rule of $\widetilde{G}_0$ ($r_G := 0$ if none), and assign to each typed nonterminal $X = A_{\mathbf{t}}$ a tuple $\iota(X) \in (\Sigma^+)^{\mathrm{fan}(A)}$ with $\widehat{s}^{(\mathrm{fan}(A))}(\iota(X)) = \mathbf{t}$ and a context $\chi(X) \in \mathcal{E}_{\mathrm{fan}(A)}$, with $\chi(X) = \square_1$ if $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Suppose K contains $\chi(X)[\iota(X)]$ for each X and $\chi(X)[\rho(\iota(Y_1), \ldots, \iota(Y_q))]$ for each typed rule $X \to \rho(Y_1, \ldots, Y_q)$, $q \ge 0$. Then*

$$L(G) \subseteq L(\mathcal{B}^{\le r_{\max}}_{f,s}(K)) \quad (r_{\max} \ge \max\{1, r_G\}), \qquad L(G) \subseteq L(\mathcal{B}_{f,s}(K)).$$

Neither substitutability nor $\iota(X) \in L_X$ is assumed in Proposition 3.11; $\iota = \omega$ is the case $\mathsf{W}(\widetilde{G}_0) \subseteq K$.

Exact reconstruction needs a good presentation of the target; the following normalization controls size, fan-out and rank, but not thickness. The rule thickness of G_{good} may exceed that of G_{in}, so all data bounds below are stated relative to a given reduced good presentation and are not transferred through normalization.

Proposition 3.12 (Good normalization with rank preservation)**.** *Let G_{in} be a finite standard f-MCFG of rule rank at most $r_{\max}$. There is a finite good f-MCFG G_{good} of rule rank at most $r_{\max}$ with $L(G_{\mathrm{good}}) = L(G_{\mathrm{in}}) \setminus \{\varepsilon\}$ and, for fixed $f, r_{\max}$, $\|G_{\mathrm{good}}\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|)$. If $L(G_{\mathrm{in}}) \cap \Sigma^+ \neq \emptyset$, G_{good} is reduced.*

Theorem 3.13 (Exact reconstruction for a fixed typing)**.** *Fix $f \geq 1$ and a typing s, and let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. If $L \cap \Sigma^+ = \emptyset$ put $\mathcal{F}_L := L$ and $r_G := 0$. Otherwise let G be a reduced good fan-out-$\leq f$ presentation of $L \cap \Sigma^+$ (Proposition 3.12), let r_G be as in Proposition 3.11, and put $\mathcal{F}_L := \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$. Then $\mathcal{F}_L \subseteq L$ is finite, and every finite K with $\mathcal{F}_L \subseteq K \subseteq L$ satisfies*

$$L(\mathcal{B}_{f,s}(K)) = L, \qquad L(\mathcal{B}^{\leq r_{\max}}_{f,s}(K)) = L \quad \text{for every } r_{\max} \geq \max\{1, r_G\}.$$

If $L \in \mathsf{MCFL}(f, r)$, G can be chosen of rank $\leq r$, so $r_{\max} = \max\{1, r\}$ suffices (also when $r = 0$).

Theorem 3.14 (Polynomial construction under a rule-rank cap)**.** *Fix $f, r_{\max} \geq 1$ and s. The encoding of $\widehat{G}_{\leq r_{\max}}(K)$ is computable in time $(1 + \|K\|_+)^{O(f r_{\max})}$; its fan-out is at most f and its rank at most $r_{\max}$.*

Definition 3.15 (Conservative sequential learner)**.** The learner $\mathcal{A}_{f,s}$ (respectively $\mathcal{A}^{\leq r_{\max}}_{f,s}$) starts with H_0, keeps its hypothesis on $\perp$ and on every datum the hypothesis generates, and otherwise replaces it by $\mathcal{B}_{f,s}(K)$ (respectively $\mathcal{B}^{\leq r_{\max}}_{f,s}(K)$), where K is the content read so far. This is the policy of [14, Algorithm 1].

Corollary 3.16 (Gold identification)**.** *Fix $f \geq 1$ and s.*

(i) *$\mathcal{A}_{f,s}$ TxtEx-identifies $\mathcal{C}^{\mathrm{mcf}}_{f,s}$.*

(ii) *For fixed $r_{\max}$, $\mathcal{A}^{\leq r_{\max}}_{f,s}$ TxtEx-identifies $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r_{\max})$, and processes each datum in time polynomial in $\|K\|_+$ for the content K read so far.*

Proofs are in Appendix D (Proposition 3.12: Appendix F). None of these statements bounds the length of characteristic data or the stage of convergence.

3.4 Typed versus Untyped Classes

Regular path languages give simple members of every class.

Lemma 3.17 (Path languages)**.** *Let R be the set of labels of paths from a vertex p to a vertex p' in a finite directed graph in which every letter labels exactly one edge. If $\vec{x}, \vec{y} \in (\Sigma^+)^d$ and $E \in \mathcal{E}_d$ satisfy $E[\vec{x}], E[\vec{y}] \in R$, then $F[\vec{x}] \in R \iff F[\vec{y}] \in R$ for every $F \in \mathcal{E}_d$.*

Proof. A word of Σ^+ lies in R iff the target of the edge of each letter is the source of the edge of the next one, the first edge starts at p and the last ends at p'. The h-th hole of E is preceded by the last letter of v_{h-1}, or by the vertex p when $h = 1$ and $v_0 = \varepsilon$, and is followed likewise; internal separators are nonempty. So E determines the vertices at both ends of each hole, and x_h and y_h label paths between the same two vertices. Replacing every x_h by y_h in $F[\vec{x}]$ therefore preserves all three conditions, and conversely. □

The class grows strictly with the typing already at fan-out two, and outside the context-free languages; the untyped operator then over-generalizes.

Proposition 3.18 (A typed class beyond the untyped one)**.** *Let* $\Sigma = \{\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}, \#_1, \#_2, \#_3\}$*, let* $s : \Sigma^+ \twoheadrightarrow \mathbb{Z}_2$ *count the letters* a *and* c *modulo* 2*, and let*

$$L_{\mathrm{sep}} := \{\mathtt{a}^m \#_1 \mathtt{b}^n \#_2 \mathtt{c}^{m'} \#_3 \mathtt{d}^n : m, n \geq 0,\ m' \in \{m, m-1\},\ m' \geq 0\}.$$

(i) L_{sep} *is a* 2*-MCFL of rule rank two and is not context-free.*

(ii) L_{sep} *is* (f, s)*-tuple-substitutable for every* $f \geq 1$*; hence* $L_{\mathrm{sep}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ *for every* $f \geq 2$*.*

(iii) L_{sep} *is not* (f, s_{triv})*-tuple-substitutable for any* $f \geq 1$*: the words* a *and* aa *share the context* $\square_1 \#_1 \#_2 \mathtt{c} \#_3$ *but not the context* $\square_1 \#_1 \#_2 \#_3$*. Accordingly, on the sample* $K = \{\mathtt{a}\#_1\#_2\mathtt{c}\#_3,\ \mathtt{aa}\#_1\#_2\mathtt{c}\#_3,\ \mathtt{a}\#_1\#_2\#_3\} \subseteq L_{\mathrm{sep}}$ *the untyped operator* $\mathcal{B}_{f,s_{\mathrm{triv}}}$ *derives* $\mathtt{aa}\#_1\#_2\#_3 \notin L_{\mathrm{sep}}$*, whereas* $\mathcal{B}_{f,s}$ *does not link* a *with* aa*.*

Proof. Let $\psi : \Sigma^* \to \mathbb{Z}^2$ be the morphism with $\psi(\mathtt{a}) = (1,0)$, $\psi(\mathtt{c}) = (-1,0)$, $\psi(\mathtt{b}) = (0,1)$, $\psi(\mathtt{d}) = (0,-1)$ and $\psi(\#_i) = (0,0)$, and let $R := \mathtt{a}^*\#_1\mathtt{b}^*\#_2\mathtt{c}^*\#_3\mathtt{d}^*$. Then $L_{\mathrm{sep}} = R \cap \psi^{-1}(P)$ with $P := \{(0,0),(1,0)\}$, and R is the path language from 0 to 3 of the graph on $\{0,1,2,3\}$ with loops $\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}$ at $0, 1, 2, 3$ and edges $\#_h : h-1 \to h$; every letter labels exactly one edge.

(i) The rules $E \to (\varepsilon, \varepsilon) \mid (\mathtt{a}\xi^1_1, \mathtt{c}\xi^2_1)(E)$, $D \to (\varepsilon, \varepsilon) \mid (\mathtt{b}\xi^1_1, \mathtt{d}\xi^2_1)(D)$ and

$$S \to (\xi^1_1\#_1\xi^1_2\#_2\xi^2_1\#_3\xi^2_2)(E, D) \mid (\mathtt{a}\xi^1_1\#_1\xi^1_2\#_2\xi^2_1\#_3\xi^2_2)(E, D)$$

form a standard MCFG of fan-out two and rule rank two generating L_{sep}, since E derives the pairs $(\mathtt{a}^k, \mathtt{c}^k)$ and D the pairs $(\mathtt{b}^n, \mathtt{d}^n)$. The words of L_{sep} in which a and c occur an even number of times altogether are exactly those with $m' = m$; this is an intersection with a regular language, and erasing the markers maps the result onto $\{\mathtt{a}^m\mathtt{b}^n\mathtt{c}^m\mathtt{d}^n : m, n \geq 0\}$, which is not context-free. Context-free languages are closed under both operations.

(ii) Let $1 \leq d \leq f$, let $\vec{x}, \vec{y} \in (\Sigma^+)^d$ have equal componentwise s-types, and let $E \in \mathcal{E}_d$ accept both. Put $(\delta_1, \delta_2) := \sum_i (\psi(x_i) - \psi(y_i)) = \psi(E[\vec{x}]) - \psi(E[\vec{y}]) \in P - P$, so $\delta_2 = 0$ and $\delta_1 \in \{-1, 0, 1\}$. The first coordinate of ψ is congruent to s modulo 2 on every word, so δ_1 is even; hence $\psi(F[\vec{x}]) = \psi(F[\vec{y}])$ for every $F \in \mathcal{E}_d$. By Lemma 3.17, $F[\vec{x}] \in R$ iff $F[\vec{y}] \in R$. So F accepts $\vec{x}$ iff it accepts $\vec{y}$. With (i), $L_{\mathrm{sep}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ for $f \geq 2$.

(iii) The two contexts give $\mathtt{a}\#_1\#_2\mathtt{c}\#_3,\ \mathtt{aa}\#_1\#_2\mathtt{c}\#_3 \in L_{\mathrm{sep}}$ ($m' = m$ and $m' = m-1$), $\mathtt{a}\#_1\#_2\#_3 \in L_{\mathrm{sep}}$ and $\mathtt{aa}\#_1\#_2\#_3 \notin L_{\mathrm{sep}}$. On K, the first context gives the Link rule $[(\mathtt{a})] \to [(\mathtt{aa})]$ for the trivial typing, and $((\mathtt{a}\#_1\#_2\#_3), (\xi^1_1\#_1\#_2\#_3), ((\mathtt{a})))$ is a composition witness of rank one, so $\widehat{S} \to [(\mathtt{a}\#_1\#_2\#_3)] \to [(\mathtt{a})]\#_1\#_2\#_3 \to [(\mathtt{aa})]\#_1\#_2\#_3 \to \mathtt{aa}\#_1\#_2\#_3$. Under s the types of a and aa differ, so no such Link rule exists. □

The positive image of s is the group $\mathbb{Z}_2$, a local group, so Theorem A gives L_{sep} characteristic data of length $O(\|G\|^2\bar{\theta}_G)$ for each of its reduced good presentations, at every fan-out $f \geq 2$, while L_{sep} lies outside the class of Yoshinaka's untyped learner [14] at every fan-out.

4 Characteristic-Data Bounds

Throughout this section f, $r_{\max} \geq 1$, Σ and the typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ are fixed; $G = (V, \Sigma, P, S, \mathrm{fan})$ is a reduced good MCFG of fan-out $\leq f$ and rule rank $\leq r_{\max}$, and $\widetilde{G}_0$ is its trimmed $\widehat{s}$-refinement. Under a rank cap, Theorem 3.13 already supplies $O(\|G\|)$ characteristic words; the task is to bound their total length *in terms of* G. All bounds follow one scheme.

(1) *Witness length* (Lemma 4.2). If, for an auxiliary finite-monoid morphism λ refining s, every tuple of every nonterminal can be exchanged for a tuple of the same nonterminal and the same λ-vector of total length at most β, then $\mathsf{W}(\widetilde{G}_0)$ is characteristic and has total length $O(\|G\|(\bar{\theta}_G + \beta))$.

(2) *Minimal counterexamples* (Section 5.1). In a shortest derivation with a prescribed root λ-vector, a large antichain of heavy subtrees would admit a simultaneous replacement preserving that vector; so antichains are small, and Lemma 5.1 turns this into $\beta = O(\|G\|\bar{\theta}_G)$ or $O(\|G\|^2\bar{\theta}_G)$.

(3) *Regimes.* Only λ and the replacement step change: window typings keep marked boundary letters; window–group typings in addition preserve ordered group products (Proposition 5.2); local-group typings reduce to these through sliding-factor coordinates (Section 5.3).

The full yield-type refinement is not used as the primary thickness measure, since it may be exponentially thicker than G (Proposition 4.7). The proofs of this section are in Appendix E.

4.1 Thickness Scales and Witness Length

For a rule R of G let

$$\theta_G(R) := \min\{\, |w| : w \in L(G) \text{ has a successful derivation using } R \,\},$$

the length of the derived *sentence*, not of the derivation. Put $\theta_G := \max_{R\in P} \theta_G(R)$, $\bar{\theta}_G := \max\{1, \theta_G\}$ and $t_G := \max_{A\in V} \min\{\|\vec{w}\|_1 : \vec{w} \in L_A(G)\}$ (rule and nonterminal thickness; θ_G is the thickness of [14, Section 4.3]).

If a target L contains ε, every statement below is applied to $L \cap \Sigma^+$ and the datum ε is added, as in Theorem 3.13; this adds one to the encoded length.

Lemma 4.1 (Thickness lemma)**.** *Let G be reduced and good.*

(i) *Every rule writes at most θ_G terminal symbols explicitly.*

(ii) *$t_G \le \theta_G$: every $A \in V$ has some $\vec{u} \in L_A(G)$ with $\|\vec{u}\|_1 \le \theta_G$.*

(iii) *$\theta_G \le 2\|G\| \max\{1, t_G\}$.*

(iv) *Every $A \in V$ of fan-out d has a context $E_A \in \mathcal{E}_d$ with $|E_A|_{\mathrm{fix}} \le \theta_G$ that results from a successful derivation by deleting one occurrence of an A-subtree; $E_S = \square_1$, and $E_A[\vec{u}] \in L(G)$ for all $\vec{u} \in L_A(G)$. For every typed nonterminal $X = A_{\mathbf{t}}$, the choice $\chi(X) := E_A$ meets Definition 3.9 and is* derivational *for X: it results from a successful derivation of $\widetilde{G}_0$ by deleting an occurrence of an X-subtree.*

Thus θ_G and t_G are polynomially equivalent on reduced good presentations; Yoshinaka's efficiency criterion measures data in target size and rule thickness [14, Definition 1]; see also [10].

For a finite-monoid morphism $\lambda : \Sigma^* \to \mathsf{N}$ we write $s \preceq \lambda$ if $\ker(\lambda|_{\Sigma^+}) \subseteq \ker s$; since $\widehat{s}$ separates ε, this is the relation of Definition 2.3 whenever λ also separates ε from Σ^+. For $\beta \ge 0$ consider

($\mathrm{R}^{\lambda}_{\beta}$) for every $B \in V$ and $\vec{u} \in L_B(G)$ there is $\vec{u}' \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}') = \lambda^{(\mathrm{fan}(B))}(\vec{u})$ and $\|\vec{u}'\|_1 \le \beta$.

Put

$$C_s(G) := |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}} \;\le\; |\mathsf{S}^\sharp|^{f r_{\max}} \|G\|,$$

using $|V| + |P| \le \|G\|$; it bounds the numbers of typed nonterminals and typed rules together, since a typed rule is determined by its source rule and the type vectors of its at most $r_{\max}$ children.

Lemma 4.2 (Witness length)**.** *Let $\lambda : \Sigma^* \to \mathsf{N}$ be a finite-monoid morphism with $s \preceq \lambda$.*

(i) (Type-preserving exchange.) *If $X = B_{\mathbf{t}}$ is a typed nonterminal, $\vec{u} \in L_X$, $\vec{u}' \in L_B(G)$ and $\lambda^{(\mathrm{fan}(B))}(\vec{u}') = \lambda^{(\mathrm{fan}(B))}(\vec{u})$, then $\vec{u}' \in L_X$. Hence under $(\mathrm{R}^{\lambda}_{\beta})$ there are canonical tuples with $\|\omega(X)\|_1 \le \beta$ for every X.*

(ii) (Length.) *Let $\chi(X) := E_B$ for $X = B_{\mathbf{t}}$, as in Lemma 4.1(iv), and let ι assign to every typed X a tuple $\iota(X)$ of nonempty words of its arity with $\|\iota(X)\|_1 \le b$. Let W_ι consist of the words $\chi(X)[\iota(X)]$ and $\chi(X)[\rho(\iota(Y_1), \dots, \iota(Y_q))]$ for typed rules $X \to \rho(Y_1, \dots, Y_q)$, $q \ge 0$; thus $\mathsf{W}_\omega = \mathsf{W}(\widetilde{G}_0)$. Every word of W_ι has length at most $2\bar{\theta}_G + r_{\max} b$, and*

$$|\mathsf{W}_\iota| \le C_s(G), \qquad \|\mathsf{W}_\iota\|_+ \le C_s(G)\,(1 + 2\bar{\theta}_G + r_{\max} b).$$

(iii) (Refined thickness.) *In (ii), each rule word $\chi(X)[\rho(\omega(Y_1), \dots, \omega(Y_q))]$ is the yield of a successful derivation of $\widetilde{G}_0$ using the typed rule $X \to \rho(Y_1, \dots, Y_q)$; hence $\theta_{\widetilde{G}_0} \le 2\bar{\theta}_G + r_{\max} b$.*

(iv) (Characteristic sample.) *Let $L(G) = L \cap \Sigma^+$ for some $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, and in* (ii) *let $\iota(X) \in \mathrm{Cl}_{L,s}(X)$ for every X (Lemma 3.8), for instance $\iota = \omega$. Then $\mathsf{W}_\iota \subseteq L$, and $\mathsf{W}_\iota \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for $\mathcal{B}^{\le r_{\max}}_{f,s}$ and for $\mathcal{B}_{f,s}$.*

Items (i)–(iii) of Lemma 4.2 do not use substitutability.

4.2 Typed Thickness and Characteristic Data

Recall that $\widetilde{G}_0$ is the trimmed full yield-type refinement of G (Definition 3.5). Define

$$t_{\mathrm{typ}}(G, s) := \max_X \min_{\vec{u} \in L_X} \|\vec{u}\|_1,$$

where X ranges over the non-start typed nonterminals, and let $N_{\mathrm{typ}}(G, s)$ denote their number plus the number of non-start typed rules. In particular $N_{\mathrm{typ}} \le C_s(G) \le |\mathsf{S}^\sharp|^{f r_{\max}} \|G\|$.

Theorem 4.3 (Characteristic data from typed thickness)**.** *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and G be a reduced good fan-out-$\le f$, rank-$\le r_{\max}$ grammar with $L(G) = L \cap \Sigma^+$. Take shortest canonical tuples and the derivational contexts E_B of Lemma 4.1(iv). Then $K = \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for both $\mathcal{B}^{\le r_{\max}}_{f,s}$ and $\mathcal{B}_{f,s}$, with*

$$|\mathsf{W}(\widetilde{G}_0)| \le N_{\mathrm{typ}}, \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \le N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}).$$

For the full sample K, the length bound has the additional term $\mathbf{1}_{\{\varepsilon \in L\}}$.

The numerical length estimate holds for any typing; substitutability is used only for the characteristic-sample claim.

For the trivial typing, every nonterminal of the ε-free G has one realized type vector, so $t_{\mathrm{typ}} = t_G$ and $N_{\mathrm{typ}} = |V| + |P| \le \|G\|$. The theorem recovers the $O(\|G\|\bar{\theta}_G)$ untyped scale, although Yoshinaka's original sample has the sharper bounds $|K_G| \le |P|$ and $\|K_G\| \le |P|\theta_G$ [14, Lemma 7 and Theorem 1]. A nontrivial typing may instead make t_{typ} exponential.

Presentation-independent lower bounds from rare types. For $L \subseteq \Sigma^*$ and $m \in \mathsf{S}$ put

$$\mu_m(L) := \min\{|w| : w \in L \cap \Sigma^+,\ s(w) = m\}, \qquad (\min \emptyset := \infty).$$

For a nonempty set of realized types $\mathsf{S}_L := s(L \cap \Sigma^+)$, put $\mu^\star(L,s) := \max_{m \in \mathsf{S}_L} \mu_m(L)$; set $\mu^\star(L,s) = 0$ when $\mathsf{S}_L = \emptyset$. A typed start nonterminal $S_{(m)}$ exists for each $m \in \mathsf{S}_L$, and $L_{S_{(m)}} = \{(w) : w \in L \cap \Sigma^+,\ s(w) = m\}$ (Proposition 3.6). Thus, whenever $L \cap \Sigma^+ \neq \emptyset$,

$$\mu^\star(L,s) \le t_{\mathrm{typ}}(G,s) \qquad \text{for every reduced good presentation } G. \tag{4.1}$$

Lemma 4.4 (Deleting an entire yield type)**.** *If $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and $m \in \mathsf{S}$, then $L^{\setminus m} := L \setminus s^{-1}(m)$ also belongs to $\mathcal{C}^{\mathrm{mcf}}_{f,s}$. Moreover, $L \setminus \{\varepsilon\} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$.*

Proof. Apply Proposition 2.9(iii) with the trivial group and $\mathsf{Acc} = \mathsf{S}^\sharp \setminus \{m\}$, respectively $\mathsf{Acc} = \mathsf{S}$. □

All lower bounds of this paper rest on one principle.

Lemma 4.5 (Distinguishing sub-targets)**.** *Let $L, L' \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ with $L' \subsetneq L$. Every characteristic sample of L for $\mathcal{B}^{\le r}_{f,s}$ ($r \ge 1$) or $\mathcal{B}_{f,s}$, and every characteristic sample of L for a set-driven operator having one for L' as well, contains a word of $L \setminus L'$.*

Proof. Let $F \subseteq L'$ be finite. For the canonical operators, Proposition 3.10 applied to the target L' gives $L(\mathcal{B}(F)) \subseteq L' \neq L$, so F is not characteristic for L. For the other operator, let F' be characteristic for L'; then $F \cup F' \subseteq L' \subseteq L$, so if F were characteristic for L, the output on $F \cup F'$ would be both L and L'. □

Proposition 4.6 (Rare types force positive data)**.** *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, and let $m \in \mathsf{S}_L$. Every characteristic sample of L for $\mathcal{B}^{\le r}_{f,s}$ ($r \ge 1$) or for $\mathcal{B}_{f,s}$ contains a word of type m. The same is true for any set-driven operator having characteristic samples for both L and $L^{\setminus m}$, in particular for an operator having characteristic samples for the entire class $\mathcal{C}^{\mathrm{mcf}}_{f,s}$. If $\varepsilon \in L$, its sample for either canonical operator must also contain ε; the same holds for a set-driven operator admitting a characteristic sample for $L \setminus \{\varepsilon\}$ as well. Consequently, for the canonical operators and for class-wide set-driven operators, every characteristic sample F satisfies*

$$\|F\|_+ \ge \sum_{m \in \mathsf{S}_L} (\mu_m(L) + 1) + \mathbf{1}_{\{\varepsilon \in L\}}.$$

Proposition 4.7 (Exponential blow-up of full yield-type refinement)**.** *Let $\Sigma = \{\mathtt{a}, \mathtt{c}\}$ and let $s_{\mathtt{c}}$ be the typing onto $U_1 = \{1, \zeta\}$ ($\zeta^2 = \zeta$, Section 7) with $s_{\mathtt{c}}(\mathtt{a}) = 1$ and $s_{\mathtt{c}}(\mathtt{c}) = \zeta$. For $n \ge 1$ the fan-out-one, rank-two grammar*

$$S \to SS \mid \mathtt{a} \mid \mathtt{c} \mid A_0 \mid \cdots \mid A_n, \qquad A_0 \to \mathtt{a}, \qquad A_i \to A_{i-1}A_{i-1} \mid \mathtt{c} \quad (1 \le i \le n)$$

is reduced and good, generates Σ^+, and has $\|G_n\| = O(n)$ and $\theta_{G_n} \le 2$, but $t_{\mathrm{typ}}(G_n, s_{\mathtt{c}}) \ge 2^n$ and $\theta_{\widetilde{G}_{n,0}} \ge 2^n$.

This blow-up concerns the witness construction, not the data, because the tuples used for a typed nonterminal need not be its yields. For a typed nonterminal X, Lemma 3.8 places L_X in one class $\mathrm{Cl}_{L,s}(X)$; the *representative length* is

$$\tau_X := \min\{\|\vec{v}\|_1 : \vec{v} \in \mathrm{Cl}_{L,s}(X)\}, \qquad \tau(G,s) := \max_X \tau_X \quad (\max \emptyset := 0).$$

Corollary 4.8 (Characteristic data from equivalent representatives)**.** *Under the hypotheses of Theorem 4.3,*

$$\mu^\star(L,s) \le \tau(G,s) \le t_{\mathrm{typ}}(G,s),$$

and L has a characteristic sample K for both $\mathcal{B}_{f,s}^{\le r_{\max}}$ and $\mathcal{B}_{f,s}$ with

$$\|K\|_+ \le N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max}\tau(G,s)) + \mathbf{1}_{\{\varepsilon\in L\}}.$$

In Proposition 4.7, $L = \Sigma^+$ and every admissible context accepts every tuple, so two tuples are equivalent exactly when their types agree; single letters represent both types, and $\tau(G_n, s_{\mathsf{c}}) = 1$ although $t_{\mathrm{typ}}(G_n, s_{\mathsf{c}}) \ge 2^n$. Corollary 4.8 then gives characteristic data of length $O(\|G_n\|)$. The corollary does not bound τ itself; the proofs of Section 5 bound t_{typ} and therefore work on G. Theorem 7.10 instead bounds τ directly at fan-out one, for the type of an isolated letter, and so for two-letter semilattice typings and for typings onto U_1 with one letter of type 1 (Corollary 7.11), where t_{typ} can be exponential.

5 Upper Bound: Local Groups at Every Fan-Out

This section proves Theorem A. The difficulty specific to multiple context-free grammars is that a subtree contributes several blocks to different root components, interleaved with blocks of other subtrees; for noncommutative groups the effect of replacing several subtrees at once depends on this order.

5.1 Short Representatives by Minimal Counterexamples

Let $\lambda : \Sigma^* \to \mathsf{N}$ be a finite-monoid morphism. A vector $\mathbf{t} \in \mathsf{N}^d$ is *realizable at B* (of fan-out d) if $\mathbf{t} = \lambda^{(d)}(\vec{u})$ for some $\vec{u} \in L_B(G)$. A *λ-minimal* derivation for $(B, \mathbf{t})$ is a terminal derivation from B with root λ-vector $\mathbf{t}$ that minimizes first the total yield length and then the number of nodes. For a threshold b, a node is *heavy* if its yield has total length greater than b.

Let $n_\lambda(G)$ be the number of pairs $(C, \lambda^{(\mathrm{fan}(C))}(\vec{u}))$ with $C \in V$ and $\vec{u} \in L_C(G)$, the *realized labels*; thus $n_\lambda(G) \le |V||\mathsf{N}|^f$.

Lemma 5.1 (Heavy-node count and shortening principle)**.** (i) *Let $b \ge \bar{\theta}_G$ and $c \ge 0$, and let T be λ-minimal for $(B, \mathbf{t})$. If every antichain of heavy nodes of T has at most c elements, then*

$$\|\textit{root tuple of } T\|_1 \le (r_{\max}+1)\, b\, \max\{1, c\, n_\lambda(G)\}.$$

(ii) (Shortening principle.) *Let $c, c' \ge 0$ and $b = \bar{\theta}_G$. Suppose that for every λ-minimal T and every antichain Q of heavy nodes of T there is $Q_0 \subseteq Q$ with $|Q \setminus Q_0| \le c'$ such that, if $|Q_0| > c$, the subtrees at the nodes of some nonempty $J \subseteq Q_0$ can be replaced simultaneously by derivations, from the same nonterminals, of tuples of total length at most $\bar{\theta}_G$ without changing the root λ-vector. Then every heavy antichain of a λ-minimal derivation has at most $c + c'$ elements, and $(\mathrm{R}_\beta^\lambda)$ holds with $\beta = (r_{\max}+1)\,\bar{\theta}_G \max\{1, (c+c')\, n_\lambda(G)\}$.*

Proof. (i) Label each node by its nonterminal and the λ-vector of its yield; only realized labels occur. If an ancestor and a proper descendant had the same label, replacing the ancestor's subtree by the descendant's would keep the root λ-vector (templates evaluate λ homomorphically), would not lengthen the root yield (by linearity the descendant's yield is part of the ancestor's), and would remove nodes, contradicting minimality. So every root-to-leaf path has at most $n_\lambda(G)$ nodes. If

the root is light its yield has length at most b. Otherwise the heavy nodes form an ancestor-closed subtree whose leaves form an antichain, so there are at most $c\, n_\lambda(G)$ heavy nodes. Every terminal of the root yield is written by a heavy node (at most $\bar{\theta}_G \le b$ each, Lemma 4.1(i)) or lies in a light child of a heavy node (at most $r_{\max}$ children of length at most b).

(ii) If $|Q_0| > c$, the replacement gives a derivation from B (the nodes of J are incomparable) with root λ-vector $\mathbf{t}$ and a strictly shorter root yield, since every replaced yield had total length greater than $\bar{\theta}_G$ and, by linearity and nondeletion, occurs exactly once in the root yield. This contradicts minimality, so $|Q| \le c + c'$. For $\vec{u} \in L_B(G)$ apply (i) to a λ-minimal derivation for $(B, \lambda^{(\mathrm{fan}(B))}(\vec{u}))$. □

Group constants. For a finite abelian group $\mathbb{H}$, $\mathsf{d}(\mathbb{H})$ is the maximal length of a zero-sum-free sequence over $\mathbb{H}$ (one with no nonempty subsequence summing to the identity; thus $\mathsf{d}(\mathbb{H}) + 1$ is the Davenport constant; $\mathsf{d}(\{1\}) = 0$). For a finite group $\mathbb{G}$ and $f \ge 1$ put

$$N_{\mathbb{G},f} = \max\{3, |\mathbb{G}|^f\}, \quad \mathcal{P}_f = f^{f+1} f!, \quad M_{\mathbb{G},f} = \mathcal{P}_f\, N_{\mathbb{G},f}^{2^{f^2}},$$

$$c_{\mathbb{G},f} := \begin{cases} \mathsf{d}(\mathbb{G}^f), & \mathbb{G} \text{ abelian}, \\ |\mathbb{G}| - 1, & \mathbb{G} \text{ nonabelian}, \ f = 1, \\ M_{\mathbb{G},f} - 1, & \text{otherwise}. \end{cases}$$

$\mathcal{P}_f$ bounds the number of block patterns of one owner, and $N_{\mathbb{G},f} \ge 3$ is needed by the order classification in Appendix A.1. The double exponent 2^{f^2} comes from applying the Erdős–Szekeres-type extraction of Lemma A.8 once for each of up to f^2 ordered pairs of block positions; we do not know whether it is necessary. For abelian groups no order information is needed, and $c_{\mathbb{G},f} = \mathsf{d}(\mathbb{G}^f)$ is linear in f (Remark A.3). At fan-out one every owner has a single block on a single row, no homogenization is needed, and the prefix-product pigeonhole alone gives $|\mathbb{G}| - 1$ (Lemma A.2(c)); since $\mathsf{d}(\mathbb{G}) \le |\mathbb{G}| - 1$, the abelian value is never larger.

Simultaneous block replacement. A *block-replacement family* of width f over $\mathbb{G}$ (Definition A.1) consists of $d \le f$ rows, each an ordered product in $\mathbb{G}$ of fixed constants and *blocks*; each block has an *owner* in $[m] := \{1, \dots, m\}$, an original value and a replacement value, and each owner owns at most f blocks, which may interleave with other owners' blocks arbitrarily. $\mathrm{val}_\varrho(J)$ is the value of row ϱ when exactly the blocks of owners in $J \subseteq [m]$ take their replacement values.

Proposition 5.2 (Simultaneous block replacement)**.** *In a block-replacement family of width f over a finite group $\mathbb{G}$ with $m > c_{\mathbb{G},f}$ owners there is a nonempty $J \subseteq [m]$ with $\mathrm{val}_\varrho(J) = \mathrm{val}_\varrho(\varnothing)$ for every row ϱ.*

Proof. This is Lemma A.2. For abelian $\mathbb{G}$ one selects a nonempty zero-sum subsequence of the owners' difference vectors in $\mathbb{G}^d \le \mathbb{G}^f$. At $f = 1$ a pigeonhole on prefix-product states of the single row selects an interval of owners. For arbitrary $\mathbb{G}$ and f no factors are commuted: among owners with a common block pattern (at most $\mathcal{P}_f$ patterns) one homogenizes all pairwise order comparisons (Lemma A.8), which decomposes every row into consecutive rank chunks (Lemma A.4); a pigeonhole on prefix-product boundary states then yields an interval of owners preserving all ordered row products (Lemma A.5). □

Individual block or owner values need not be preserved by the replacement, and owner contributions cannot be commuted: if the two rows are $\mathfrak{b}_{1,1}\mathfrak{b}_{2,1}$ and $\mathfrak{b}_{2,2}\mathfrak{b}_{1,2}$, with value (12) on the blocks of owner 1 and (23) on those of owner 2 in S_3, the row values are $(12)(23) \ne (23)(12)$. Example A.6 shows a replacement preserving rows of opposite owner orientation.

Lemma 5.3 (Short representatives for window–group invariants)**.** *Fix $k, \ell \geq 0$, a finite group $\mathbb{G}$ (possibly trivial) and a morphism $g : \Sigma^* \to \mathbb{G}$. Put $\lambda := (\widehat{s}_{k,\ell}, g) : \Sigma^* \to \mathsf{N} := M_{k,\ell} \times \mathbb{G}$. Then $(\mathrm{R}^{\lambda}_{\beta})$ holds with*

$$\beta = b_{k,\ell,\mathbb{G}} := (r_{\max}+1)\,\bar{\theta}_G\,(c_{\mathbb{G},f}+f(k+\ell+1))\,|V||\mathsf{N}|^f \;\leq\; (r_{\max}+1)(c_{\mathbb{G},f}+f(k+\ell+1))|\mathsf{N}|^f\,\|G\|\,\bar{\theta}_G.$$

Proof. We verify the hypothesis of Lemma 5.1(ii) with $c = c_{\mathbb{G},f}$ and $c' = f(k+\ell+1)$; since $c + c' \geq 1$ and $n_\lambda(G) \leq |V||\mathsf{N}|^f$, it gives the bound, and the second inequality uses $|V| \leq \|G\|$. Let T be λ-minimal with heavy threshold $b = \bar{\theta}_G$. In each root component of length at least $m_0 = \max\{1, k+\ell\}$ mark the first k and the last ℓ positions (one arbitrary position if $k = \ell = 0$), and in each shorter component mark all positions: at most c' marks. Let Q be an antichain of heavy nodes. Incomparable subtrees have disjoint yields, so at most c' nodes of Q contain a mark; let Q_0 be the others.

Let $|Q_0| > c_{\mathbb{G},f}$. For $v \in Q_0$ with nonterminal C_v fix a tuple of $L_{C_v}(G)$ of total length at most $\bar{\theta}_G$ (Lemma 4.1(ii)). By linearity and nondeletion each $v \in Q_0$ contributes at most f disjoint contiguous blocks to the root components, and the g-value of each root component is the ordered product of the g-values of fixed material and of these blocks. This is a block-replacement family over $\mathbb{G}$ with owners Q_0, so Proposition 5.2 gives a nonempty $J \subseteq Q_0$ whose simultaneous replacement preserves every root g-value. The replaced blocks contain no mark: short root components are untouched, and in a long component they lie strictly between the marked prefix and suffix, which keep their letters and positions relative to the ends, while the component keeps its $k+\ell$ marked letters (at least one if $k = \ell = 0$) and stays long. So the root $\widehat{s}_{k,\ell}$-vector is preserved as well, and with it the root λ-vector. □

5.2 Window and Window–Group Typings: Explicit Constants

Window–group typings are local-group typings (Proposition B.6(i)), so Theorem A covers them; this subsection is the base case of the replacement argument, with explicit constants. A typing s is a *window–group typing* if $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$ for some $k, \ell \geq 0$, a finite group $\mathbb{G}$ and a morphism $g : \Sigma^* \to \mathbb{G}$; for trivial $\mathbb{G}$ this means that s is a window typing.

Proposition 5.4 (Polynomial characteristic data for window–group typings)**.** *Let $\mathbb{G}$ be a finite group (possibly trivial), $g : \Sigma^* \to \mathbb{G}$ a morphism, and $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$; let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and $L(G) = L \cap \Sigma^+$. Then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for both $\mathcal{B}^{\leq r_{\max}}_{f,s}$ and $\mathcal{B}_{f,s}$ with $|\mathsf{W}(\widetilde{G}_0)| \leq C_s(G) = O(\|G\|)$ and*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq K_{k,\ell,\mathbb{G}}\,\|G\|^2\bar{\theta}_G,$$

$$K_{k,\ell,\mathbb{G}} := |\mathsf{S}^\sharp|^{f r_{\max}}\big(3 + r_{\max}(r_{\max}+1)(c_{\mathbb{G},f} + f(k+\ell+1))(|M_{k,\ell}||\mathbb{G}|)^f\big).$$

Proof. Lemma 5.3 and Lemma 4.2(i),(ii),(iv) with $\lambda = (\widehat{s}_{k,\ell}, g)$ and $b = b_{k,\ell,\mathbb{G}}$ give the sample and $\|\mathsf{W}\|_+ \leq C_s(G)(1 + 2\bar{\theta}_G + r_{\max} b_{k,\ell,\mathbb{G}})$; insert $C_s(G) \leq |\mathsf{S}^\sharp|^{f r_{\max}}\|G\|$, $1 + 2\bar{\theta}_G \leq 3\|G\|\bar{\theta}_G$ and $|V| \leq \|G\|$. □

The group may be noncommutative and its blocks may interleave across components. For trivial $\mathbb{G}$, i.e. $s \preceq s_{k,\ell}$, one has $c_{\mathbb{G},f} = 0$.

Remark 5.5 (Abelian groups). If $\mathbb{G}$ is abelian, Lemma A.2(a) gives $c_{\mathbb{G},f} = \mathsf{d}(\mathbb{G}^f)$, and $\mathsf{d}(\mathbb{G}^f) = O(\exp(\mathbb{G})\, f \log|\mathbb{G}|)$ (Remark A.3; $\exp(\mathbb{G})$ is the least common multiple of the element orders), so $K_{k,\ell,\mathbb{G}}$ is polynomial in f apart from the factors $|\mathsf{S}^\sharp|^{f r_{\max}}$ and $(|M_{k,\ell}||\mathbb{G}|)^f$. For nonabelian $\mathbb{G}$ and $f \geq 2$, by contrast, $c_{\mathbb{G},f} = M_{\mathbb{G},f} - 1$ grows doubly exponentially in f.

At fan-out one, polynomial time and thick-data learning for fixed windows is due to Yoshinaka [12, Theorem 1]; the bounds above are relative to a reduced good presentation and are not claimed to improve on that result after normalization. What is new is that window and ordered group values are preserved *simultaneously* across up to f interleaved components.

5.3 Local-Group Typings

LI is the pseudovariety of finite semigroups T satisfying $e\mathsf{T}e = \{e\}$ for every idempotent $e \in \mathsf{T}$, **Grp** that of finite groups, and $\mathbf{LI} \vee \mathbf{Grp}$ consists of the divisors of products $\mathsf{T} \times \mathbb{G}$ with $\mathsf{T} \in \mathbf{LI}$ and $\mathbb{G}$ a finite group [26, 27]. By Lemma B.5 (Appendix B), every typing with $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$ is a window–group typing, with k, $\mathbb{G}$ and g depending only on s, so Proposition 5.4 applies to it.

A finite semigroup is a *local group*, $\mathsf{T} \in \mathbf{LG}$, if $e\mathsf{T}e$ is a group for every idempotent $e \in \mathsf{T}$. A typing is a *local-group typing* if its positive image S is a local group; the condition is never tested on $\mathsf{S}^\sharp$. By Proposition B.6, $\mathbf{LI} \vee \mathbf{Grp} \subsetneq \mathbf{LG}$.

Definition 5.6 (Sliding-factor morphism)**.** Fix $k \geq 1$, a finite group $\mathbb{G}$ and a map $\nu : \Sigma^{k+1} \to \mathbb{G}$ (the *sliding-factor labelling*). For $|w| \geq 2k$ let $\mathrm{fac}_k(w) = (F_1, \ldots, F_{|w|-k})$ be the factors of length $k+1$ of w in order of their starting positions, and put $\gamma_{k,\nu}(w) := \nu(F_1)\nu(F_2)\cdots\nu(F_{|w|-k})$ (ordered product in $\mathbb{G}$); put $\gamma_{k,\nu}(w) := 1_{\mathbb{G}}$ for $|w| < 2k$. The *sliding-factor morphism* is $\lambda_{k,\nu}(w) := (\widehat{s}_{k,k}(w), \gamma_{k,\nu}(w))$; by Lemma B.1 it is a morphism onto a finite monoid $M_{k,\nu}$. The typing s is *local-group-determined* if $\widehat{s} \preceq \lambda_{k,\nu}$ for some k, $\mathbb{G}$ and ν.

A sliding-factor morphism can encode a group contribution determined by a sliding factor, rather than by each letter alone. For example, $\lambda_{1,\nu}$ with $\nu(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and zero on other pairs records the parity of occurrences of $\mathtt{ab}$. Even on $\{\mathtt{a}, \mathtt{b}\}$ this is not determined by any fixed window combined with a finite-group morphism (Proposition B.6(ii)). The local-group characterization below expresses this additional power without requiring a new learner or a target-specific typing.

Theorem 5.7 (Polynomial characteristic data for local-group typings)**.** *Let $\mathsf{S} \in \mathbf{LG}$. Every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G satisfies $t_{\mathrm{typ}}(G, s) = O_{f,r_{\max},s,\Sigma}(\|G\|\bar{\theta}_G)$. If moreover $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and $L(G) = L \cap \Sigma^+$, then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for both $\mathcal{B}^{\leq r_{\max}}_{f,s}$ and $\mathcal{B}_{f,s}$ with*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},s,\Sigma}(\|G\|^2\bar{\theta}_G).$$

Proof. By Theorem B.4, $\widehat{s} \preceq \lambda := \lambda_{k,\nu}$ for $k = |\mathsf{S}|$ and some $\mathbb{G}, \nu$ depending only on s. Lemma B.8 gives $(\mathrm{R}^{\lambda}_{\beta})$ with $\beta = \beta_\lambda = O(\|G\|\bar{\theta}_G)$: heavy subtrees are replaced by arbitrary short yields of their nonterminals, after the owners have been thinned so that their blocks lie at least $2k$ letters apart and each block has been enlarged by k unchanged letters on both sides (Lemma B.7); the enlarged blocks keep their window types, so the ordered sliding-factor products are controlled by Lemma B.2 and Proposition 5.2. This step uses no substitutability. Lemma 4.2(i) gives $t_{\mathrm{typ}} \leq \beta_\lambda$, and Theorem 4.3 gives $\|\mathsf{W}\|_+ \leq N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max}\beta_\lambda) = O(\|G\|^2\bar{\theta}_G)$. □

The Rees and sliding-factor coordinates are classical; their combination with owner-wise block shortening is the quantitative step. Local-group and window–group typings thus have the same quadratic dependence on $\|G\|$; Proposition 5.4 gives explicit constants for the latter.

5.4 Separating Typing Regimes on Non-Context-Free Targets

Theorem A has two ingredients beyond windows: sliding-factor information (local groups rather than window–group typings) and ordered, possibly noncommutative, group products across interleaved

components. Proposition B.6 separates the corresponding semigroups. The following targets separate them on languages: each is a non-context-free language of fan-out two that lies in the class of one typing and in the class of no typing of the weaker kind (no window–group typing, respectively no window–abelian typing), so the weaker theorem does not cover it under any typing of that kind. Both have the form $\varsigma w\$x_\times$: a flag $\varsigma = \mathtt{r}$ leaves w unconstrained, $\varsigma = \mathtt{l}$ imposes a typing-dependent condition on w, and $x_\times \in C_\times := \{\mathtt{1}^m\mathtt{2}^n\mathtt{3}^m\mathtt{4}^n : m, n \geq 1\}$ makes the target non-context-free. Let $\varphi_\times$ map $\mathtt{1}, \mathtt{3}, \mathtt{2}, \mathtt{4}$ to $(1,0), (-1,0), (0,1), (0,-1) \in \mathbb{Z}^2$ and all other letters to 0, and let $\#_{\mathtt{ab}}(w)$ be the number of occurrences of the factor $\mathtt{ab}$ in w.

Lemma 5.8 (Flag separation)**.** *Let s' be a typing, $u, v \in \Sigma^+$ with $s'(u) = s'(v)$, and $z \in \Sigma^*$. If L contains $\mathtt{r}uz$, $\mathtt{r}vz$ and $\mathtt{l}uz$ but not $\mathtt{l}vz$, then $L \notin \mathcal{C}^{\mathrm{mcf}}_{f,s'}$ for every $f \geq 1$.*

Proof. $\mathtt{r}\square_1 z \in \mathcal{E}_1$ accepts u and v, while $\mathtt{l}\square_1 z$ accepts u but not v; so $\mathcal{D}^{(1)}_L(u) \neq \mathcal{D}^{(1)}_L(v)$, and every (f, s') condition includes arity one. □

The targets. Let $\Sigma_{\mathrm{A}} = \{\mathtt{l}, \mathtt{r}, \mathtt{a}, \mathtt{b}, \mathtt{c}, \$, \mathtt{1}, \mathtt{2}, \mathtt{3}, \mathtt{4}\}$,

$$L_{\mathrm{A}} := \{\varsigma w\$x_\times : \varsigma \in \{\mathtt{l}, \mathtt{r}\},\ w \in \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}^+,\ x_\times \in C_\times,\ \varsigma = \mathtt{r} \text{ or } \#_{\mathtt{ab}}(w) \text{ even}\},$$

and $s_{\mathrm{A}} := \lambda_{1,\nu}|_{\Sigma^+_{\mathrm{A}}}$, corestricted to its positive image, with $\nu : \Sigma^2_{\mathrm{A}} \to \mathbb{Z}_2$, $\nu(\mathtt{ab}) = 1$ and $\nu = 0$ otherwise (Definition 5.6); it records the first and last letters and, for $|w| \geq 2$, the parity of $\#_{\mathtt{ab}}(w)$. Let $\Sigma_{\mathrm{B}} = \Sigma_{\mathrm{A}} \setminus \{\mathtt{c}\}$, $\alpha = (12)$, $\beta = (23) \in S_3$, $g_{\mathrm{B}} : \Sigma^*_{\mathrm{B}} \to S_3$ with $\mathtt{a} \mapsto \alpha$, $\mathtt{b} \mapsto \beta$ and all other letters $\mapsto 1$,

$$L_{\mathrm{B}} := \{\varsigma w\$x_\times : \varsigma \in \{\mathtt{l}, \mathtt{r}\},\ w \in \{\mathtt{a}, \mathtt{b}\}^+,\ x_\times \in C_\times,\ \varsigma = \mathtt{r} \text{ or } g_{\mathrm{B}}(w) = 1\},$$

and $s_{\mathrm{B}} := (\widehat{s}_{1,1}, g_{\mathrm{B}})|_{\Sigma^+_{\mathrm{B}}}$, corestricted. **Ab** is the pseudovariety of finite abelian groups, and $\mathbf{LI} \vee \mathbf{Ab}$ consists of the divisors of $\mathsf{T} \times \mathbb{G}$ with $\mathsf{T} \in \mathbf{LI}$ and $\mathbb{G}$ finite abelian.

Proposition 5.9 (A target that needs a local group)**.** (i) *$L_{\mathrm{A}} \in \mathsf{MCFL}(2,2)$ and L_{A} is not context-free.*

(ii) *$L_{\mathrm{A}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{A}}}$ for every $f \geq 2$, and $s_{\mathrm{A}}(\Sigma^+_{\mathrm{A}}) \in \mathbf{LG}$.*

(iii) *If $\widehat{s'} \preceq (\widehat{s}_{k,\ell}, g')$ for some k, ℓ and a morphism g' into a finite group (in particular if $s'(\Sigma^+_{\mathrm{A}}) \in \mathbf{LI} \vee \mathbf{Grp}$), then $L_{\mathrm{A}} \notin \mathcal{C}^{\mathrm{mcf}}_{f,s'}$ for all $f \geq 1$.*

Hence $s_{\mathrm{A}}(\Sigma^+_{\mathrm{A}}) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{Grp})$, and L_{A} is covered by Theorem 5.7 but, for no typing, by Proposition 5.4, even with Lemma B.5.

Proposition 5.10 (A target that needs a noncommutative group)**.** (i) *$L_{\mathrm{B}} \in \mathsf{MCFL}(2,2)$ and L_{B} is not context-free.*

(ii) *$L_{\mathrm{B}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{B}}}$ for every $f \geq 2$, and s_{B} is a window–group typing with group S_3.*

(iii) *If $\widehat{s'} \preceq (\widehat{s}_{k,\ell}, g')$ for some k, ℓ and a morphism g' into a finite* abelian *group (in particular if $s'(\Sigma^+_{\mathrm{B}}) \in \mathbf{LI} \vee \mathbf{Ab}$), then $L_{\mathrm{B}} \notin \mathcal{C}^{\mathrm{mcf}}_{f,s'}$ for all $f \geq 1$.*

Hence L_{B} is covered by Proposition 5.4 with the nonabelian constant $c_{S_3,f} = M_{S_3,f} - 1$, but for no typing by its trivial-group or abelian-group cases (Remark 5.5).

The proofs, with the explicit separating words, are in Appendix C. Both propositions concern class membership, not data length: they show that the typing regimes of Theorem A differ on non-context-free targets, not that its exponent is sharp. Each target is a regular flagged prefix followed by the fixed non-context-free part $C_\times$; the separation comes from the prefix. The multiple context-free content of Theorem A lies in the block-replacement argument for arbitrary presentations.

6 Lower Bound: The Linear Factor at Every Fan-Out

This section proves Theorem B.

The upper bound of Theorem 4.3 grows with $\|G\|$, and the rare-type lower bound of Proposition 4.6 does not. Nevertheless the product $\|G\|\bar{\theta}_G$ is necessary at every fan-out, under every typing over an alphabet of at least $4f+1$ letters (two at fan-out one) and for presentations within a constant factor of the smallest.

Proposition 6.1 (The linear factor is necessary at every fan-out)**.** *Let $f \geq 1$, let Σ contain $4f+1$ distinct letters $\mathtt{a}_1, \ldots, \mathtt{a}_{2f}, \mathtt{d}_1, \ldots, \mathtt{d}_{2f}, \mathtt{z}$, and let s be any typing. For integers $q \geq n \geq 1$ put $W := 2fq + 2^n + 2f$ and*

$$w_i := \mathtt{a}_1^i \mathtt{d}_1 \mathtt{a}_2^i \mathtt{d}_2 \cdots \mathtt{a}_{2f}^i \mathtt{d}_{2f} \mathtt{z}^{2f(q-i)+2^n} \quad (0 \leq i \leq q), \qquad L_{q,n}^{(f)} := \{w_0, \ldots, w_q\}.$$

(i) *Every subset of $L_{q,n}^{(f)}$ belongs to $\mathcal{C}_{f,s_{\mathrm{triv}}}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$.*

(ii) *For $\mathcal{B}_{f,s}$, for $\mathcal{B}_{f,s}^{\leq r}$ ($r \geq 1$), and for every set-driven operator with characteristic samples throughout $\mathcal{C}_{f,s}^{\mathrm{mcf}}$, the only characteristic sample of $L_{q,n}^{(f)}$ is $L_{q,n}^{(f)}$ itself.*

(iii) *$L_{q,n}^{(f)}$ has a reduced good presentation $G_{q,n}^{(f)}$ of fan-out f and rule rank two with $\|G_{q,n}^{(f)}\| = (7f+11)q + 6n + 3f + 10 \leq (7f+17)(q+1)$ and $\bar{\theta}_{G_{q,n}^{(f)}} = W$. Every reduced good fan-out-$\leq f$ presentation G of $L_{q,n}^{(f)}$ has $\bar{\theta}_G = W$ and $\|G\| \geq q+1$.*

(iv) *For $f = 1$,* (i)–(iii) *hold over any alphabet containing two letters $\mathtt{a}, \mathtt{d}$, for the words $w_i = \mathtt{a}^i\mathtt{d}\mathtt{a}^i\mathtt{d}\mathtt{a}^{2(q-i)+2^n}$ obtained by identifying $\mathtt{a}_1, \mathtt{a}_2, \mathtt{z}$ with $\mathtt{a}$ and $\mathtt{d}_1, \mathtt{d}_2$ with $\mathtt{d}$, and the grammar identified in the same way.*

Consequently every characteristic sample F in (ii) *satisfies*

$$\|F\|_+ = (q+1)(W+1) \;\geq\; \tfrac{1}{7f+17}\, \|G_{q,n}^{(f)}\|\, \bar{\theta}_{G_{q,n}^{(f)}}.$$

Each word w_i is determined by the position of one marker and by the distance of two, which gives (i) (Appendix K). With $q = n$ one has $\|G_{n,n}^{(f)}\| = \Theta(n)$ and $\bar{\theta}_{G_{n,n}^{(f)}} = 2^n + \Theta(n)$; with n fixed, $\bar{\theta}_{G_{q,n}^{(f)}} = \Theta(\|G_{q,n}^{(f)}\|)$. By (iii) the presentations are within a constant factor of the smallest reduced good ones, so the bound is not obtained by padding G; with the trivial typing the proposition shows that Yoshinaka's bound $\|K_G\| \leq |P|\theta_G$ is attained up to a constant factor. All yields of a nonterminal of $G_{q,n}^{(f)}$ have one total length, so $t_{\mathrm{typ}} = \bar{\theta}_G = W$ for this family, and it says nothing about the further factor $\|G\|$ in Proposition 5.4 and Theorem 5.7. The proof is in Appendix K.

Corollary 6.2 (The trivial typing)**.** *Fix f and $r_{\max} \geq 1$. For the trivial typing, every $L \in \mathcal{C}_{f,s_{\mathrm{triv}}}^{\mathrm{mcf}}$ and every reduced good rank-$\leq r_{\max}$ presentation G of $L \cap \Sigma^+$, the operator $\mathcal{B}_{f,s_{\mathrm{triv}}}$ has a characteristic sample of total length $O(\|G\|\bar{\theta}_G)$, as for Yoshinaka's sample [14, Lemma 7 and Theorem 1]. Suppose moreover that $r_{\max} \geq 2$ and that Σ has at least $4f+1$ letters, or that $r_{\max} \geq 2$, $f = 1$ and Σ has at least two letters. Then this bound is attained up to a constant factor: for the targets $L_{q,n}^{(f)}$ and presentations $G_{q,n}^{(f)}$ of Proposition 6.1, every characteristic sample has length at least $\|G_{q,n}^{(f)}\|\bar{\theta}_{G_{q,n}^{(f)}}/(7f+17)$.*

Proof. Every typed nonterminal is an untyped one, so $t_{\mathrm{typ}}(G, s_{\mathrm{triv}}) \leq \bar{\theta}_G$ by Lemma 4.1, and Theorem 4.3 gives $O(\|G\|\bar{\theta}_G)$. Proposition 6.1 gives the second assertion; the presentations $G_{q,n}^{(f)}$ have rule rank two, which is why $r_{\max} \geq 2$ is needed there. □

A characteristic sample stays characteristic when finitely many words of the target are added, so only the length of a *shortest* characteristic sample is meaningful; the corollary bounds it from above for every target and shows that the bound is attained by one family.

7 The Boundary at Fan-Out One

This section proves Theorem C.

A three-letter doubling family. Let $\mathtt{a}, \mathtt{b}, \mathtt{c}$ be distinct letters, let $U_1 = \{1, \zeta\}$ with identity 1 and $\zeta^2 = \zeta$, and let $s_U : \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}^+ \twoheadrightarrow U_1$ send $\mathtt{a}, \mathtt{b}$ to 1 and $\mathtt{c}$ to ζ; thus $s_U(w) = 1$ exactly for the nonempty words without $\mathtt{c}$. For $n \geq 1$ let G_n be the fan-out-one grammar

$$S \to E_n, \qquad E_0 \to \mathtt{a}, \qquad E_i \to \mathtt{b}E_{i-1}E_{i-1}\mathtt{b} \mid \mathtt{c}E_{i-1}\mathtt{c} \quad (1 \leq i \leq n),$$

put $L^{(i)} := L_{E_i}(G_n)$, so $L^{(0)} = \{\mathtt{a}\}$ and $L^{(i)} = \mathtt{b}L^{(i-1)}L^{(i-1)}\mathtt{b} \cup \mathtt{c}L^{(i-1)}\mathtt{c}$, put $L_n := L^{(n)} = L(G_n)$, $W_0 := \mathtt{a}$ and $W_i := \mathtt{b}W_{i-1}W_{i-1}\mathtt{b}$. This is a three-letter variant of the bracketed doubling family of the companion CFG manuscript [23, Proposition 6.4], in which the opening and closing brackets are different letters; all its properties are proved here.

Proposition 7.1 (Three-letter doubling family)**.** *For every $n \geq 1$:*

(i) *G_n is reduced and good, of fan-out one and rule rank two, with $\|G_n\| = 13n + 7$ and $\bar{\theta}_{G_n} = 4n$.*

(ii) *L_n is Clark–Eyraud substitutable: for all $x, y \in \Sigma^+$, $\mathcal{D}^{(1)}_{L_n}(x) \cap \mathcal{D}^{(1)}_{L_n}(y) \neq \emptyset$ implies $\mathcal{D}^{(1)}_{L_n}(x) = \mathcal{D}^{(1)}_{L_n}(y)$. Hence $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$ for every typing s on every alphabet $\Sigma \supseteq \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$; in particular $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$.*

(iii) *W_n is the only word of L_n without $\mathtt{c}$, hence the only one of s_U-type 1, and $|W_n| = 3 \cdot 2^n - 2$.*

(iv) *Every characteristic sample F of L_n for $\mathcal{B}^{\leq r}_{1,s_U}$ ($r \geq 1$) or $\mathcal{B}_{1,s_U}$, or for any set-driven operator having characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{1,s_U}$, contains W_n; hence $\|F\|_+ \geq 3 \cdot 2^n - 1$.*

The proof is in Appendix G: (iii) holds because a derivation without $\mathtt{c}$ is the complete binary tree, and (iv) follows from (iii) and Proposition 4.6.

Definition 7.2 (Uniform polynomial characteristic-data length)**.** Let $f, r_{\max} \geq 1$. A pseudovariety $\mathbf{V}$ of finite semigroups has *uniform polynomial characteristic-data length at fan-out f and rank $r_{\max}$* if for every nonempty finite alphabet Σ and every surjective typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ with $\mathsf{S} \in \mathbf{V}$ there is a polynomial $p = p_{f,r_{\max},s,\Sigma}$ such that for every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation G of $L \cap \Sigma^+$, the operator $\mathcal{B}^{\leq r_{\max}}_{f,s}$ has a characteristic sample F for L with $\|F\|_+ \leq p(\|G\|, \bar{\theta}_G)$.

The quantifiers are ordered as written: p may depend on f, $r_{\max}$, Σ and the typing s, but not on the target L or the presentation G.

Theorem 7.3 (Fan-out one: a pseudovariety boundary for data length)**.** *Let $r_{\max} \geq 2$ and let $\mathbf{V}$ be a pseudovariety of finite semigroups.*

(i) *$\mathbf{V}$ has uniform polynomial characteristic-data length at fan-out one and rank $r_{\max}$ if and only if $\mathbf{V} \subseteq \mathbf{LG}$.*

(ii) *If* $\mathbf{V} \subseteq \mathbf{LG}$*, one may take* $p(\|G\|, \bar{\theta}_G) = O_{r_{\max},s,\Sigma}(\|G\|^2 \bar{\theta}_G)$ *in Definition 7.2.*

(iii) *If* $\mathbf{V} \not\subseteq \mathbf{LG}$*, the typing* s_U *of Proposition 7.1 has positive image* $U_1 \in \mathbf{V}$*, and the targets* $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$ *have reduced good rank-two presentations* G_n *with* $\|G_n\| = 13n + 7$ *and* $\bar{\theta}_{G_n} = 4n$*, while every characteristic sample of* L_n *has total encoded length at least* $3 \cdot 2^n - 1 \geq 2^{(\|G_n\| + \bar{\theta}_{G_n})/17}$*. This lower bound holds for* $\mathcal{B}^{\leq r_{\max}}_{1,s_U}$*, for* $\mathcal{B}_{1,s_U}$*, and for every set-driven operator having characteristic samples throughout* $\mathcal{C}^{\mathrm{mcf}}_{1,s_U}$*.*

Proof. (ii), and hence the *if* direction of (i), is Theorem 5.7 with $f = 1$, since every $\mathsf{S} \in \mathbf{V}$ then lies in $\mathbf{LG}$. For (iii), pick $\mathsf{S} \in \mathbf{V} \setminus \mathbf{LG}$. By Theorem B.4(ii), S has a subsemigroup isomorphic to U_1, and $U_1 \in \mathbf{V}$ because pseudovarieties are closed under subsemigroups. The typing s_U maps onto U_1; the remaining claims are Proposition 7.1(i) and (iv), using $\|G_n\| = O(n)$ and $\theta_{G_n} \leq 4n$. Since $3 \cdot 2^n - 1$ is not bounded by any polynomial in $\|G_n\|$ and $\bar{\theta}_{G_n}$, the *only if* direction of (i) follows. □

Remark 7.4 (Scope of Theorem 7.3). (a) The lower bound uses rank two: the doubling rules $E_i \to \mathtt{b} E_{i-1} E_{i-1} \mathtt{b}$ are binary. For targets represented by linear CFGs the companion manuscript gives, for every fixed typing, characteristic data polynomial in the grammar size alone [23, Theorem 8.3]. (b) The theorem concerns pseudovarieties. Which individual typings have polynomial data is not characterized here, except for the typings onto U_1 (Corollary 7.12). (c) The upper bound is for the fixed operator $\mathcal{B}^{\leq r_{\max}}_{1,s}$, whereas the lower bound holds for every set-driven operator with characteristic samples throughout the class.

Polynomial time and thick data. Eyraud, Heinz and Yoshinaka [10, Definition 2.12], following [12], call a class identifiable in polynomial time and thick data (IPTtD) if some algorithm returns, in time polynomial in the size of a given sample, a hypothesis consistent with it, and every representation R of a target has a characteristic sample of size polynomial in the size of R and its thickness. By Lemma 4.1(ii),(iii), their thickness (the longest shortest yield of a nonterminal, t_G) and our rule thickness are polynomially equivalent given $\|G\|$. The sample sizes $\sum_{w \in K} |w|$ and $\|K\|_+$ differ by at most the number of words. Their definition asks the algorithm to return a hypothesis in a fixed class of representations; the raw output of $\mathcal{B}^{\leq r_{\max}}_{f,s}$ may contain the rule $\widehat{S} \to \varepsilon$, so we first remove it.

Lemma 7.5 (Reduced good hypotheses)**.** *Fix* f*,* $r_{\max} \geq 1$ *and* s*, let* $K \subseteq \Sigma^*$ *be finite with* $K \cap \Sigma^+ \neq \emptyset$*, and let* G^+_K *be* $\mathcal{B}^{\leq r_{\max}}_{f,s}(K)$ *without the rule* $\widehat{S} \to \varepsilon$*. Then* G^+_K *is a reduced good MCFG of fan-out* $\leq f$ *and rank* $\leq r_{\max}$*,* $L(G^+_K) = L(\mathcal{B}^{\leq r_{\max}}_{f,s}(K)) \setminus \{\varepsilon\}$*, and* G^+_K *is computable from* K *in time* $(1 + \|K\|_+)^{O(f r_{\max})}$*.*

Proof. Start and Link rules have identity templates, Const rules write tuples of nonempty words, and Comp rules have good templates (Definition 3.3); so every rule of G^+_K is good, and no nonterminal derives a tuple with an empty component. Hence ε is derived only by the removed rule, which gives the language. Every $[\vec{x}]$ is productive by its Const rule, and $\widehat{S}$ is productive since K has a nonempty word. For reachability let $\vec{x} \in \mathrm{Obs}_d(K)$, say $w = E[\vec{x}] \in K$ with $E = v_0 \Box_1 v_1 \cdots \Box_d v_d \in \mathcal{E}_d$. Then $w \neq \varepsilon$, $(w) \in \mathrm{Obs}_1(K)$ with the context $\Box_1$, and the template $\rho = (v_0 \xi^1_1 v_1 \cdots \xi^d_1 v_d)$ is good: it is ε-free, uses each variable once and in order, and is nonmerging since $v_1, \ldots, v_{d-1}$ are nonempty. So $((w), \rho, (\vec{x}))$ is a composition witness of rank one, and $\widehat{S} \to [(w)] \to \rho([\vec{x}])$ reaches $[\vec{x}]$. Fan-out, rank and running time are those of Theorem 3.14. □

Corollary 7.6 (Identification in polynomial time and thick data)**.** *Fix* $f \geq 1$*,* $r_{\max} \geq 1$ *and a typing* s*. Let* $\mathcal{R}$ *consist of the pairs* (G, z) *of a bit* $z \in \{0, 1\}$ *and either a reduced good fan-out-*$\leq f$*, rank-*$\leq r_{\max}$ *MCFG* G *or a fixed grammar* $G_\emptyset$ *without rules; the pair represents* $L(G) \cup \{\varepsilon : z = 1\}$*. The*

members of $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r_{\max})$ *are represented by* $(G, [\varepsilon \in L])$ *with* G *a reduced good fan-out-*$\leq f$*, rank-*$\leq r_{\max}$ *presentation of* $L \cap \Sigma^+$*, or* $G = G_\emptyset$ *if* $L \cap \Sigma^+ = \emptyset$*.*

(i) *If* $s(\Sigma^+) \in \mathbf{LG}$*, the algorithm returning* $(G^+_K, [\varepsilon \in K])$ *of Lemma 7.5, or* $(G_\emptyset, [\varepsilon \in K])$ *if* $K \cap \Sigma^+ = \emptyset$*, identifies* $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r_{\max})$ *in polynomial time and thick data for the class* $\mathcal{R}$*.*

(ii) *If* $f = 1$ *and* $r_{\max} \geq 2$*, no algorithm, whatever its running time, satisfies the data condition of IPTtD for* $\mathcal{C}^{\mathrm{mcf}}_{1,s_U} \cap \mathsf{MCFL}(1, r_{\max})$ *with respect to the class* $\mathcal{R}$*.*

(iii) *Consequently, for* $r_{\max} \geq 2$ *and a pseudovariety* $\mathbf{V}$*, the following are equivalent:* $\mathbf{V} \subseteq \mathbf{LG}$*; for every typing* s *with* $s(\Sigma^+) \in \mathbf{V}$*, the class* $\mathcal{C}^{\mathrm{mcf}}_{1,s} \cap \mathsf{MCFL}(1, r_{\max})$ *is IPTtD for the class* $\mathcal{R}$*.*

Proof. (i) By Lemma 7.5 the hypothesis lies in $\mathcal{R}$, is computed in time polynomial in $\|K\|_+$, and represents $L(\mathcal{B}^{\leq r_{\max}}_{f,s}(K))$, which contains K (each $w \in K \cap \Sigma^+$ is derived by its Start and Const rules). Theorem 5.7 gives a sample F of length $O(\|G\|^2 \bar{\theta}_G)$ with $L(\mathcal{B}^{\leq r_{\max}}_{f,s}(K)) = L$ whenever $F \subseteq K \subseteq L$, so the hypothesis then represents L; and $\bar{\theta}_G \leq 2\|G\| \max\{1, t_G\}$. (ii) In [10, Definition 2.12] an algorithm reads a sample, which is a finite set of strings, so it is a set-driven operator. Such an algorithm has a characteristic sample for every member of the class, since every member L with $L \cap \Sigma^+ \neq \emptyset$ has a reduced good presentation of rank $\leq r_{\max}$ (Proposition 3.12). The targets L_n of Proposition 7.1 belong to the class, and so does $L_n \setminus \{W_n\}$: it lies in $\mathcal{C}^{\mathrm{mcf}}_{1,s_U}$ by Lemma 4.4 and is finite. By Proposition 4.6, every characteristic sample of L_n contains W_n, of length $3 \cdot 2^n - 2$, while $\|G_n\| = O(n)$ and $t_{G_n} \leq \theta_{G_n} \leq 4n$. (iii) follows from (i) with $f = 1$ and from (ii), since a pseudovariety not contained in $\mathbf{LG}$ contains U_1 (Theorem 7.3). □

The hypotheses lie in $\mathcal{R}$ on every sample; on samples that contain no characteristic sample their languages need not belong to the class, as for the learner of [14]. Item (i) holds at every fan-out. Item (ii) is proved only at fan-out one; at fan-out two its analogue is part of Open Problem 9.1, and the family L_n cannot prove it (Theorem 8.2).

Individual typings onto U_1. Theorem 7.3 concerns pseudovarieties. For an individual typing at fan-out one, what matters is which words realize a copy $\{e, z\}$ of U_1: letter-disjoint words of type e force exponential data, while an isolated letter (Definition 7.9), whose type is realized only by its own powers, gives polynomial data (Theorem 7.10 below). For typings onto U_1 these two cases are exhaustive (Corollary 7.12).

Corollary 7.7 (A typing-level obstruction at fan-out one)**.** *Let* $r_{\max} \geq 2$ *and let* $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ *be a typing. Suppose that distinct idempotents* $e, z \in \mathsf{S}$ *satisfy* $ez = ze = z$*, and that words* $u, v, w \in \Sigma^+$ *satisfy* $s(u) = s(v) = e$ *and* $s(w) = z$ *and are* letter-disjoint*: no letter occurs twice in* uvw*. Let* H_n *be the image of the language* L_n *of Proposition 7.1, over three fresh letters, under* $\mathtt{a} \mapsto u$*,* $\mathtt{b} \mapsto v$*,* $\mathtt{c} \mapsto w$*, and let* G''_n *be the image of* G_n*, with rules* $S \to E_n$*,* $E_0 \to u$ *and* $E_i \to vE_{i-1}E_{i-1}v \mid wE_{i-1}w$*. Then* $H_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$*,* G''_n *is a reduced good rank-two presentation of* H_n *with*

$$\|G''_n\| = (2|v| + 2|w| + 9)n + |u| + 6, \qquad \bar{\theta}_{G''_n} \leq 2|u| + 2|v| + 4n|w|,$$

and every characteristic sample of H_n *for* $\mathcal{B}^{\leq r_{\max}}_{1,s}$*, for* $\mathcal{B}_{1,s}$*, or for any set-driven operator having characteristic samples throughout* $\mathcal{C}^{\mathrm{mcf}}_{1,s}$ *has total encoded length greater than* $(|u| + 2|v|)2^n - 2|v|$*. So for this single typing no polynomial bounds the characteristic-data length at fan-out one and rank* $r_{\max}$ *as in Definition 7.2.*

Proof. L_n is finite and Clark–Eyraud substitutable (Proposition 7.1), so H_n is Clark–Eyraud substitutable by Lemma I.2; it is finite, hence $H_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$. The grammar is good, and reduced since $E_i \Rightarrow^* w^i u w^i$. The size follows from Definition 2.5: the start rule has size 3, the base rule $1 + |u|$, and the two rules of each level $2|v| + 5$ and $2|w| + 3$. Completing every other choice by the rules $E_j \to wE_{j-1}w$ shows that each rule has a successful use of length at most $|u| + 2n|w|$ (start and unary rules) or $2|u| + 2|v| + (2n + 2i - 4)|w|$ (the binary rule of level i). In $\{e, z\}$, e is an identity and z is absorbing, so a word of H_n has type e iff it is the image of a word of L_n without $\mathtt{c}$, that is, of W_n. The image $h(W_n)$ has length $(|u| + 2|v|)2^n - 2|v|$ by $|h(W_i)| = 2|h(W_{i-1})| + 2|v|$, and Proposition 4.6 forces it into every such sample. □

With u, v, w letters this is the case of two letters of type e; in particular it applies to every typing onto U_1 with at least two letters of type 1, since surjectivity gives a letter of type ζ. It also applies when no letter has type e: for $\mathsf{S} = \{1, g, z\}$ with identity 1, $g^2 = 1$ and zero z, and the typing $\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d} \mapsto g$, $\mathtt{t} \mapsto z$, take $u = \mathtt{ab}$, $v = \mathtt{cd}$, $w = \mathtt{t}$ and $e = 1$. Whether two non-commuting words of type e always suffice, without letter-disjointness, is open in general; since two words commute iff they are powers of one word, their absence means that all words of type e are powers of one primitive word. For typings onto U_1 the question does not arise: with two letters of type 1 the corollary applies, and with one, $\mathtt{a}$ say, the words of type 1 are the powers of $\mathtt{a}$ and the data are polynomial (Corollary 7.12).

Let $\Sigma = \{\mathtt{a}, \mathtt{b}\}$ and let $s_{U,2} : \Sigma^+ \twoheadrightarrow U_1$ send $\mathtt{a}$ to 1 and $\mathtt{b}$ to ζ (the binary typing). For this typing, with a single letter of type 1, Corollary 7.7 does not apply; still, its data can be quadratic in $\bar{\theta}_G$ for presentations of logarithmic size.

Proposition 7.8 (Quadratic data for the binary typing at fan-out one)**.** *For $m, n \geq 1$ let $L_{m,n} \subseteq \{\mathtt{a}, \mathtt{b}\}^+$ be the set of all concatenations of 2^n words from $\{\mathtt{a}^{2^m}, \mathtt{b}\}$.*

(i) *$L_{m,n} \in \mathcal{C}^{\mathrm{mcf}}_{1,s_{U,2}}$.*

(ii) *$L_{m,n}$ has a reduced good fan-out-one presentation $G_{m,n}$ of rule rank two with $\|G_{m,n}\| = 6m + 6n + 9$ and $\bar{\theta}_{G_{m,n}} = 2^m + 2^n - 1$.*

(iii) *For $\mathcal{B}_{1,s_{U,2}}$, for $\mathcal{B}^{\leq r}_{1,s_{U,2}}$ ($r \geq 1$), and for every set-driven operator with characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{1,s_{U,2}}$, every characteristic sample F of $L_{m,n}$ has $\|F\|_+ > 2^{m+n}$.*

Consequently, for $m = n$ one has $\|G_{n,n}\| = 12n + 9$, $\bar{\theta}_{G_{n,n}} = 2^{n+1} - 1$ and $\|F\|_+ > \bar{\theta}^2_{G_{n,n}}/4$; in particular, for this typing at fan-out one there is no bound of the form $p(\|G\|)\,\bar{\theta}_G$ with p a polynomial, in contrast with Theorem 5.7.

The proof is in Appendix I. The pure word $\mathtt{a}^{2^{m+n}}$ is the product of the block length and the number of blocks, while $\bar{\theta}_G$ is about their sum, since a shortest sentence using the long block fills every other position with $\mathtt{b}$. Words containing a letter other than $\mathtt{a}$ and $\mathtt{b}$ have empty distribution in $L_{m,n}$, so the proposition holds verbatim for every typing onto U_1 in which $\mathtt{a}$ is the only letter of type 1 and $\mathtt{b}$ has type ζ. The data for these typings at fan-out one are nevertheless polynomial (Corollary 7.11 below), and for $s_{U,2}$ so they are at every fan-out $f \geq 2$ (Corollary 8.7).

Isolated letters and two letters. The family of Theorem 7.3(iii) uses three letters, two of them of type 1. A letter whose type is realized only by its own powers already yields polynomial data, over any alphabet.

Definition 7.9 (Isolated letter)**.** A letter $\mathtt{a} \in \Sigma$ is *isolated* for a typing s if $s^{-1}(s(\mathtt{a})) = \mathtt{a}^+$ and all words that contain $\mathtt{a}$ but do not lie in $\mathtt{a}^+$ have one common type.

For instance, let $\Sigma = \{\mathtt{a}, \mathtt{b}\}$ and let c be the *content typing*, $c(w)$ the set of letters of w, onto the three-element semilattice $\{\{\mathtt{a}\}, \{\mathtt{b}\}, \Sigma\}$ under union; both letters are isolated for c, since a word that contains $\mathtt{a}$ and lies outside $\mathtt{a}^+$ has content Σ. If $s : \Sigma^+ \twoheadrightarrow U_1$ and $\mathtt{a}$ is the only letter of type 1, then $s^{-1}(1) = \mathtt{a}^+$ and all other words have type ζ, so $\mathtt{a}$ is isolated, over any alphabet. Over three or more letters no letter is isolated for the content typing, since $\mathtt{ab}$ and $\mathtt{ac}$ have different contents.

Theorem 7.10 (Isolated letters: short pure representatives at fan-out one)**.** *Let* $\mathtt{a}$ *be isolated for a typing* $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$*, let* $L \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$*, and let* G *be a reduced good presentation of* $L \cap \Sigma^+$ *of fan-out one and any rank. For every nonterminal* A *of* G*, or of its binarization* G' *(Appendix H), that has a yield in* $\mathtt{a}^+$*, the typed nonterminal* $A_{s(\mathtt{a})}$ *satisfies*

$$\tau_{A_{s(\mathtt{a})}} \leq C(\|G\| + 2)^{22}\bar{\theta}_G^8,$$

where $C < 10^{43}$ *is an absolute constant, independent of* Σ*,* s *and the rank. In particular, if* L *contains a word of* $\mathtt{a}^+$*, its shortest one has length at most* $C(\|G\| + 2)^{22}\bar{\theta}_G^8$*.*

The theorem bounds only the type of the isolated letter. The other types are bounded, for the two cases of the next corollary, by Lemma H.10, and the assumption that all words containing $\mathtt{a}$ outside $\mathtt{a}^+$ have one type is essential: it fails for the content typing over three or more letters, where the data can be exponential (see below).

Corollary 7.11 (Polynomial data from isolated letters)**.** *Let* $r_{\max} \geq 2$ *and let* s *be*

(i) *a typing over* $\Sigma = \{\mathtt{a}, \mathtt{b}\}$ *whose positive image is a semilattice (for instance* c *or* $s_{U,2}$*), or*

(ii) *a typing* $s : \Sigma^+ \twoheadrightarrow U_1$*, over any finite alphabet, in which exactly one letter has type* 1*.*

Every $L \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$ *with a reduced good fan-out-one presentation* G *of* $L \cap \Sigma^+$*, of any rank, has a common characteristic sample* K *for* $\mathcal{B}^{\leq r_{\max}}_{1,s}$ *and* $\mathcal{B}_{1,s}$ *with*

$$\|K\|_+ \leq C' r_{\max}(\|G\| + 2)^{23}\bar{\theta}_G^8 + \mathbf{1}_{\{\varepsilon \in L\}},$$

where $C' < 10^{45}$ *is an absolute constant.*

Corollary 7.12 (Typings onto U_1 at fan-out one)**.** *Let* $r_{\max} \geq 2$ *and let* $s : \Sigma^+ \twoheadrightarrow U_1$ *be a typing. Then the bound required of each typing in Definition 7.2 holds for* s *at fan-out one and rank* $r_{\max}$ *if and only if exactly one letter has type* 1*. More precisely:*

(i) *if exactly one letter has type* 1*, the data are bounded as in Corollary 7.11, for presentations of every rank, and by Proposition 7.8 not by any* $p(\|G\|)\bar{\theta}_G$ *with* p *a polynomial;*

(ii) *if at least two letters have type* 1*, the targets of Corollary 7.7 need data exponential in the size and the thickness of their reduced good rank-two presentations.*

Proof. Since s is onto and U_1 is generated by the types of the letters, some letter has type 1 and some letter has type ζ. (i) is Corollary 7.11(ii) together with Proposition 7.8 and the remark following it, applied with $\mathtt{b}$ a letter of type ζ. For (ii) let $\mathtt{a} \neq \mathtt{b}$ have type 1 and $\mathtt{c}$ type ζ; the idempotents $e := 1$ and $z := \zeta$ are distinct with $ez = ze = z$, and $u := \mathtt{a}$, $v := \mathtt{b}$, $w := \mathtt{c}$ are letter-disjoint with $s(u) = s(v) = e$ and $s(w) = z$, so Corollary 7.7 applies. □

The proofs of Theorem 7.10 and Corollary 7.11 are in Appendix H. The idea is as follows. Take a node-minimal derivation T of a shortest yield $\mathtt{a}^n$ of a nonterminal. Two occurrences of one label at distance d give, by cancellation of pure blocks, a base z_0 outside $\mathtt{a}^+$ that commutes with $\mathtt{a}^d$ and satisfies $z_0 z_0 \sim \mathtt{a}^{c_0} z_0$. Short yields outside $\mathtt{a}^+$ of the labels on the heavy path of T, padded by at most d letters on each side, collapse to $\mathtt{a}^e z_0$; isolation is used exactly here, since it makes all words that contain $\mathtt{a}$ and lie outside $\mathtt{a}^+$ comparable. The set of exponents f with $\mathtt{a}^f z_0$ in a given distribution class is a residue class modulo some p_0; if p_0 exists, then $\mathtt{a} \sim \mathtt{a}^{1+p_0}$, so pure words have short equivalent representatives. Otherwise, membership is decided by the values of the codes. Each code is chosen to exceed the sum of the previous ones, which costs a factor of at most 3 per code, and the permutations of these codes are counted against the size of the grammar, as for the doubling families.

Corollary 7.11 does not contradict Theorem 7.3(i), whose quantification is over all alphabets: the semilattices form a pseudovariety not contained in **LG**, and the failure occurs over three letters. Since the positive image of every two-letter typing onto a semilattice is a quotient of the image of c, part (i) of the corollary covers all of them. Over three or more letters the failure of isolation for the content typing c_Σ is necessary: $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,c_\Sigma}$ for $\Sigma \supseteq \{\mathtt{a},\mathtt{b},\mathtt{c}\}$ by Proposition 7.1(ii), and W_n is its only word of content $\{\mathtt{a},\mathtt{b}\}$, so Proposition 4.6 forces data of length at least $3 \cdot 2^n - 1$. In the proof the failure occurs when the codes are collapsed: a short yield of content $\{\mathtt{a},\mathtt{c}\}$ and a yield of content $\{\mathtt{a},\mathtt{b}\}$ of one nonterminal need not be equivalent. Thus, at fan-out one, a typing onto U_1 has polynomial data if and only if exactly one letter has type 1 (Corollary 7.12); the data are then polynomial but not of the form $p(\|G\|)\,\bar{\theta}_G$ (Proposition 7.8).

Table 2 collects, for each main claim, the fixed parameters, the operators and the side conditions.

8 Fan-Out Two: Non-Lifting and Two Letters

This section shows that the exponential family of Theorem C(ii), and its images under letter-disjoint morphisms, do not carry over to fan-out two (Theorem 8.2), reduces Open Problem 9.1 to content typings (Proposition 8.5), and settles two-letter alphabets at every fan-out $f \geq 2$ (Theorem 8.6).

Lemma 8.1 (Power transfer across two tuple components)**.** *Let $f \geq 2$, $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, $u \in \Sigma^+$, and let $i_0, p \geq 1$ be an eventual index and period of $s(u)$, i.e. $s(u)^{t+p} = s(u)^t$ for $t \geq i_0$. Suppose $i, j \geq i_0$, $\delta \in p\mathbb{Z}_{>0}$, and some word of L contains the factor $u^{i+\delta+j+1}$. Then for all $w_{\mathrm{L}}, w_{\mathrm{R}} \in \Sigma^*$ and $w_{\mathrm{M}} \in \Sigma^+$,*

$$w_{\mathrm{L}} u^i w_{\mathrm{M}} u^{\delta+j} w_{\mathrm{R}} \in L \quad \Longleftrightarrow \quad w_{\mathrm{L}} u^{i+\delta} w_{\mathrm{M}} u^j w_{\mathrm{R}} \in L.$$

Proof. Write the given word as $y u^{i+\delta+j+1} z$. The tuples $\vec{x} = (u^i, u^{\delta+j})$ and $\vec{x}' = (u^{i+\delta}, u^j)$ have the same componentwise s-types by eventual periodicity. They also have the common admissible accepting context $E = y\Box_1 u \Box_2 z$, since $E[\vec{x}] = E[\vec{x}'] = y u^{i+\delta+j+1} z \in L$. The (f,s)-condition thus equates their two-component distributions. Test them in the admissible context $w_{\mathrm{L}}\Box_1 w_{\mathrm{M}} \Box_2 w_{\mathrm{R}}$; the condition $w_{\mathrm{M}} \neq \varepsilon$ is essential. □

Theorem 8.2 (The doubling families do not lift)**.** *Let $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ be any fixed finite-semigroup typing, let $u_{\mathtt{a}}, u_{\mathtt{b}}, u_{\mathtt{c}} \in \Sigma^+$ be letter-disjoint, let φ be the morphism $\mathtt{a} \mapsto u_{\mathtt{a}}$, $\mathtt{b} \mapsto u_{\mathtt{b}}$, $\mathtt{c} \mapsto u_{\mathtt{c}}$, and let $H_n := \varphi(L_n)$ for the family L_n of Proposition 7.1 (so $H_n = L_n$ when φ is the identity). Let $n \geq 2|\mathsf{S}| + 2$ and $f \geq 2$. Every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ containing $\varphi(W_n)$ contains another word with the same type, the same Parikh vector and no letter of $u_{\mathtt{c}}$. In particular $H_n \notin \mathcal{C}^{\mathrm{mcf}}_{f,s}$, and in no member of $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ is $\varphi(W_n)$ the only word of its type.*

Table 2: Scope of the main claims. "Canonical" means both $\mathcal{B}_{f,s}^{\le r_{\max}}$ and $\mathcal{B}_{f,s}$ (with one common sample); "class-wide" means every set-driven operator with characteristic samples throughout the class. Unless stated otherwise, G is a reduced good presentation of $L \cap \Sigma^+$ of fan-out $\le f$ and rank $\le r_{\max}$.

Claim	Fixed	Operators	Bound and conditions
Theorem A (Theorem 5.7)	f, $r_{\max}$, Σ, s with $s(\Sigma^+) \in \mathbf{LG}$	canonical	upper $O(\|G\|^2 \bar{\theta}_G)$, every $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$ and every G
Theorem B(i) (Proposition 6.1)	f, any s	canonical ($r \ge 1$), class-wide	lower $\Omega(\|G\| \bar{\theta}_G)$ for finite targets; needs $\|\Sigma| \ge 4f+1$ ($\|\Sigma| \ge 2$ if $f = 1$); presentations of rank two
Theorem B(ii) (Corollary 6.2)	f, $r_{\max} \ge 1$, s_{triv}	$\mathcal{B}_{f,s_{\mathrm{triv}}}$	upper $O(\|G\| \bar{\theta}_G)$ (Yoshinaka [14]); attained if $r_{\max} \ge 2$ and $\|\Sigma| \ge 4f+1$ (≥ 2 if $f = 1$)
Theorem C(i) (Theorem 7.3)	$f = 1$, $r_{\max} \ge 2$, a pseudovariety	$\mathcal{B}_{1,s}^{\le r_{\max}}$ (Definition 7.2)	uniform polynomial data iff $\mathbf{V} \subseteq \mathbf{LG}$
Theorem C(ii)	$f = 1$, s_U on three letters	canonical, class-wide	lower $3 \cdot 2^n - 1$ for presentations of size and thickness $O(n)$
Corollary 7.7	$f = 1$, $r_{\max} \ge 2$, a typing with letter-disjoint u, v, w	canonical, class-wide	lower $(\|u| + 2\|v|)2^n - 2\|v|$ for presentations of size and thickness $O(n)$
Proposition 7.8	$f = 1$, $s_{U,2}$, or any s onto U_1 with one letter of type 1	canonical ($r \ge 1$), class-wide	lower $\bar{\theta}_G^2/4$ with $\|G\| = O(\log \bar{\theta}_G)$
Corollary 7.11	$f = 1$, $r_{\max} \ge 2$; $\Sigma = \{\mathtt{a}, \mathtt{b}\}$ and s onto a semilattice, or any Σ and s onto U_1 with one letter of type 1	canonical	upper $O(r_{\max}(\|G\| + 2)^{23} \bar{\theta}_G^8)$ with an absolute constant, every $L \in \mathcal{C}_{1,s}^{\mathrm{mcf}}$ and every G of any rank
Corollary 7.12	$f = 1$, $r_{\max} \ge 2$, s onto U_1	canonical, class-wide	polynomial data iff exactly one letter has type 1
Corollary 8.7 (Theorem 8.6)	$f \ge 2$, $r_{\max} \ge 2$, $\Sigma = \{\mathtt{a}, \mathtt{b}\}$, s onto a semilattice	canonical	upper $O_{f,r_{\max}}(\|G\|^3 \bar{\theta}_G^4 \log \bar{\theta}_G)$; $O_{f,r_{\max}}(\|G\|^2 \bar{\theta}_G)$ if not Parikh-saturated
Proposition 8.5	f, $r_{\max}$	$\mathcal{B}_{f,s}^{\le r_{\max}}$	the boundary moves at f iff every c_Σ has uniform polynomial data at f
Corollary 7.6	f, $r_{\max}$, s	set-driven	IPTtD for $\mathbf{LG}$ at every f; at $f = 1$ exactly for $\mathbf{V} \subseteq \mathbf{LG}$
Theorem 8.2	any $f \ge 2$, any s	none (membership)	the family of Theorem C(ii) and its letter-disjoint images leave $\mathcal{C}_{f,s}^{\mathrm{mcf}}$, and their forced word is never the only one of its type there; no data bound at $f \ge 2$
Propositions 5.9, 5.10	the alphabets and targets L_{A}, L_{B}	none (membership)	non-context-free, fan-out two, rank two; in $\mathcal{C}_{f,s_{\mathrm{A}}}^{\mathrm{mcf}}$, $\mathcal{C}_{f,s_{\mathrm{B}}}^{\mathrm{mcf}}$ for $f \ge 2$, and for every $f \ge 1$ outside the class of every window–group, respectively window–abelian, typing; no data bound
Open			whether $\mathbf{LG}$ is the boundary at $f \ge 2$ (Open Problem 9.1), equivalently whether every c_Σ with $\|\Sigma| \ge 3$ has uniform polynomial data at f

Proof. The morphism φ is injective, and a word of H_n without the letters of $u_{\mathtt{c}}$ is the image of a word of L_n without $\mathtt{c}$, so $\varphi(W_n)$ is the only such word. Choose an eventual index $i_0 \geq 1$ and period $p \geq 1$ for $s(u_{\mathtt{b}})$ with $i_0 + p \leq |\mathsf{S}| + 1$; then $n \geq 2i_0 + p + 1$. Put $i = i_0$, $\delta = p$ and $j = n - i - \delta - 1 \geq i_0$. The word $\varphi(W_n) = u_{\mathtt{b}}\,\varphi(W_{n-1})\,\varphi(W_{n-1})\,u_{\mathtt{b}}$ starts with $u_{\mathtt{b}}^n u_{\mathtt{a}}$, so it contains $u_{\mathtt{b}}^{i+\delta+j+1}$. Let $u_{\mathtt{b}}^i$ be the i copies of $u_{\mathtt{b}}$ after the first one, and let $u_{\mathtt{b}}^{\delta+j}$ be the $\delta + j \leq n - 1$ copies immediately before the first $u_{\mathtt{a}}$ of the second copy of $\varphi(W_{n-1})$, which is preceded by at least $n - 1$ copies of $u_{\mathtt{b}}$. Then

$$\varphi(W_n) = w_{\mathrm{L}} u_{\mathtt{b}}^i w_{\mathrm{M}} u_{\mathtt{b}}^{\delta+j} w_{\mathrm{R}}$$

with $w_{\mathrm{L}} = u_{\mathtt{b}}$ and $w_{\mathrm{M}} \neq \varepsilon$ containing the first $u_{\mathtt{a}}$. Let $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$ contain $\varphi(W_n)$. By Lemma 8.1, $w' := w_{\mathrm{L}} u_{\mathtt{b}}^{i+\delta} w_{\mathrm{M}} u_{\mathtt{b}}^j w_{\mathrm{R}} \in L$. The two filled tuples have equal componentwise types, so $s(w') = s(\varphi(W_n))$; w' has the Parikh vector of $\varphi(W_n)$ and no letter of $u_{\mathtt{c}}$, but begins with $u_{\mathtt{b}}^{n+\delta} u_{\mathtt{a}}$, not $u_{\mathtt{b}}^n u_{\mathtt{a}}$, so $w' \neq \varphi(W_n)$ since $u_{\mathtt{a}}$ and $u_{\mathtt{b}}$ have no common letter. For $L = H_n$ this contradicts the uniqueness of $\varphi(W_n)$ shown above. □

Remark 8.3. Power transfer preserves the total length and Parikh vector of the word, so it does not show that rare-type representatives are short; it only excludes the family of Proposition 7.1 and its letter-disjoint images, also after words are added.

Reduction to semilattices. Let **SL** be the pseudovariety of finite semilattices (commutative idempotent semigroups). For a nonempty finite alphabet Σ, the *content typing* c_Σ maps $w \in \Sigma^+$ to the set of letters occurring in w, a morphism onto the nonempty subsets of Σ under union.

Lemma 8.4 (Coarser typings inherit characteristic samples)**.** *Let $s \preceq t$ be typings, $f, r \geq 1$, and $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$. For every finite $K \subseteq L$,*

$$L(\mathcal{B}_{f,t}^{\leq r}(K)) \subseteq L(\mathcal{B}_{f,s}^{\leq r}(K)) \subseteq L,$$

and likewise for $\mathcal{B}_{f,t}$ and $\mathcal{B}_{f,s}$. Hence every characteristic sample of L for $\mathcal{B}_{f,t}^{\leq r}$ (resp. $\mathcal{B}_{f,t}$) is one for $\mathcal{B}_{f,s}^{\leq r}$ (resp. $\mathcal{B}_{f,s}$).

Proof. Both grammars have the same nonterminals and the same Start, Const and Comp rules (Definition 3.4). A Link rule of the t-grammar joins observed tuples with a common context and equal t-types; since $\ker t \subseteq \ker s$, their s-types are equal, so it is a Link rule of the s-grammar. This gives the first inclusion; the second is Proposition 3.10. If F is characteristic for the t-operator and $F \subseteq K \subseteq L$, the left-hand side equals L, hence so does the middle term. □

Proposition 8.5 (Reduction of the fan-out boundary to semilattices)**.** *Fix $f \geq 1$ and $r_{\max} \geq 1$, and read uniform polynomial characteristic-data length at fan-out f and rank $r_{\max}$ (Definition 7.2).*

(i) *Every pseudovariety $\mathbf{V} \not\subseteq \mathbf{LG}$ contains* **SL***. Consequently, the pseudovarieties with uniform polynomial characteristic-data length are exactly those contained in* **LG** *if and only if* **SL** *does not have it.*

(ii) **SL** *has uniform polynomial characteristic-data length if and only if, for every nonempty finite alphabet Σ, the content typing c_Σ satisfies the condition of Definition 7.2.*

Proof. (i) A finite semigroup $\mathsf{S} \notin \mathbf{LG}$ is the positive image of its evaluation typing over the alphabet S, so it has a subsemigroup isomorphic to U_1 (Theorem B.4); hence $U_1 \in \mathbf{V}$. Every finite semilattice H embeds into U_1^{H} via $x \mapsto (\chi_z(x))_{z \in \mathsf{H}}$, where $\chi_z(x) = 1$ if $zx = z$ and $\chi_z(x) = \zeta$ otherwise (a

morphism, since $zxy = z$ iff $zx = z$ and $zy = z$), so $\mathbf{SL} \subseteq \mathbf{V}$. The property of Definition 7.2 passes from a pseudovariety to its subpseudovarieties, because fewer typings are quantified. If $\mathbf{SL}$ lacks it, so does every $\mathbf{V} \not\subseteq \mathbf{LG}$, while every $\mathbf{V} \subseteq \mathbf{LG}$ has it by Theorem 5.7. Conversely, $U_1 \notin \mathbf{LG}$, so $\mathbf{SL} \not\subseteq \mathbf{LG}$.

(ii) The image of c_Σ is a semilattice, which gives one direction. For the other, let $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ with $\mathsf{S} \in \mathbf{SL}$. Since S is commutative and idempotent, $s(w)$ is the product of $s(a)$ over the letters a occurring in w, so $s \preceq c_\Sigma$ and $\mathcal{C}^{\mathrm{mcf}}_{f,s} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,c_\Sigma}$ (item (a) after Definition 2.8). For $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and a presentation G, a characteristic sample for $\mathcal{B}^{\leq r_{\max}}_{f,c_\Sigma}$ of length at most $p_{c_\Sigma}(\|G\|, \bar{\theta}_G)$ is one for $\mathcal{B}^{\leq r_{\max}}_{f,s}$ by Lemma 8.4; take $p_s := p_{c_\Sigma}$. □

By Theorem 7.3, $\mathbf{SL}$ lacks the property at fan-out one when $r_{\max} \geq 2$, although over two letters every typing in $\mathbf{SL}$ has polynomial data there (Corollary 7.11). At fan-out two the question is open (Open Problem 9.1), and the next theorem settles two-letter alphabets at every fan-out $f \geq 2$. Rare-type lower bounds transfer to content typings: for $s \preceq c_\Sigma$ every content type contained in an s-type m has shortest word at least $\mu_m(L)$, computed for s.

Two letters. Call L *Parikh-saturated* if membership of a word of Σ^+ in L depends only on its Parikh vector.

Theorem 8.6 (Content typings on two letters at every fan-out $f \geq 2$)**.** *Let $\Sigma = \{\mathtt{a}, \mathtt{b}\}$, $c := c_\Sigma$, $f \geq 2$, $r_{\max} \geq 2$, and put $D_f := (6f+1)3^f$ and $C_{f,r_{\max}} := (r_{\max}+1)\big(4r_{\max}(3f+1) + D_f\big) + 3$, and let $H_m := \sum_{k=1}^{m} 1/k \leq 1 + \ln m$ ($H_0 := 0$). For every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,c}$ and every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation G of $L \cap \Sigma^+$, $\tau(G,c) \leq C_{f,r_{\max}} |V|^2 \bar{\theta}_G^4 H_{\bar{\theta}_G}$, and the operators $\mathcal{B}^{\leq r_{\max}}_{f,c}$ and $\mathcal{B}_{f,c}$ have a common characteristic sample K with*

$$\|K\|_+ \leq 4^{f r_{\max}} \big(3 + r_{\max} C_{f,r_{\max}}\big) \|G\|^3 \bar{\theta}_G^4 H_{\bar{\theta}_G} + \mathbf{1}_{\{\varepsilon \in L\}}.$$

If L is not Parikh-saturated, then $t_{\mathrm{typ}}(G,c) \leq D_f(r_{\max}+1)|V|\bar{\theta}_G$ and one may take $\|K\|_+ \leq 4^{f r_{\max}}\big(3 + r_{\max} D_f(r_{\max}+1)\big)\|G\|^2 \bar{\theta}_G + \mathbf{1}_{\{\varepsilon \in L\}}$.

Corollary 8.7 (Semilattice typings on two letters)**.** *Let $f \geq 2$, $r_{\max} \geq 2$ and let $s : \{\mathtt{a}, \mathtt{b}\}^+ \twoheadrightarrow \mathsf{S}$ be a typing with $\mathsf{S} \in \mathbf{SL}$, for instance $s_{U,2}$ ($\mathtt{a} \mapsto 1$, $\mathtt{b} \mapsto \zeta$). Then the bound required of each typing in Definition 7.2 holds for this fixed typing at fan-out f and rank $r_{\max}$, with the bound of Theorem 8.6, for $\mathcal{B}^{\leq r_{\max}}_{f,s}$ and for $\mathcal{B}_{f,s}$.*

Proof. As in the proof of Proposition 8.5(ii), $s \preceq c$ and $\mathcal{C}^{\mathrm{mcf}}_{f,s} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,c}$; apply Theorem 8.6 and Lemma 8.4. □

The proof is in Appendix J. It uses the conditions of arities one and two, which hold at every fan-out $f \geq 2$, through runs of one letter: once x^4 occurs in a word of L for a letter x, membership depends only on the x-skeleton (each maximal run of x replaced by one x) and the number of letters x, and two runs of the other letter next to a run xx either force Parikh saturation or merge into a forbidden longer run. Without saturation this bounds the antichains of heavy nodes in a shortest typed derivation; with saturation, node-minimal derivations of words with a prescribed Parikh vector are analysed directly. The theorem does not separate fan-outs: by Corollary 7.11, two-letter semilattice typings have polynomial data at fan-out one as well, so over two letters they have polynomial data at every fan-out. For three or more letters a long component of content $\{\mathtt{a}, \mathtt{b}\}$ need not contain a long run, and the question is open.

9 Conclusion and an Open Problem

Theorem 4.3 and Corollary 4.8 reduce characteristic data to the shortest typed yield and to the representative length. Local-group typings have data $O(\|G\|^2\bar{\theta}_G)$ at every fan-out (Theorem A), the product $\|G\|\bar{\theta}_G$ is necessary (Theorem B), and at fan-out one **LG** is the exact boundary for pseudovarieties, with exponential failure outside it (Theorem C, Corollary 7.6). For typings onto semilattices this failure needs three letters, and at fan-out one a typing onto U_1 has polynomial data exactly when one letter has type 1 (Corollaries 7.11, 8.7 and 7.12). At fan-out two the exponential fan-out-one family does not lift (Theorem 8.2).

Open Problem 9.1 (The boundary at fan-out two). Fix $f \geq 2$ and $r_{\max} \geq 2$. Is **LG** the pseudovariety boundary at fan-out f, that is, does every pseudovariety of finite semigroups not contained in **LG** fail to have uniform polynomial characteristic-data length at fan-out f and rank $r_{\max}$ (Definition 7.2)?

Pseudovarieties contained in **LG** have uniform polynomial characteristic-data length at every fan-out (Theorem 5.7). Every pseudovariety not contained in **LG** contains U_1 (Theorem B.4), so the answer is positive as soon as one typing onto U_1 needs superpolynomial data at fan-out f; by Proposition 4.6 a family of targets whose shortest word of some type is superpolynomially long in $\|G\|$ and $\bar{\theta}_G$ would suffice, and Theorem 8.2 excludes the doubling family. A negative answer would show that the boundary depends on the fan-out. By Proposition 8.5 the problem asks whether the content typings c_Σ have uniform polynomial data at fan-out f; for $|\Sigma| = 2$ they have it at every fan-out (Corollary 7.11 and Theorem 8.6), so the problem concerns alphabets of at least three letters.

For a related quantity, the size of finite-monoid observations preserving guarded tuple substitution, the arity at which it stabilizes over a pseudovariety is 1, 2 or ∞ [24].

Tightness of the exponents. In $O(\|G\|^2\bar{\theta}_G)$, for window–group and local-group typings alike, one factor $\|G\|$ counts the witness words and the other bounds root-to-leaf paths in a λ-minimal derivation by the number of node labels (Lemma 5.1). The path factor is genuine for the witness construction: the grammars $S \to SS \mid \mathtt{a} \mid \mathtt{b} \mid A_0 \mid \cdots \mid A_n$, $A_0 \to \mathtt{b}$, $A_i \to \mathtt{a} \mid A_{i-1}\mathtt{b}$ $(1 \leq i \leq n)$ generate $\{\mathtt{a}, \mathtt{b}\}^+$ with $\|G_n\| = 10n + 16$ and $\bar{\theta}_{G_n} = 2$, but $\mathtt{b}^{n+1}$ is the only yield of A_n of $s_{1,1}$-type $\langle \mathtt{b}, \mathtt{b} \rangle$, so $t_{\mathrm{typ}}(G_n, s_{1,1}) \geq n + 1$. For this target, however, $\tau(G_n, s_{1,1}) \leq 2$ (Corollary 4.8), so the example bounds the witness method, not the data. For the trivial typing the upper bound $O(\|G\|\bar{\theta}_G)$ is attained by the marker families (Corollary 6.2); for every typing with $\mathsf{S} \in \mathbf{LG}$ the remaining gap is a single factor $\|G\|$, a gap between the known examples and the bounds rather than a proved ratio. Neither lower-bound argument closes it: for a fixed typing the rare-type argument yields at most $|\mathsf{S}|(t_{\mathrm{typ}} + 1)$, and the marker families consist of $O(\|G\|)$ words of length $\bar{\theta}_G = t_{\mathrm{typ}}$. For noncommutative groups the constant $M_{\mathbb{G},f}$ is doubly exponential in f because of the Ramsey-type homogenization (Lemma A.8); $|\mathbb{G}|$ owners suffice at $f = 1$, and whether $\mathrm{poly}(|\mathbb{G}|^f)$ owners suffice for $f \geq 2$, as for abelian groups, is open. At fan-out one, for isolated letters, the exponents $(23, 8)$ of Corollary 7.11 come from the counting argument of Appendix H and are far from the lower bound $\bar{\theta}_G^2/4$ of Proposition 7.8.

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Data availability

No empirical dataset is associated with this theoretical paper. All proofs are given in full.

A Simultaneous Block Replacement in Finite Groups

This appendix proves Proposition 5.2, used in Lemmas 5.3 and B.8. The abelian case is a zero-sum argument. The general case never commutes factors; it has three parts: an order classification for homogeneous point configurations (Lemma A.4), an interval selection by boundary states (Lemma A.5), and a Ramsey-type extraction of homogeneous indices (Lemmas A.7 and A.8).

Roadmap of the general case. The proof of Lemma A.2(b), at the end of this appendix, applies the steps below in this order; each step keeps the owners it does not discard unchanged and absorbs the blocks of discarded owners into the constants (they keep their original values in every row).

Step	Input → output	Invariant gained
1. Common pattern	$m \geq M_{\mathbb{G},f}$ owners → $m/\mathcal{P}_f$ owners	same number of blocks, lines and within-owner order
2. Lemmas A.7, A.8	→ $N_{\mathbb{G},f} = \max\{3, \lvert\mathbb{G}\rvert^f\}$ owners	each comparison "block t of i left of block u of j" is constant over $i < j$
3. Lemma A.4	all lines → at most f class regions in total, each a run of $N_{\mathbb{G},f}$ chunks	one chunk per owner, increasing or decreasing; no foreign points inside a region
4. Lemma A.5	regions → an interval $J = (a, b]$ of owners	the states $(\eta_p^{\mathcal{Q}})_{\mathcal{Q}}$, $\eta_p^{\mathcal{Q}} = P_p Q_p^{-1}$, agree at a and b, so every ordered row product is preserved

For $f = 2$ and $\mathbb{G} = S_3$, Example A.6 shows the output of Step 3 (one increasing region on row 1, one decreasing region on row 2) and the use of η in Step 4: the replaced owners do not preserve their own contributions, only the two ordered row products are preserved.

Definition A.1 (Block-replacement family)**.** Let $\mathbb{G}$ be a finite group and $f \geq 1$. A *block-replacement family* of width f over $\mathbb{G}$ with index set $[m]$ consists of $d \leq f$ *lines*; on line ϱ, a left-to-right sequence of blocks $\mathfrak{b}_{\varrho,1}, \ldots, \mathfrak{b}_{\varrho,q_\varrho}$ and constants $a_{\varrho,0}, \ldots, a_{\varrho,q_\varrho} \in \mathbb{G}$; for every block $\mathfrak{b}$ an *owner* $o(\mathfrak{b}) \in [m]$ and two values $x_{\mathfrak{b}}, x'_{\mathfrak{b}} \in \mathbb{G}$; every index owns at most f blocks. For $J \subseteq [m]$ put

$$\mathrm{val}_\varrho(J) := a_{\varrho,0}\, y_{\varrho,1}\, a_{\varrho,1} \cdots y_{\varrho,q_\varrho}\, a_{\varrho,q_\varrho}, \qquad y_{\varrho,j} := \begin{cases} x'_{\mathfrak{b}_{\varrho,j}}, & o(\mathfrak{b}_{\varrho,j}) \in J, \\ x_{\mathfrak{b}_{\varrho,j}}, & o(\mathfrak{b}_{\varrho,j}) \notin J. \end{cases}$$

Lemma A.2 (Block replacement)**.** *In a block-replacement family of width f over $\mathbb{G}$ there is a nonempty $J \subseteq [m]$ with $\mathrm{val}_\varrho(J) = \mathrm{val}_\varrho(\emptyset)$ for every line ϱ, provided that either* (a) *$\mathbb{G}$ is abelian and $m > \mathsf{d}(\mathbb{G}^f)$, or* (b) *$\mathbb{G}$ is arbitrary and $m \geq M_{\mathbb{G},f}$, or* (c) *$\mathbb{G}$ is arbitrary, $f = 1$ and $m \geq |\mathbb{G}|$, with d and $M_{\mathbb{G},f}$ as in Section 5.1.*

Proof of (a). Write $\mathbb{G}$ additively. By commutativity $\mathrm{val}_\varrho(J) - \mathrm{val}_\varrho(\emptyset) = \sum_{i \in J} \delta_{i,\varrho}$, where $\delta_{i,\varrho}$ is the sum of $x'_{\mathfrak{b}} - x_{\mathfrak{b}}$ over the blocks $\mathfrak{b}$ of line ϱ owned by i. The sequence $(\delta_i)_{i \in [m]}$ with $\delta_i := (\delta_{i,\varrho})_\varrho \in \mathbb{G}^d$ has length $m > \mathsf{d}(\mathbb{G}^f) \geq \mathsf{d}(\mathbb{G}^d)$ ($\mathbb{G}^d$ embeds in $\mathbb{G}^f$), so it has a nonempty zero-sum subsequence $(\delta_i)_{i \in J}$. □

Remark A.3 (Size of the zero-sum constant). The prefix-sum pigeonhole gives $\mathsf{d}(\mathbb{G}^f) \leq |\mathbb{G}|^f - 1$, but much more is true. Write $\mathbb{G} \cong C_{q_1} \oplus \cdots \oplus C_{q_\ell}$ and $q = \exp(\mathbb{G}) \geq 2$. Then

$$f \sum_{i=1}^{\ell} (q_i - 1) \;\leq\; \mathsf{d}(\mathbb{G}^f) \;\leq\; (q-1) + q \log \frac{|\mathbb{G}|^f}{q},$$

where the lower bound is attained by a basis sequence and the upper bound is due to van Emde Boas and Kruyswijk [32, Proposition 5.1.8 and Theorem 5.5.5]; for a p-group the lower bound is an equality [32, Theorem 5.5.9]. Hence $\mathsf{d}(\mathbb{G}^f) = O(\exp(\mathbb{G})\, f \log |\mathbb{G}|)$; for $\mathbb{G} = C_2$ the antichain bound drops from $2^f - 1$ to f. Since zero-sum-free sequences of length $\mathsf{d}(\mathbb{G}^d)$ exist, the argument cannot do better; we do not claim that a grammar realizes them.

A.1 Arbitrary Finite Groups: Order Homogenization

Lemma A.4 (A homogeneous block-order lemma)**.** *Let $\mathcal{T}$ be a finite set of types and $n \geq 3$, and let (t,i), $t \in \mathcal{T}$, $i \in [n]$, be distinct points on a line. Suppose that* (i) *the order of (t,i) and (u,i) does not depend on i, and* (ii) *for each ordered pair (t,u), including $t = u$, the order of (t,i) and (u,j) is the same for all $i < j$. Then $\mathcal{T}$ is partitioned into classes such that: the points of each class occupy a consecutive region containing no other points; each class has a direction (increasing or decreasing in i); and its region consists of n successive* chunks*, one per index in that direction, each containing exactly one point of every type of the class, in an order independent of the index.*

Proof. By (ii) with $t = u$, the points of each type are monotone in i. For distinct t, u let X mean $(t,i) < (u,j)$ and Y mean $(u,i) < (t,j)$ for $i < j$; both are constant by (ii).

Both increasing. $X \wedge Y$ puts all points of index i of both types before those of index $j > i$ (*interleaved by rank*). Under $X \wedge \neg Y$ all comparisons of different indices put t left of u, and so do equal indices, since $(u,i) < (t,i)$ would give $(t,1) < (t,2) < (u,1) < (t,1)$; so t and u are *separated*. $\neg X \wedge Y$ is symmetric, and $\neg X \wedge \neg Y$ gives $(t,1) < (t,2) < (u,1) < (u,2) < (t,1)$.

t increasing, u decreasing. $X \wedge Y$ gives $(t,2) < (u,3) < (u,1) < (t,2)$ and $\neg X \wedge \neg Y$ gives $(t,3) < (u,2) < (t,1) < (t,3)$. Under $X \wedge \neg Y$ the comparisons of different indices put t left of u, and $(u,i) < (t,i)$ would give $(t,1) < (u,2) < (u,1) < (t,1)$; so they are separated, and $\neg X \wedge Y$ is symmetric. Reversing indices covers two decreasing types.

Thus two types are separated, or have the same direction and are interleaved. Interleaving is transitive: if t, u and u, v are interleaved (all increasing, say) but t, v are separated, then $(v,1) < (u,2) < (t,3) < (v,1)$ if t is left of v, and $(t,1) < (u,2) < (v,3) < (t,1)$ otherwise. So interleaving is an equivalence relation. A type outside a class is separated from every type of the class and lies on the same side of all of them, since otherwise two interleaved types would be separated; hence class regions contain no other points. Inside a class, interleaving gives the chunks, and (i) gives the within-chunk order. □

Lemma A.5 (Simultaneous finite-group interval shortening)**.** *Let $\mathbb{G}$ be a finite group and consider a block-replacement family of width f over $\mathbb{G}$ with $n \geq \max\{3, |\mathbb{G}|^f\}$ indices, in which all indices have the same number of blocks, on the same lines, in the same within-index order, and in which, for every two block positions t, u on a common line, the truth value of "block t of index i lies left of block u of index j" is the same for all $i < j$. Then a nonempty interval J of indices satisfies $\mathrm{val}_\varrho(J) = \mathrm{val}_\varrho(\emptyset)$ for every line ϱ.*

Proof. Apply Lemma A.4 on each line; there are $c \leq f$ class regions in total. Fix a region $\mathcal{Q}$ and number its chunks $1, \ldots, n$ from left to right. Assign constants between chunks to the chunk on their left, and absorb fixed constants at the ends of the region into its first or last chunk. Each chunk has original and replacement products with the *same* fixed constants in the same order. Let P_p be the original value of chunks $1, \ldots, p$ and Q_p their value when all of them are replaced ($P_0 = Q_0 = 1_\mathbb{G}$), and put $\eta_p^\mathcal{Q} := P_p Q_p^{-1}$. Replacing exactly chunks $p+1, \ldots, p'$ gives the ordered value $P_p\,(Q_p^{-1}Q_{p'})\,(P_{p'}^{-1}P_n)$, which equals P_n iff $P_pQ_p^{-1} \cdot Q_{p'}P_{p'}^{-1} = 1_\mathbb{G}$, that is, iff $\eta_p^\mathcal{Q} = \eta_{p'}^\mathcal{Q}$; no factors are commuted. The order in $\eta_p^\mathcal{Q} = P_pQ_p^{-1}$ is forced by the position of the replaced chunks between an original prefix and an original suffix; the element $Q_p^{-1}P_p$ need not detect this in a noncommutative group. If $\mathcal{Q}$ is increasing, the owners in $(a, b]$ occupy exactly chunks $(a, b]$; if it is decreasing, chunks $(n-b, n-a]$. For $j = 0, \ldots, n$ let the state be the vector in $\mathbb{G}^c$ whose $\mathcal{Q}$-entry is $\eta_j^\mathcal{Q}$ for increasing and $\eta_{n-j}^\mathcal{Q}$ for decreasing regions. Since $n+1 > |\mathbb{G}|^c$, two states coincide, at $a < b$. Replacing the owners $a+1, \ldots, b$ then preserves every region value, and nothing outside the regions changes. □

Example A.6 (Opposite orientations and noncommuting values). In S_3 let $\sigma_1 = (12)$, $\sigma_2 = (23)$. Three owners have one block B_i on row 1 and one block W_i on row 2, so the family has width two:

$$\text{row 1: } B_1\,\sigma_1\,B_2\,\sigma_1\,B_3, \qquad \text{row 2: } W_3\,\sigma_2\,W_2\,\sigma_2\,W_1,$$

so row 1 is one increasing class and row 2 one decreasing class. Originally $B_i = W_i = 1$, and both rows have value 1. Let owner 1 have the trivial replacement $B_1' = B_1$, $W_1' = W_1$, and replace exactly owners $J = \{2, 3\}$ by $B_2' = \sigma_2$, $W_2' = \sigma_2\sigma_1\sigma_2$, $B_3' = \sigma_1\sigma_2\sigma_1$, $W_3' = \sigma_1$. Then row 1 becomes $\sigma_1 \cdot \sigma_2\sigma_1 \cdot \sigma_1\sigma_2\sigma_1 = 1$ and row 2 becomes $\sigma_1\sigma_2 \cdot \sigma_2\sigma_1\sigma_2 \cdot \sigma_2 = 1$, computed in order. In the notation of the proof, $(a, b] = (1, 3]$ and $\eta_1 = \eta_3 = 1$ on row 1, while $\eta_{3-b} = \eta_0 = 1 = \eta_2 = \eta_{3-a}$ on row 2: one owner interval matches two different boundary-index orientations, although neither replaced owner preserves its own contribution. The example illustrates the mechanism; with three owners it is far below the threshold of Proposition 5.2, and the good owner interval is found by direct computation.

Lemma A.7 (Two monotone sequences)**.** *Let $m \geq 3$, $m' \geq (m-1)^2 + 1$ and $\mathrm{lt} : \{(i, j) : 1 \leq i < j \leq m'\} \to \{0, 1\}$ be monotone in each argument with fixed directions: for each fixed i it is monotone in j, for each fixed j monotone in i, and each of these two directions is the same for all values of the fixed argument. Then* lt *is constant on the pairs $i < j$ of some $I \subseteq [m']$ with $|I| = m$.*

Proof. Replacing lt by $1 - \mathrm{lt}$ if necessary, rows are nondecreasing in j. It suffices to find an index set of size at least m, since constancy persists in subsets.

If columns are nonincreasing in i, let $\mathrm{nx}(i) := \min\{j > i : \mathrm{lt}(i, j) = 1\}$ (∞ if none); nx is nondecreasing. The chain $i_1 = 1$, $i_{r+1} = \mathrm{nx}(i_r)$ satisfies $\mathrm{lt}(i_a, i_b) = 1$ for $a < b$; if it has m terms we are done. Otherwise its at most $m-1$ terms cut $[m']$ into at most $m-1$ intervals $[i_r, i_{r+1})$, $[i_L, m']$, one of which has at least $m'/(m-1) > m-1$, hence at least m, indices, and inside it $j < \mathrm{nx}(i_r) \leq \mathrm{nx}(i)$ gives $\mathrm{lt}(i, j) = 0$.

If columns are nondecreasing, let i^* be the least index with $\mathrm{lt}(i^*, i^*+1) = 1$ ($i^* = m'$ if none). Then $\mathrm{lt}(i, j) = 1$ for $i^* \leq i < j$ and $\mathrm{lt}(i, j) \leq \mathrm{lt}(j-1, j) = 0$ for $i < j \leq i^*$, and one of $[1, i^*]$, $[i^*, m']$ has at least $(m'+1)/2 \geq m$ indices because $(m-1)^2 + 1 \geq 2m - 1$ for $m \geq 3$. □

Lemma A.8 (Polynomial order homogenization)**.** *Let $n_0 \geq 3$, $m_0 := n_0$ and $m_{q+1} := (m_q - 1)^2 + 1$, so $m_q \leq n_0^{2^q}$. Let $\mathcal{T}$ be a finite set of types, each assigned to a line, and for $i \in [n]$ and $t \in \mathcal{T}$ let $p_t(i)$ be a point on the line of t, the points on each line being distinct. Let $\mathcal{O}$ be the set of ordered pairs (t, u), $t = u$ allowed, of types on a common line, If $n \geq m_{|\mathcal{O}|}$, there is $I \subseteq [n]$ with $|I| = n_0$ such that for every $(t, u) \in \mathcal{O}$ the truth value of $p_t(i) < p_u(j)$ is the same for all $i < j$ in I.*

Proof. Treat the pairs one at a time, the pairs (t, t) first; each step passes from an index set of size at least $(m-1)^2 + 1$ to a subset of size at least m, and earlier homogeneity persists in subsets. For (t, t), Erdős–Szekeres gives a monotone subsequence of length m. For $t \neq u$ (both now monotone), relabel the current indices $1, \ldots, m'$ and put $\mathrm{lt}(i, j) := [p_t(i) < p_u(j)]$ for $i < j$. For fixed i, lt is monotone in j in the direction of p_u; for fixed j, monotone in i in the direction opposite to p_t. Since $m \geq m_0 = n_0 \geq 3$, Lemma A.7 applies. □

Proof of Lemma A.2(c)*.* An index owning no block can be chosen alone, so let every index own one block; at $f = 1$ all blocks lie on the single line. Renumber the indices by the positions of their blocks, so that they form one increasing region with one block per chunk. As in the proof of Lemma A.5, replacing exactly the indices $a + 1, \ldots, b$ preserves the line value iff $\eta_a = \eta_b$, and the $m + 1 > |\mathbb{G}|$ values $\eta_0, \ldots, \eta_m \in \mathbb{G}$ cannot all be distinct. □

Proof of Lemma A.2(b)*.* An index owning no block can be chosen alone, so assume every index owns a block. The *pattern* of an index records the number of its blocks, the line of each, and their order on shared lines; there are at most $f \cdot f^f \cdot f! = \mathcal{P}_f$ patterns, so at least $N_{\mathbb{G},f}^{2^{f^2}}$ indices share one. Absorb the blocks of all other indices into the constants: these indices are never replaced, so their blocks keep their original values in every row. The blocks of the common pattern give at most f types and at most f^2 ordered pairs on common lines, so Lemma A.8 with $n_0 = N_{\mathbb{G},f}$ yields $N_{\mathbb{G},f}$ indices with homogeneous order comparisons. The blocks of the indices discarded in this extraction are absorbed into the constants in the same way. The common pattern fixes the order of the blocks of one index, which is hypothesis (i) of Lemma A.4, and Lemma A.8 gives hypothesis (ii); a comparison with $i > j$ is the negation of the comparison for the reversed pair, since the points are distinct. The remaining $N_{\mathbb{G},f}$ indices then satisfy the hypotheses of Lemma A.5, whose interval is the required J. □

For fixed f, $M_{\mathbb{G},f} = \mathcal{P}_f \max\{3, |\mathbb{G}|^f\}^{2^{f^2}}$ is polynomial in $|\mathbb{G}|$; its degree $f \cdot 2^{f^2}$ is exponential in f^2, so $M_{\mathbb{G},f}$ itself is doubly exponential in f.

B Local-Group Coordinates and MCFG Block Shortening

LG is always tested on the positive image $\mathsf{S} = \widehat{s}(\Sigma^+)$. The classical identity $\mathbf{LG} = \mathbf{Grp} * \mathbf{D}$ places it within the sliding-window and delay theory [34, 35], and the exclusion of $U_1 = \{1, \zeta\}$ ($\zeta^2 = \zeta$) is [33, Proposition 2.3]. Write x^ω for the idempotent power of an element x of a finite semigroup. Equivalently, **LG** is defined by the single pseudo-identity $(x^\omega y x^\omega)^\omega = x^\omega$: for an idempotent $e = x^\omega$ the finite monoid $e\mathsf{S}e = \{eye : y \in \mathsf{S}\}$ is a group exactly when e is its only idempotent, and every idempotent has the form x^ω. The coordinates below are a self-contained implementation of this algebra, not a new characterization; what is specific to MCFG learning is the factorization of group values into fixed separators and replaceable blocks (Lemma B.2). Throughout, $k \geq 1$, $\mathbb{G}$ is a finite group, $\lambda_{k,\nu}$ is as in Definition 5.6, a word is *short* if $|w| < 2k$ (recorded exactly by $\widehat{s}_{k,k}$) and *long* otherwise, and for $x, y \in \Sigma^k$ we put $\Delta_\nu(x, y) := \prod_{i=1}^{k} \nu((xy)[i..i+k])$, the ordered product over the k factors of length $k + 1$ of xy that start in x.

Pumping. For a finite semigroup T, every product of at least $|\mathsf{T}|$ elements lies in $\mathsf{T}^1E(\mathsf{T})\mathsf{T}^1$ [33, Lemma 2.2], [25]: either two of its first $|\mathsf{T}|$ prefix products coincide, $t_i = t_j$ with $i < j$, and then $t_i = t_i x^\omega$ for the product x of the factors $i+1, \dots, j$; or these prefix products exhaust T and one of them is idempotent. All three uses below (Lemmas B.3 and B.5, Theorem B.4) rest on this fact.

Lemma B.1 (Sliding-factor maps are morphisms)**.** *$\lambda_{k,\nu}(uv)$ is a function of $\lambda_{k,\nu}(u)$ and $\lambda_{k,\nu}(v)$; hence $\lambda_{k,\nu}$ is a morphism onto a finite monoid $M_{k,\nu}$ with at most $|\Sigma|^{<2k} + |\Sigma|^{2k}|\mathbb{G}|$ elements. For long u, v,*

$$\lambda_{k,\nu}(uv) = \big(\langle \mathrm{pref}_k\, u, \mathrm{suf}_k\, v\rangle,\ \gamma_{k,\nu}(u)\, \Delta_\nu(\mathrm{suf}_k\, u, \mathrm{pref}_k\, v)\, \gamma_{k,\nu}(v)\big).$$

Proof. The window component is a morphism. For long u, v the factors of length $k+1$ of uv are, in order, those of u, the k factors starting in the last k positions of u (inside $\mathrm{suf}_k(u)\,\mathrm{pref}_k(v)$), and those of v. If u is short and v long, the factors starting in u lie inside $u\,\mathrm{pref}_k(v)$ and the others are those of v; the case of long u and short v is symmetric; and if both are short, uv is determined exactly. □

Lemma B.2 (Block decomposition)**.** *Let $w = x_0\mathfrak{b}_1x_1\cdots\mathfrak{b}_qx_q$ and $w' = x_0\mathfrak{b}'_1x_1\cdots\mathfrak{b}'_qx_q$ with $\widehat{s}_{k,k}(\mathfrak{b}'_j) = \widehat{s}_{k,k}(\mathfrak{b}_j)$ for all j. Then $\widehat{s}_{k,k}(w') = \widehat{s}_{k,k}(w)$, and if w is long there are $z_0, \dots, z_q \in \mathbb{G}$, depending only on the x_j and the window types of the $\mathfrak{b}_j$, with*

$$\gamma_{k,\nu}(w) = z_0\,\gamma_{k,\nu}(\mathfrak{b}_1)\,z_1\cdots\gamma_{k,\nu}(\mathfrak{b}_q)\,z_q, \qquad \gamma_{k,\nu}(w') = z_0\,\gamma_{k,\nu}(\mathfrak{b}'_1)\,z_1\cdots\gamma_{k,\nu}(\mathfrak{b}'_q)\,z_q.$$

Proof. The window claim holds because $\widehat{s}_{k,k}$ is a morphism. A short block equals its replacement and is absorbed into the fixed words, so let all blocks be long. A factor of length $k+1$ either lies inside some block or meets every block it touches in a prefix or suffix of length at most k; factors of the second kind are spelled by the fixed words and the first and last k letters of the blocks, so they are the same in w and w' and occur in the same order. Ordered by starting position, the factors inside $\mathfrak{b}_j$ form one run with product $\gamma_{k,\nu}(\mathfrak{b}_j)$; let z_j be the ordered product of the factors between consecutive runs. □

Lemma B.3 (Sliding-factor coordinates)**.** *Let s be a typing whose positive image S has all its idempotents in its minimal ideal $\mathsf{K} = \mathcal{M}(\mathbb{H}; I, \Lambda; \mathsf{P})$, a Rees matrix semigroup over a group $\mathbb{H}$ with sandwich matrix $\mathsf{P} = (p_{ji})_{j\in\Lambda, i\in I}$, whose elements $(i,\gamma,j) \in I\times\mathbb{H}\times\Lambda$ multiply by $(i,\gamma,j)(i',\gamma',j') = (i, \gamma p_{ji'}\gamma', j')$. Let $k \geq 1$ satisfy $\mathsf{S}^k \subseteq \mathsf{K}$ (by pumping, $\mathsf{S}^{|\mathsf{S}|} \subseteq \mathsf{S}^1E(\mathsf{S})\mathsf{S}^1 \subseteq \mathsf{K}$, so $k = |\mathsf{S}|$ will do; the least such k gives the smallest windows). There is $\nu : \Sigma^{k+1} \to \mathbb{H}$ such that every w with $|w| \geq 2k$ satisfies*

$$s(w) = \big(\mathrm{row}(s(\mathrm{pref}_k\, w)),\ \mathrm{grp}(s(\mathrm{pref}_k\, w))\,\gamma_{k,\nu}(w),\ \mathrm{col}(s(\mathrm{suf}_k\, w))\big), \tag{$\star$}$$

where row, grp, col *are the three coordinates of an element of K. In particular $s(w)$ is determined by its first and last k letters and an ordered product in $\mathbb{H}$.*

Proof. (1) Coordinates. Fix $i_0 \in I$; for $j \in \Lambda$ the element $r_j := (i_0, p_{ji_0}^{-1}, j)$ is a right identity for the elements of K with Λ-coordinate j. For $y \in \mathsf{S}$ write $r_jy = (i', \gamma', j')$ and $\mathrm{act}(j,y) := p_{ji'}\gamma'$; then $(i,\gamma,j)y = (i, \gamma\,\mathrm{act}(j,y), j')$, so right multiplication fixes the I-coordinate. Dually, left multiplication fixes the Λ-coordinate of an element of K: for any $j'' \in \Lambda$ the idempotent $e' := (i, p_{j''i}^{-1}, j'') \in \mathsf{K}$ satisfies $e'(i,\gamma,j) = (i,\gamma,j)$, so $y(i,\gamma,j) = (ye')(i,\gamma,j)$ with $ye' \in \mathsf{K}$, and the Rees product of two elements of K has the Λ-coordinate of its right factor.

(2) Group coordinate. Let $w = a_1\cdots a_m$ with letters a_t, $m \geq 2k$, and $s(a_1\cdots a_k) = (i,\gamma,j_k)$. For $t \geq k$ the Λ-coordinate j_t of $s(a_1\cdots a_t)$ equals that of $s(a_{t-k+1}\cdots a_t)$ by $\mathsf{S}^k \subseteq \mathsf{K}$ and (1). Put $\nu(F) := \mathrm{act}\big(\mathrm{col}(s(F[1..k])), s(F[k+1])\big)$. Applying the right-action identity for $t = k, \dots, m-1$ in order gives $s(w) = (i,\ \gamma\prod_{t=k}^{m-1}\nu(a_{t-k+1}\cdots a_{t+1}),\ j_m)$, and the product is $\gamma_{k,\nu}(w)$. □

Theorem B.4 (Local-group criteria in sliding-factor coordinates)**.** *For a typing s with positive image S the following are equivalent:* (i) $\mathsf{S} \in \mathbf{LG}$*;* (ii) S *has no subsemigroup isomorphic to* U_1*;* (iii) *every idempotent of* S *lies in the minimal ideal of* S*;* (iv) s *is local-group-determined, and one may take* $k = |\mathsf{S}|$*, or any* k *with* $\mathsf{S}^k = \mathsf{K}$*.*

Proof. By [33, Proposition 2.3, items (1), (3) and (4)], (i) and (ii) are each equivalent to $\mathsf{S}^m = \mathsf{K}$ for some m. If $\mathsf{S}^m = \mathsf{K}$ then $e = e^m \in \mathsf{K}$ for each idempotent e; conversely, if $E(\mathsf{S}) \subseteq \mathsf{K}$ then $\mathsf{S}^k \subseteq \mathsf{S}^1 E(\mathsf{S}) \mathsf{S}^1 \subseteq \mathsf{K}$ for $k = |\mathsf{S}|$ by pumping, and $\mathsf{S}^k \supseteq \mathsf{K}$ since S^k is an ideal. So (i)–(iii) are equivalent.

(iii)⇒(iv): Lemma B.3 with $k = |\mathsf{S}|$ (or any k with $\mathsf{S}^k = \mathsf{K}$) determines $s(w)$ for $|w| \geq 2k$ from $\lambda_{k,\nu}(w)$, and $\widehat{s}_{k,k}$ records shorter words and ε exactly; so $\widehat{s} \preceq \lambda_{k,\nu}$.

(iv)⇒(i): S is a homomorphic image of $\mathsf{T} := \lambda_{k,\nu}(\Sigma^+)$ and $\mathbf{LG}$ is closed under homomorphic images, so it suffices that $\mathsf{T} \in \mathbf{LG}$. No short element is idempotent (squaring changes the exactly recorded word or makes it long). Let $e = (\langle p, q\rangle, \gamma_e)$ be idempotent. By Lemma B.1, every element of $e\mathsf{T}e$ has window $\langle p, q\rangle$, and long elements with this window multiply by $(\langle p, q\rangle, \gamma)(\langle p, q\rangle, \gamma') = (\langle p, q\rangle, \gamma\Delta\gamma')$, $\Delta := \Delta_\nu(q, p)$. This operation makes $\{\langle p, q\rangle\} \times \mathbb{G}$ a group isomorphic to $\mathbb{G}$ via $\gamma \mapsto \gamma\Delta$, and $e\mathsf{T}e$ is a finite nonempty subsemigroup of it, hence a subgroup. □

Lemma B.5 (Locally trivial–group typings admit window–group refinements)**.** *Let $\Sigma \neq \emptyset$. If $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$, there are $k \geq 1$, a finite group $\mathbb{G}$ and a morphism $g : \Sigma^* \to \mathbb{G}$ with $\widehat{s} \preceq (\widehat{s}_{k,k}, g)$; if $\mathsf{S} \in \mathbf{LI}$, $\mathbb{G}$ can be trivial.*

Proof. Choose a finite $\mathsf{S}' \in \mathbf{LI}$, a finite group $\mathbb{G}$, a subsemigroup $\mathsf{T} \leq \mathsf{S}' \times \mathbb{G}$ and a surjective morphism $\vartheta : \mathsf{T} \to \mathsf{S}$ (for $\mathsf{S} \in \mathbf{LI}$: $\mathsf{S}' = \mathsf{S}$, $\mathbb{G} = \{1\}$). Choosing for every letter a preimage in T of its s-image defines morphisms $s' : \Sigma^+ \to \mathsf{S}'$ and g with $\vartheta(s'(w), g(w)) = s(w)$; put $g(\varepsilon) = 1_{\mathbb{G}}$. The subsemigroup $\mathsf{S}_0 := s'(\Sigma^+) \leq \mathsf{S}'$ is locally trivial; let $k := |\mathsf{S}_0|$. For idempotents e, e' of S_0 and $y \in \mathsf{S}_0^1$ one has $eye' = ey(e'ee') = (e(ye')e)e' = ee'$. For $|w| \geq 2k$ write $w = pxz$ with $|p| = |z| = k$. By pumping, $s'(p) = y_1 e y_2$ and $s'(z) = y_3 e' y_4$ with $y_1, \ldots, y_4 \in \mathsf{S}_0^1$ and e, e' idempotent, so $s'(w) = y_1 e\,(y_2\, s'(x)\, y_3)\, e' y_4 = y_1 e e' y_4$ (with $s'(\varepsilon) := 1$), which depends only on $p = \mathrm{pref}_k(w)$ and $z = \mathrm{suf}_k(w)$. Shorter words are recorded exactly by $\widehat{s}_{k,k}$, which also separates ε. Hence equal $(\widehat{s}_{k,k}, g)$-values imply equal (s', g)-values and equal s-values. □

Proposition B.6 (Window–group typings are local-group typings, strictly)**.** (i) *If $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$ for a morphism g into a finite group, then $\widehat{s} \preceq \lambda_{\hat{k},\nu}$ with $\hat{k} = \max\{k, \ell, 1\}$ and $\nu(F) := g($last letter of $F)$. Moreover $\mathbf{LI} \vee \mathbf{Grp} \subseteq \mathbf{LG}$.*

(ii) *Already for $\Sigma = \{\mathtt{a}, \mathtt{b}\}$ and the typing $s_{\mathtt{ab}} := \lambda_{1,\nu}|_{\Sigma^+}$ with $\nu(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and $\nu = 0$ otherwise, $s_{\mathtt{ab}}(\Sigma^+) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{Grp})$.*

Proof. (i) For $|w| \geq 2\hat{k}$, $g(w) = g(\mathrm{pref}_{\hat{k}}\, w)\, \gamma_{\hat{k},\nu}(w)$, and $\mathrm{pref}_{\hat{k}}, \mathrm{suf}_{\hat{k}}$ determine $\widehat{s}_{k,\ell}(w)$; shorter words are recorded exactly. The inclusion holds because $\mathbf{LG}$ is a pseudovariety containing $\mathbf{LI}$ ($e\mathsf{T}e = \{e\}$) and $\mathbf{Grp}$. (ii) Membership in $\mathbf{LG}$ follows from Theorem B.4. Let $k, \ell \geq 0$ and let $g' : \Sigma^* \to \mathbb{G}'$ map into an arbitrary finite group. Choose $m \geq k + \ell + 1$ and $n \geq 1$ divisible by the order of $g'(\mathtt{a})$, and put $u = \mathtt{b}^m \mathtt{a}^n \mathtt{b}^m$ and $v = \mathtt{b}^{2m}$. They have equal length-k prefix and length-ℓ suffix, are long for $s_{k,\ell}$, and satisfy $g'(u) = g'(\mathtt{b})^{2m} = g'(v)$, but $\#_{\mathtt{ab}}(u) = 1$ and $\#_{\mathtt{ab}}(v) = 0$. Thus $s_{\mathtt{ab}}$ is not a window–group typing, and Lemma B.5 excludes $\mathbf{LI} \vee \mathbf{Grp}$. □

Lemma B.7 (Thinned and padded owners)**.** *Fix $k \geq 1$. Consider pairwise disjoint nonempty intervals in at most f words (*rows*), each interval belonging to one of m owners and each owner holding at most f intervals, and suppose that no interval meets the first or last k letters of its row.*

Then there is a set Q_1 of at least $m/(4kf+1)$ owners with the following property. For arbitrary nonempty replacement words of the intervals of the owners in Q_1, there are pairwise disjoint enlarged blocks*, at most f per owner, such that* (a) *each enlarged block contains intervals of exactly one owner in Q_1, and every interval of an owner in Q_1 lies in exactly one enlarged block;* (b) *replacing, inside an enlarged block, the intervals of its owner yields a word of length at least $2k+1$ with the same first and last k letters as the block, which itself has length at least $2k+1$; and* (c) *for every $J \subseteq Q_1$, replacing the intervals of the owners in J equals replacing their enlarged blocks as in* (b)*.*

Proof. Join two distinct owners if some interval of one and some interval of the other lie in the same row with fewer than $2k$ letters between them. A neighbouring interval to the right of an interval $[l, r)$ (in cut positions) starts at one of the $2k$ cuts $r, \dots, r+2k-1$, and disjointness and nonemptiness allow at most one interval per cut; the same holds on the left. So each owner has at most $4kf$ neighbours, and greedily choosing an owner and deleting its closed neighbourhood gives an independent set Q_1 of at least $m/(4kf+1)$ owners.

Enlarge every interval of an owner in Q_1 by k letters on each side; the boundary hypothesis keeps the enlargement inside its row. Intervals of distinct owners in Q_1 are at least $2k$ letters apart, so their enlargements have disjoint interiors; merge overlapping enlargements of the same owner. This gives (a) and does not increase the number of blocks per owner. Let $[l, r)$ be the union of the merged enlargements and l_0 the smallest left end of an interval in it, so $l = l_0 - k$. No interval of an owner in Q_1 meets $[l, l_0)$: one of the same owner would have been merged with a smaller left end, and one of another owner in Q_1 would end at least $2k$ letters before l_0. Symmetrically for the last k letters. Replacing only the owner's intervals therefore keeps the first and last k letters, and the replacement contains at least one nonempty word besides these $2k$ letters, which gives (b). Letters of owners outside Q_1 inside a block are never replaced, so (c) holds for every $J \subseteq Q_1$. □

Lemma B.8 (Short tuples with prescribed local-group type)**.** *Let G be reduced and good of fan-out $\le f$ and rank $\le r_{\max}$, and $\lambda = \lambda_{k,\nu}$ with group $\mathbb{G}$. Put $C^{\mathrm{buf}}_{k,\mathbb{G},f} := 2kf + (4kf+1)c_{\mathbb{G},f}$ and $\bar{n}_\lambda := |V||M_{k,\nu}|^f$. For every $B \in V$ and every realizable $\mathbf{t} \in M_{k,\nu}^{\mathrm{fan}(B)}$ there is $\vec{u} \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$ and*

$$\|\vec{u}\|_1 \le \beta_\lambda := (r_{\max}+1)\,\bar{\theta}_G\, C^{\mathrm{buf}}_{k,\mathbb{G},f}\, \bar{n}_\lambda.$$

Proof. We verify the hypothesis of Lemma 5.1(ii) with $c = (4kf+1)c_{\mathbb{G},f}$ and $c' = 2kf$, so $c + c' = C^{\mathrm{buf}}_{k,\mathbb{G},f} \ge 1$; with $n_\lambda(G) \le \bar{n}_\lambda$ it gives the bound. Let T be λ-minimal with heavy threshold $b = \bar{\theta}_G$. In each root component of length at least $2k$ mark the first and the last k letters, and in each shorter component mark every letter: at most $2kf$ marks. Let Q be an antichain of heavy nodes. By linearity and nondeletion their components occupy disjoint nonempty intervals of the root components, at most f per node, and at most $2kf$ nodes contain a mark; let Q_0 be the others, so their intervals avoid the first and last k letters of every root component and lie in components of length at least $2k$.

Let $|Q_0| > (4kf+1)c_{\mathbb{G},f}$. Lemma B.7 gives a set $Q_1 \subseteq Q_0$ of more than $c_{\mathbb{G},f}$ owners. For $v \in Q_1$ with nonterminal C_v fix a tuple of $L_{C_v}(G)$ of total length at most $\bar{\theta}_G$ (Lemma 4.1(ii)); its components are nonempty, and no window or group condition is imposed on it. By Lemma B.7(b) every enlarged block and its replacement are long and have the same first and last k letters, hence the same $\widehat{s}_{k,k}$-type. So Lemma B.2, applied with the enlarged blocks, writes $\gamma_{k,\nu}$ of every long root component as an ordered product of the block values and constants that do not depend on which owners are replaced. This is a block-replacement family of width f over $\mathbb{G}$ with more than $c_{\mathbb{G},f}$ owners, and Proposition 5.2 gives a nonempty $J \subseteq Q_1$ preserving every root $\gamma_{k,\nu}$-value.

Replace the subtrees at the nodes of J by derivations of the chosen tuples. This is a derivation from B because J is an antichain, and by Lemma B.7(c) its root components are those of the

enlarged-block replacement. Short root components contain no replaced letter and are unchanged; long ones keep their marked first and last k letters and stay long, so their $\widehat{s}_{k,k}$-types are unchanged, and their $\gamma_{k,\nu}$-values are preserved by the choice of J. Hence the root λ-vector is unchanged. □

Substitutability is not used in Lemma B.8.

C Proofs for the Separating Targets

This appendix proves Propositions 5.9 and 5.10. A language $L \subseteq \Sigma^*$ is *strictly 2-local* if there is a set P of pairs over $\Sigma \cup \{\vdash, \dashv\}$ such that $w \in L$ if and only if every pair of adjacent symbols of $\vdash w \dashv$ lies in P.

Lemma C.1 (Strictly 2-local languages are boundary typed)**.** *Every strictly 2-local L is $(f, s_{1,1})$-tuple-substitutable for every f; in particular $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{1,1}}$, and $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ whenever $\widehat{s}_{1,1} \preceq \widehat{s}$.*

Proof. Let $\vec{x}, \vec{y} \in (\Sigma^+)^d$ have equal $s_{1,1}$-types, so corresponding components are identical letters or share first and last letters, and let $E_0[\vec{x}], E_0[\vec{y}] \in L$. Suppose $E[\vec{x}] \in L$. Since the internal separators of E are nonempty, every adjacent pair of $\vdash E[\vec{y}] \dashv$ either lies inside some y_i, and then occurs in $E_0[\vec{y}] \in L$, or joins a first or last letter of some y_i to a symbol of $\vdash E \dashv$, or lies in $\vdash E \dashv$, and then occurs in $\vdash E[\vec{x}] \dashv$. So $E[\vec{y}] \in L$; symmetry and $\widehat{s}_{1,1} \preceq \widehat{s}$ finish the proof (regular languages are MCFLs). □

Proof of Propositions 5.9 and 5.10. Write $\diamond \in \{\mathrm{A}, \mathrm{B}\}$, let $\Sigma^\circ_\diamond$ be $\{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$ or $\{\mathtt{a}, \mathtt{b}\}$, and let $P_\diamond(w)$ be "$\#_{\mathtt{ab}}(w)$ even" or "$g_{\mathrm{B}}(w) = 1$".

(i) Let W be a right-linear (fan-out-one, rank-one, good) grammar for the regular language $\{\varsigma w : \varsigma \in \{\mathtt{1}, \mathtt{r}\}, w \in (\Sigma^\circ_\diamond)^+, \varsigma = \mathtt{r} \text{ or } P_\diamond(w)\}$; it tracks the last letter and the parity, respectively the value in S_3. Add

$$C \to (\xi^1_1\$\mathtt{1}, \mathtt{3})(W) \mid (\xi^1_1\mathtt{1}, \xi^2_1\mathtt{3})(C), \quad D \to (\mathtt{2}, \mathtt{4}) \mid (\xi^1_1\mathtt{2}, \xi^2_1\mathtt{4})(D), \quad S \to (\xi^1_1\xi^1_2\xi^2_1\xi^2_2)(C, D).$$

Then $L_C = \{(\varsigma w\$\mathtt{1}^m, \mathtt{3}^m)\}$, $L_D = \{(\mathtt{2}^n, \mathtt{4}^n)\}$, and S generates $L_\diamond$; the grammar is good (adjacent variables in the start template belong to different children). Intersecting with the regular language $\mathtt{ra}\$\mathtt{1}^+\mathtt{2}^+\mathtt{3}^+\mathtt{4}^+$ and erasing $\mathtt{ra}\$$ gives $C_\times$, which is not context-free; context-free languages are closed under both operations, so $L_\diamond$ is not context-free.

(ii) $L_\diamond = Q_\diamond \cap \Pi_\diamond \cap \varphi_\times^{-1}(0)$ with $Q_\diamond = \{\mathtt{1}, \mathtt{r}\}(\Sigma^\circ_\diamond)^+\$\mathtt{1}^+\mathtt{2}^+\mathtt{3}^+\mathtt{4}^+$ and $\Pi_\diamond$ the set of nonempty words that begin with $\mathtt{r}$, or begin with $\mathtt{1}$ and satisfy $P_\diamond$ for the whole word. On $Q_\diamond$ the condition $P_\diamond$ of the whole word is that of w: no factor $\mathtt{ab}$ meets ς, \$ or the digits, and g_{B} is trivial on all other letters. $Q_\diamond$ is strictly 2-local (its pairs are $\vdash \varsigma$, ςx, xy, $x\$$, $\$\mathtt{1}$, $\mathtt{11}, \mathtt{12}, \mathtt{22}, \mathtt{23}, \mathtt{33}, \mathtt{34}, \mathtt{44}$, $\mathtt{4} \dashv$ for $\varsigma \in \{\mathtt{1}, \mathtt{r}\}$, $x, y \in \Sigma^\circ_\diamond$) and $\widehat{s}_{1,1} \preceq \widehat{s}_\diamond$, so Lemma C.1 applies. $\Pi_\diamond$ is a union of $s_\diamond$-classes: $s_\diamond$ records the first letter, and for words of length at least two the parity of $\#_{\mathtt{ab}}$, respectively the value of g_{B} (the one-letter words $\mathtt{1}, \mathtt{r}$ lie in $\Pi_\diamond$); so item (c) after Definition 2.8 applies. Proposition 2.9 handles $\varphi_\times^{-1}(0)$ and the intersection; with (i), $L_\diamond \in \mathcal{C}^{\mathrm{mcf}}_{f,s_\diamond}$ for $f \geq 2$. s_{A} is local-group-determined, so its positive image lies in **LG** by Theorem B.4.

(iii) for A. Let $\hat{k} = \max\{k, \ell, 1\}$, $q \geq 2$ an even multiple of the exponent of the group of g', $u = \mathtt{a}^{\hat{k}}(\mathtt{abc})^q\mathtt{a}^{\hat{k}}$ and $v = \mathtt{a}^{\hat{k}}(\mathtt{bac})^q\mathtt{a}^{\hat{k}}$. Both are long for $\widehat{s}_{k,\ell}$ with prefix $\mathtt{a}^k$ and suffix $\mathtt{a}^\ell$, and $g'(u) = g'(\mathtt{a})^{2\hat{k}} = g'(v)$; so $s'(u) = s'(v)$. But $\#_{\mathtt{ab}}(u) = q$ is even and $\#_{\mathtt{ab}}(v) = 1$ (only at $\mathtt{a}^{\hat{k}}\mathtt{b}$). With $z = \$\mathtt{1234}$, Lemma 5.8 applies. A typing with positive image in $\mathbf{LI} \vee \mathbf{Grp}$ satisfies the hypothesis by Lemma B.5.

(iii) for B. Let $\hat{k} \geq \max\{k, \ell, 1\}$ be even, $u = \mathtt{a}^{\hat{k}}(\mathtt{ab})^3\mathtt{a}^{\hat{k}}$ and $v = \mathtt{a}^{\hat{k}}(\mathtt{ab})^2\mathtt{ba}\,\mathtt{a}^{\hat{k}}$. They have the same prefix $\mathtt{a}^k$, suffix $\mathtt{a}^\ell$ and Parikh vector, so $s'(u) = s'(v)$ for abelian g'. Since $\alpha^2 = \beta^2 = 1$, $\alpha\beta$ has order 3 and $\beta\alpha = (\alpha\beta)^{-1}$, we get $g_{\mathrm{B}}(u) = (\alpha\beta)^3 = 1$ and $g_{\mathrm{B}}(v) = (\alpha\beta)^2(\alpha\beta)^{-1} = \alpha\beta \neq 1$. With $z = \$\mathtt{1234}$, Lemma 5.8 applies. For $\mathbf{LI} \vee \mathbf{Ab}$, the proof of Lemma B.5 with an abelian group factor gives an abelian g'.

Consequences. If $s_{\mathrm{A}}(\Sigma_{\mathrm{A}}^+)$ were in $\mathbf{LI} \vee \mathbf{Grp}$, (iii) with $s' = s_{\mathrm{A}}$ would contradict (ii). Proposition 5.4 and its trivial-group and abelian cases assume a typing of the excluded kind. □

D Auxiliary Verification of Qualitative Reconstruction

Proof of Lemma 3.1. The identity holds because substitutions commute. By linearity and nondeletion each $\square_\kappa$ occurs exactly once in E_j, and by nonpermutation in the order $\kappa = 1, \ldots, d_j$, since E places $\alpha_1, \ldots, \alpha_{d_0}$ from left to right. Consider ξ_j^κ and $\xi_j^{\kappa+1}$. If both lie in one α_i, nonmerging puts a terminal or a variable $\xi_i^{\kappa'}$ with $i \neq j$ between them, which becomes a nonempty word. Otherwise they lie in $\alpha_i, \alpha_{i'}$ with $i < i'$, and the material between them contains the separator $v_i \in \Sigma^+$ of E. So every internal separator of E_j is nonempty. □

Proof of Lemma 3.2. Put $\vec{z}_j := \rho(\vec{u}_1, \ldots, \vec{u}_j, \vec{x}_{j+1}, \ldots, \vec{x}_r)$. For $E \in \mathcal{E}_{d_0}$, Lemma 3.1 gives an admissible E_j with $E[\vec{z}_{j-1}] = E_j[\vec{x}_j]$ and $E[\vec{z}_j] = E_j[\vec{u}_j]$; since $\vec{u}_j \equiv_{L,s} \vec{x}_j$, E accepts $\vec{z}_{j-1}$ iff it accepts $\vec{z}_j$. So $\mathcal{D}_L(\vec{z}_0) = \mathcal{D}_L(\vec{z}_r)$. The types agree because $\widehat{s}^{(d_0)}(\rho(\cdots)) = \mathrm{out}_\rho$ of the children's types. □

Proof of Proposition 3.10. Induction on the height of a derivation from $[\vec{x}]$. Const derives $\vec{x}$. Link $[\vec{x}] \to [\vec{y}]$ has same-type $\vec{x}, \vec{y}$ with a common context in $K \subseteq L$, so $\vec{x} \equiv_{L,s} \vec{y}$ by substitutability; combine with the hypothesis for $[\vec{y}]$. For Comp $[\vec{z}] \to \rho([\vec{u}_1], \ldots, [\vec{u}_r])$ with $\vec{z} = \rho(\vec{u}_1, \ldots, \vec{u}_r)$, the children derive tuples equivalent to the $\vec{u}_j$, and Lemma 3.2 transfers this to $\vec{z}$. For $\widehat{S} \to [(w)]$ with $w \in K \subseteq L$, a derived (v) satisfies $(v) \equiv_{L,s} (w)$, and $\square_1$ accepts (w), so $v \in L$; $\widehat{S} \to \varepsilon$ occurs only if $\varepsilon \in K \subseteq L$. The uncapped grammar is the case of cap $\|K\|_+$. □

Proof of Proposition 3.11. This is the argument of [15, Lemma 3] for the concrete grammar. Each $\iota(X)$ and each $\vec{z}_R := \rho(\iota(Y_1), \ldots, \iota(Y_q))$ for a typed rule $R : X \to \rho(Y_1, \ldots, Y_q)$ is observed, via $\chi(X)$. They have the same type (out_ρ of the children's types is the type of X) and the common context $\chi(X)$, so Link gives $[\iota(X)] \to [\vec{z}_R]$. If $q = 0$, Const gives $[\vec{z}_R] \to \vec{z}_R$; otherwise $(\vec{z}_R, \rho, (\iota(Y_j))_j)$ is a composition witness of rank $q \leq r_G$ (and $q \leq \|K\|_+$ by (3.1)), so Comp gives $[\vec{z}_R] \to \rho([\iota(Y_1)], \ldots, [\iota(Y_q)])$. By induction, $X \Rightarrow^*_{\widetilde{G}_0} \vec{u}$ implies $[\iota(X)] \Rightarrow^* \vec{u}$. For $w \in L(G) = L(\widetilde{G}_0)$ (Proposition 3.6(iii)) a successful derivation starts with a start link $\widetilde{S} \to X$; then $\chi(X) = \square_1$, so $\iota(X) = (w')$ with $w' \in K$, and Start gives $\widehat{S} \to [\iota(X)] \Rightarrow^* (w)$. □

Proof of Theorem 3.13. If $L \cap \Sigma^+ = \emptyset$, then $K = L \subseteq \{\varepsilon\}$ and Definition 3.4 gives the claim. Otherwise $\mathcal{F}_L \subseteq L$ is finite, Proposition 3.11 with $\iota = \omega$ (canonical tuples have the right type by Proposition 3.6(i)) gives $L \cap \Sigma^+ \subseteq L(\mathcal{B}(K))$, Proposition 3.10 gives $L(\mathcal{B}(K)) \subseteq L$, and $\varepsilon \in L(\mathcal{B}(K))$ iff $\varepsilon \in K$ iff $\varepsilon \in L$. Rank-preserving normalization gives the last sentence. □

Proof of Theorem 3.14 and Corollary 3.16. As in [14, Lemma 6]: an occurrence is a sample word with at most $2f$ cut positions, and a composition witness of rank $\leq r_{\max}$ adds at most $2fr_{\max}$ more, so there are $(1 + \|K\|_+)^{O(fr_{\max})}$ candidate rules, each checked in polynomial time. Identification is the conservative-learner argument of [14, Corollary 1 and Theorem 1]: once the characteristic sample of Theorem 3.13 has appeared, every new hypothesis is correct, and an earlier hypothesis that is

never changed is sound by Proposition 3.10 and complete because every later datum is generated by it. Membership is decidable in polynomial time for fixed f and $r_{\max}$ [14, Proposition 4]. None of this bounds the stage of convergence. □

E Proofs for Typed Witnesses and Characteristic Data

Proof of Proposition 3.6. At every rule application the morphism property gives

$$\widehat{s}^{(\mathrm{fan}(A))}(\rho(\vec{u}_1,\dots,\vec{u}_r)) = \mathrm{out}_\rho(\widehat{s}^{(d_1)}(\vec{u}_1),\dots,\widehat{s}^{(d_r)}(\vec{u}_r)).$$

Induction on height gives (i) and the existence and uniqueness of the lift in (ii), since each typed label is forced by its yield. Erasing labels gives $L(G^{\widehat{s}}) \subseteq L(G)$, (ii) gives the converse, and trimming keeps every rule of every successful derivation. For (iv), insert the terminal derivation at an occurrence of X in a successful derivation of $\widetilde{G}_0$: typed rules depend only on typed symbols, so the result is a successful derivation of $G^{\widehat{s}}$, and trimming retains all its rules. (A rule of $G^{\widehat{s}}$ whose child has no terminal derivation, such as $A_{(0)} \to \mathtt{a}B_{(0)}$ when B derives only words of another type, is not retained; it occurs in no terminal derivation.) □

Definitions 2.8, 3.4 and 3.5 apply verbatim with any finite-monoid morphism $q : \Sigma^* \to \mathsf{N}$ in place of $\widehat{s}$ (tuple types being $q^{(d)}$); write $\mathcal{C}^{\mathrm{mcf}}_{f,q}$ for the class. The class and the operator use types of nonempty components only, so they depend only on $q|_{\Sigma^+}$.

Lemma E.1 (Monoid implementations)**.** *Let $q : \Sigma^* \to \mathsf{N}$ be a finite-monoid morphism, $\mathsf{S}_0 := q(\Sigma^+)$ and $s_0 := q|_{\Sigma^+}$.*

(i) *On $(\Sigma^+)^d$, equality of $q^{(d)}$-, $s_0^{(d)}$- and $\widehat{s_0}^{(d)}$-values coincide. Hence $\mathcal{C}^{\mathrm{mcf}}_{f,q} = \mathcal{C}^{\mathrm{mcf}}_{f,s_0}$, and the operators built with q and with $\widehat{s_0}$ return the same grammar on every sample.*

(ii) *For every good MCFG G, the trimmed refinements of G under q and under $\widehat{s_0}$ coincide (the untrimmed ones need not); hence the same canonical tuples, contexts and witness words may be chosen.*

Proof. (i) The three maps agree on Σ^+. (ii) By ε-freeness and Proposition 3.6(i), every typed nonterminal of a successful derivation under either morphism is $A_{\mathbf{t}}$ with $\mathbf{t} \in \mathsf{S}_0^{\mathrm{fan}(A)}$, and every start link used is $\widetilde{S} \to S_{(m)}$ with $m \in \mathsf{S}_0$. For children indexed in S_0, each nonempty template component is evaluated in S_0 identically by both morphisms, so both refinements have the same rules among S_0-indexed symbols, which carry all successful derivations. □

Proof of Lemma 3.7. Take a successful typed derivation containing X and follow the path from the root to an occurrence of X, starting with $\Box_1$; at each rule apply Lemma 3.1 with the (nonempty) sibling yields fixed. This gives $\chi(X) \in \mathcal{E}_d$, and replacing the X-subtree by any X-derivation gives a successful derivation of $\chi(X)[\vec{u}]$. If $\widetilde{S} \to X$ is a rule, take the path of length zero. □

Proof of Lemma 3.8. For $\vec{u}, \vec{v} \in L_X$, Proposition 3.6(i) gives equal types, $\chi(X)$ accepts both, and the arity is at most f; substitutability gives $\mathcal{D}_L(\vec{u}) = \mathcal{D}_L(\vec{v})$. □

Proof of Lemma 4.1. For a rule R with left-hand side A fix a successful derivation D_R of length $\theta_G(R)$ that uses R; it exists because G is reduced. By nondeletion every terminal written by R, and every component of the tuple produced at that occurrence of A, survives exactly once in the sentence. This gives (i), and (ii) because every productive A has a rule.

(iv) For $A = S$ take $\Box_1$. Otherwise delete the A-subtree of D_R and put ordered holes in place of its components; iterating Lemma 3.1 along the path to the root (siblings have nonempty yields) shows $E_A \in \mathcal{E}_d$, and its fixed letters belong to the sentence of D_R. Inserting any A-derivation gives a successful derivation, so E_A accepts $L_A(G) \supseteq L_X$ (Proposition 3.6(i)). A typed nonterminal with a start link has the form $S_{\mathbf{t}}$, and $E_S = \Box_1$. For derivationality, insert the erasure of any $\widetilde{G}_0$-derivation D_X from X into the hole; the unique lift of the resulting successful derivation (Proposition 3.6(ii)) labels every node by its nonterminal and the $\widehat{s}$-vector of its yield, as D_X does, so it contains D_X at the hole, and deleting that X-subtree leaves E_A.

(iii) Follow a shortest dependency path from S to A; it repeats no nonterminal, so its rules are distinct and differ from R. Complete every sibling on the path and every child of R by a shortest terminal tuple (length $\leq t_G$) and apply R. The distinct rules used write at most $\|G\|$ terminals and have at most $\|G\|$ child occurrences, so the sentence has length at most $\|G\| + \|G\| t_G \leq 2\|G\| \max\{1, t_G\}$. $\square$

Proof of Lemma 4.2. (i) All components are nonempty because G is ε-free, so $s \preceq \lambda$ and Proposition 3.6(i) give $\widehat{s}^{(\mathrm{fan}(B))}(\vec{u}') = \widehat{s}^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$. By Proposition 3.6(ii) a G-derivation of $\vec{u}'$ lifts to a $G^{\widehat{s}}$-derivation from X, so $\vec{u}' \in L_X$ by Proposition 3.6(iv). For the second claim apply $(\mathrm{R}^{\lambda}_{\beta})$ to some $\vec{u} \in L_X \neq \emptyset$.

(ii) A word of W_ι is $\chi(X)[\iota(X)]$ or a rule word. The context has at most $\bar{\theta}_G$ fixed letters, a typed rule has the template of its source and so writes at most $\bar{\theta}_G$ letters (Lemma 4.1(i)), and there are at most $r_{\max}$ child tuples. There is one word per typed nonterminal and per typed rule.

(iii) Apply the typed rule to $\widetilde{G}_0$-derivations of the $\omega(Y_j) \in L_{Y_j}$, erase the labels, and insert the result into the successful G-derivation from which E_B was obtained. As in the proof of Lemma 4.1(iv), the unique lift contains the typed derivation at the hole, and every rule of a successful derivation of $G^{\widehat{s}}$ is retained by the definition of $\widetilde{G}_0$. A start link $\widetilde{S} \to S_{\mathbf{t}}$ is used by the successful derivation of $\omega(S_{\mathbf{t}})$, of length at most b.

(iv) Since $\chi(X)$ accepts every tuple of L_X and $\iota(X)$ is equivalent to them, $\chi(X)[\iota(X)] \in L$. For a typed rule $X \to \rho(Y_1, \ldots, Y_q)$ and $\vec{u}_j \in L_{Y_j}$, Lemma 3.2 gives $\rho(\iota(Y_1), \ldots, \iota(Y_q)) \equiv_{L,s} \rho(\vec{u}_1, \ldots, \vec{u}_q) \in L_X$, so the rule word lies in L as well. Equivalent tuples have equal types, so $\widehat{s}^{(\mathrm{fan}(B))}(\iota(X)) = \mathbf{t}$ for $X = B_{\mathbf{t}}$. Hence Proposition 3.11 ($r_G \leq r_{\max}$, and $\chi(S_{\mathbf{t}}) = E_S = \Box_1$) and Proposition 3.10 give $L(\mathcal{B}(K)) \cap \Sigma^+ = L \cap \Sigma^+$ for every finite K with $\mathsf{W}_\iota \subseteq K \subseteq L$, for both operators; the Start rule handles ε. $\square$

Proof of Theorem 4.3. Apply Lemma 4.2(ii),(iv) with $\iota = \omega$ and $b = t_{\mathrm{typ}}$, counting the non-start typed nonterminals and rules instead of using the upper bound $C_s(G)$. $\square$

Proof of Proposition 4.6. Apply Lemma 4.5 to $L' = L^{\setminus m}$, which lies in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ by Lemma 4.4, and, if $\varepsilon \in L$, to $L' = L \setminus \{\varepsilon\}$; the canonical operators have characteristic samples for both by Theorem 3.13. Distinct positive types demand distinct sample words, and a word of type m costs at least $\mu_m(L) + 1$. $\square$

Proof of Proposition 4.7. All templates are nonempty and use each variable once in order, so G_n is good; it is reduced and generates Σ^+ via $S \to SS \mid \mathtt{a} \mid \mathtt{c}$. Each rule has a successful use of length at most 2: complete every A_i by $\mathtt{c}$ ($i \geq 1$) or $\mathtt{a}$ ($i = 0$). A yield of A_n has type 1 iff it contains no $\mathtt{c}$, so the unique yield of $(A_n)_{(1)}$ is $\mathtt{a}^{2^n}$, and $t_{\mathrm{typ}}(G_n, s_{\mathtt{c}}) \geq 2^n$; the typed rule $S_{(1)} \to (A_n)_{(1)}$ is retained (via $S \to A_n$), and every successful derivation using it has length at least 2^n. $\square$

Proof of Corollary 4.8. $L_X \subseteq \mathrm{Cl}_{L,s}(X)$ gives $\tau \leq t_{\mathrm{typ}}$. Apply Lemma 4.2(ii),(iv) with $\iota(X)$ a shortest tuple of $\mathrm{Cl}_{L,s}(X)$ and $b = \tau$, counting as in Theorem 4.3; $K := \mathsf{W}_\iota \cup (\{\varepsilon\} \cap L)$. Finally, for

$m \in \mathsf{S}_L$ the typed start nonterminal $X = S_{(m)}$ has $\chi(X) = \square_1$, so $\square_1 \in \mathcal{D}_L(\iota(X))$; thus $\iota(X) = (v)$ with $v \in L$ and $s(v) = m$, and $\tau_X \geq \mu_m(L)$. □

F Details of the Good Normalization

The four conversions below give Proposition 3.12; each satisfies a tuple-language identity from which the size bound is read off. Correctness of the nondeleting and nonmerging conversions is standard [14,28,30]; the size bounds are not stated there and are derived here. For $\vec{w} = (w_1, \ldots, w_d)$ and nonempty $I \subseteq [d]$, $\pi_I(\vec{w})$ keeps the components indexed by I in order, $\sigma\vec{w}$ permutes components and $\psi(\vec{w})$ concatenates consecutive blocks of components; so $\|\pi_I(\vec{w})\|_1 \leq \|\vec{w}\|_1 = \|\sigma\vec{w}\|_1 = \|\psi(\vec{w})\|_1$.

Lemma F.1 (Stagewise tuple languages)**.** *Let G_0 be a reduced f-MCFG of rank at most $r_{\max}$. Each conversion below preserves linearity, does not increase fan-out, rank or any template, and multiplies the numbers of nonterminals and rules by a factor depending only on f and $r_{\max}$.*

(a) Nondeletion *(construction (f3) in the proof of [28, Lemma 2.2]). Introduce A^I for nonempty $I \subseteq [\mathrm{fan}(A)]$. For $R : A \to \rho(B_1, \ldots, B_q)$ let J_j be the components of B_j used by the I-indexed outputs of ρ; keep these outputs, use $B_j^{J_j}$ if $J_j \neq \emptyset$ and omit B_j otherwise. The rules are nondeleting and $L_{A^I} = \pi_I(L_A(G_0))$.*

(b) Nonpermutation. *Decorate a fan-out-d nonterminal by $\sigma \in \mathfrak{S}_d$ (the symmetric group on $\{1, \ldots, d\}$), the child's permutation being determined by the order in which the parent template uses its variables. The decorated nonterminal generates σL_A.*

(c) Empty components. *Let $\mathsf{Pat}(A)$ be the set of support patterns of L_A, computed by saturation. For realizable child patterns, replace variables of empty components by ε, delete empty parent components, omit all-empty children, and introduce A^I for each nonempty $I \in \mathsf{Pat}(A)$. Then $L_{A^I} = \pi_I\{\vec{u} \in L_A : \mathrm{supp}(\vec{u}) = I\}$, the result is ε-free, and $S^{\{1\}}$ generates the nonempty part of the language.*

(d) Nonmerging *(construction in the proof of [14, Section 2.3, Lemma 1]). Decorate a nonterminal by a composition ψ of its fan-out, the child's composition being given by the maximal runs of its consecutive variables in the parent template. The decorated nonterminal generates the image $\psi(L_A)$.*

Stages (b)–(d) preserve the properties established by earlier stages.

Proof. (a) For any child tuples, $\pi_I(\rho(\vec{v}_1, \ldots, \vec{v}_q)) = \rho^I((\pi_{J_j}\vec{v}_j)_{J_j \neq \emptyset})$; induction gives $\subseteq$, and for $\supseteq$ the omitted children are completed by terminal derivations, which exist because G_0 is reduced. There are at most 2^f copies of each nonterminal and rule. (b) Linearity and nondeletion make the child permutations well defined; induction gives σL_A. There are at most $f!$ copies. (c) Retained variables occur once and in order; induction gives the identity, and omitted all-empty children are completed by derivations of their empty pattern. A retained child B^J has only the components indexed by J, and each of them is used by the parent template exactly once and in the original order, so nondeletion and nonpermutation persist. There are at most 2^f patterns per nonterminal and $2^{f r_{\max}}$ choices per rule. (d) Nonpermutation makes the runs compatible with each child's order and nondeletion makes them cover its components; then $\rho^\psi(\psi_1(\vec{v}_1), \ldots, \psi_q(\vec{v}_q)) = \psi(\rho(\vec{v}_1, \ldots, \vec{v}_q))$, and induction gives the image identity (merging need not be injective, so no inverse-image claim is made). Concatenating nonempty components yields nonempty components, and each merged block of a child is used once and in order, so ε-freeness, nondeletion and nonpermutation persist. The size is linear for fixed $f, r_{\max}$ [14, Section 2.3, after Example 2]. □

Proof of Proposition 3.12. If $L(G_{\text{in}}) \cap \Sigma^+ = \emptyset$, take G_{good} to consist of the start symbol alone, with no rules; it is good and generates $\emptyset = L(G_{\text{in}}) \setminus \{\varepsilon\}$. Otherwise apply (a)–(d) to the trimmed input and trim again. Each stage multiplies the size by a factor depending only on $f, r_{\max}$, so $\|G_{\text{good}}\| = O_{f,r_{\max}}(\|G_{\text{in}}\|)$, and $L(G_{\text{good}}) = L(G_{\text{in}}) \setminus \{\varepsilon\}$ by (c). If this language is nonempty, the final trimming leaves a reduced grammar, and trimming preserves goodness. □

G The Three-Letter Doubling Family

This appendix proves Proposition 7.1; the proof of (ii) adapts the deterministic-parser argument of [23, Appendix C]. Fix $n \geq 1$, and for $x, y \in \Sigma^*$ write $x \equiv y$ if $\{(u,v) : uxv \in L_n\} = \{(u,v) : uyv \in L_n\}$; this is a monoid congruence.

The parser. A *stack* is a word over $E_0, \dots, E_n$ and the closing markers $\hat{\mathtt{b}}_i, \hat{\mathtt{c}}_i$ $(1 \leq i \leq n)$, with its top on the left; ε is the empty stack. For every stack $\mathcal{Z}$ and $1 \leq i \leq n$ the moves are

$$E_0\mathcal{Z} \xrightarrow{\mathtt{a}} \mathcal{Z}, \quad E_i\mathcal{Z} \xrightarrow{\mathtt{b}} E_{i-1}E_{i-1}\hat{\mathtt{b}}_i\mathcal{Z}, \quad E_i\mathcal{Z} \xrightarrow{\mathtt{c}} E_{i-1}\hat{\mathtt{c}}_i\mathcal{Z},$$
$$\hat{\mathtt{b}}_i\mathcal{Z} \xrightarrow{\mathtt{b}} \mathcal{Z}, \quad \hat{\mathtt{c}}_i\mathcal{Z} \xrightarrow{\mathtt{c}} \mathcal{Z},$$

and there are no others; in particular no move leaves ε. Write $\mathcal{X} \xrightarrow{w} \mathcal{Y}$ if reading w from $\mathcal{X}$ ends in $\mathcal{Y}$. Three properties are used.

(D1) *Determinism.* The top symbol and the letter determine the move, so every word acts as a partial map on stacks.

(D2) *Weight.* Give E_i weight 4^i and each marker weight 1, and let $\mathrm{wt}(\mathcal{X})$ be the total weight. Every move lowers wt, since $4^i > 2 \cdot 4^{i-1} + 1$; hence $\mathcal{X} \xrightarrow{w} \mathcal{X}$ only for $w = \varepsilon$.

(D3) *Locality.* Moves read and rewrite only the top symbol, and markers are only popped. Hence $\mathcal{X} \xrightarrow{w} \mathcal{Y}$ implies $\mathcal{X}\mathcal{Z} \xrightarrow{w} \mathcal{Y}\mathcal{Z}$. Conversely, if $\mathcal{X}_1 \neq \varepsilon$ and $\mathcal{X}_1\mathcal{Z} \xrightarrow{w} \mathcal{Y}$, then either (a) the run reaches $\mathcal{Z}$: $w = w_1w_2$ with $\mathcal{X}_1 \xrightarrow{w_1} \varepsilon$ and $\mathcal{Z} \xrightarrow{w_2} \mathcal{Y}$, so $\mathrm{wt}(\mathcal{Y}) \leq \mathrm{wt}(\mathcal{Z})$; or (b) it does not: $\mathcal{Y} = \mathcal{Y}_1\mathcal{Z}$ with $\mathcal{X}_1 \xrightarrow{w} \mathcal{Y}_1 \neq \varepsilon$, so $\mathrm{wt}(\mathcal{Y}) > \mathrm{wt}(\mathcal{Z})$, and $\mathcal{Y}_1$ ends with the last symbol of $\mathcal{X}_1$ whenever that symbol is a marker. For fixed $\mathcal{Z}$, comparing $\mathrm{wt}(\mathcal{Y})$ with $\mathrm{wt}(\mathcal{Z})$ decides the case.

A letter $\mathtt{b}$ or $\mathtt{c}$ both opens and closes, so a word need not act injectively on all stacks; injectivity holds on the *reachable* stacks, those $\mathcal{X}$ with $E_n \xrightarrow{u} \mathcal{X}$ for some u.

Lemma G.1. (i) *For $0 \leq i \leq n$ and every stack $\mathcal{Z}$, $E_i\mathcal{Z} \xrightarrow{w} \mathcal{Z}$ if and only if $w \in L^{(i)}$.*

(ii) *Put $\mathcal{F}_i := \{E_{i-1}\hat{\mathtt{b}}_i, \hat{\mathtt{b}}_i, \hat{\mathtt{c}}_i\}$. Every reachable stack is ε or $\mathcal{Y}F_{j+1}\cdots F_n$ with $F_i \in \mathcal{F}_i$ and $\mathcal{Y} \in \{E_j, \hat{\mathtt{b}}_j, \hat{\mathtt{c}}_j\}$ ($\mathcal{Y} = E_j$ with $0 \leq j \leq n$, or a marker with $1 \leq j \leq n$). Every word acts as a partial injection on reachable stacks.*

Proof. (i) A run $E_i\mathcal{Z} \xrightarrow{w} \mathcal{Z}$ is in case (a) of (D3), since case (b) ends in $\mathcal{Y}_1\mathcal{Z} \neq \mathcal{Z}$; so $w = w_1w_2$ with $E_i \xrightarrow{w_1} \varepsilon$ and $\mathcal{Z} \xrightarrow{w_2} \mathcal{Z}$, and $w_2 = \varepsilon$ by (D2). With the lifting in (D3), it suffices to show $E_i \xrightarrow{w} \varepsilon$ iff $w \in L^{(i)}$, by induction on i. From E_0 only the $\mathtt{a}$-move applies, and ε admits none. For $i \geq 1$ the first letter is $\mathtt{b}$ or $\mathtt{c}$. After $\mathtt{c}$, the stack $E_{i-1}\hat{\mathtt{c}}_i$ must be emptied; case (a) of (D3) with $\mathcal{X}_1 = E_{i-1}$, $\mathcal{Z} = \hat{\mathtt{c}}_i$ splits the rest as zz' with $E_{i-1} \xrightarrow{z} \varepsilon$ and $z' = \mathtt{c}$. After $\mathtt{b}$, two such splittings of $E_{i-1}E_{i-1}\hat{\mathtt{b}}_i$ give $z_1z_2\mathtt{b}$ with $E_{i-1} \xrightarrow{z_j} \varepsilon$.

(ii) The set of such stacks contains E_n and is closed under the moves: from $E_jF_{j+1}\cdots F_n$, $\mathtt{b}$ gives $E_{j-1}(E_{j-1}\hat{\mathtt{b}}_j)F_{j+1}\cdots F_n$ and $\mathtt{c}$ gives $E_{j-1}\hat{\mathtt{c}}_jF_{j+1}\cdots F_n$; popping the top leaves $F_{j+1}\cdots F_n$,

which is ε or again of the stated form, a first frame $E_j\hat{\mathtt{b}}_{j+1}$ being read as $\mathcal{Y} = E_j$ followed by the frame $\hat{\mathtt{b}}_{j+1}$.

For injectivity it suffices to consider letters, since moves map reachable stacks to reachable stacks. A *frame product* is a stack $F_{i+1}\cdots F_n$ with $F_k \in \mathcal{F}_k$ (the empty stack if $i = n$). Each frame F_k contains exactly one marker, of index k, so a frame product determines i: it is n for ε and otherwise one less than the least marker index. In a frame product a symbol E is always followed by a marker $\hat{\mathtt{b}}$. The moves of a letter on reachable stacks, and the shape of their results, are:

Letter	Move (reachable source)	Result begins with	Source recovered from
$\mathtt{a}$	$E_0\mathcal{Z} \to \mathcal{Z}$	a frame product	$\mathcal{Z}$
$\mathtt{b}$, opening	$E_i\mathcal{Z} \to E_{i-1}E_{i-1}\hat{\mathtt{b}}_i\mathcal{Z}$	two symbols E	i and $\mathcal{Z}$ read off
$\mathtt{b}$, closing	$\hat{\mathtt{b}}_i\mathcal{Z} \to \mathcal{Z}$, $\mathcal{Z}$ a frame product	a marker, $E\hat{\mathtt{b}}$, or ε	i from $\mathcal{Z}$
$\mathtt{c}$, opening	$E_i\mathcal{Z} \to E_{i-1}\hat{\mathtt{c}}_i\mathcal{Z}$	E followed by $\hat{\mathtt{c}}$	i and $\mathcal{Z}$ read off
$\mathtt{c}$, closing	$\hat{\mathtt{c}}_i\mathcal{Z} \to \mathcal{Z}$, $\mathcal{Z}$ a frame product	a marker, $E\hat{\mathtt{b}}$, or ε	i from $\mathcal{Z}$

In the closing rows $\mathcal{Z}$ is a frame product because a reachable stack with a marker $\hat{\mathtt{b}}_i$ or $\hat{\mathtt{c}}_i$ on top is $\mathcal{Y}F_{i+1}\cdots F_n$ with $\mathcal{Y}$ that marker. For $\mathtt{b}$, an opening result begins with two symbols E and a closing result never does, since in a frame product a symbol E is followed by a marker; for $\mathtt{c}$, an opening result begins with $E\hat{\mathtt{c}}$, which never occurs in a frame product. So the openings and the closings of one letter have disjoint images, each row is injective by its last column, and every letter, hence every word, acts as a partial injection on reachable stacks. □

Lemma G.2. *If $\mathcal{X}$ is reachable, $w \in L^{(i)}$ and $\mathcal{X} \xrightarrow{w} \mathcal{Y}$, then $\mathcal{X} = E_i\mathcal{Y}$. Consequently $w \equiv w'$ for all $w, w' \in L^{(i)}$.*

Proof. Induct on i; for $i = 0$, $\mathtt{a}$ only pops E_0. Let $i \geq 1$; all stacks met are reachable. If $w = \mathtt{c}v\mathtt{c}$ with $v \in L^{(i-1)}$ and the first $\mathtt{c}$ opens E_j, the induction hypothesis for v at $E_{j-1}\hat{\mathtt{c}}_j\mathcal{Z}$ gives $j = i$, and the last $\mathtt{c}$ pops $\hat{\mathtt{c}}_i$. If the first $\mathtt{c}$ pops a marker, the rest is ε or a product of frames; to read v it must begin with E_{i-1} by the induction hypothesis, so its first frame is $E_{i-1}\hat{\mathtt{b}}_i$, and after v the top is $\hat{\mathtt{b}}_i$, on which $\mathtt{c}$ has no move. If $w = \mathtt{b}v_1v_2\mathtt{b}$ and the first $\mathtt{b}$ opens E_j, the induction hypothesis gives $j = i$, the words v_1, v_2 pop the two copies of E_{i-1}, and the last $\mathtt{b}$ pops $\hat{\mathtt{b}}_i$. If the first $\mathtt{b}$ pops a marker, then as before v_1 leaves the top $\hat{\mathtt{b}}_i$, from which v_2 cannot be read by the induction hypothesis. For the consequence, if $uwv \in L_n$, the run $E_n \xrightarrow{u} \mathcal{X} \xrightarrow{w} \mathcal{Y} \xrightarrow{v} \varepsilon$ has $\mathcal{X} = E_i\mathcal{Y}$, and $E_i\mathcal{Y} \xrightarrow{w'} \mathcal{Y}$ by Lemma G.1(i); so $uw'v \in L_n$, and conversely. □

Lemma G.3 (Branch synchronization)**.** *Let $\mathcal{X} \xrightarrow{x} \mathcal{Y}$ and $\mathcal{X} \xrightarrow{y} \mathcal{Y}$, where x, y are nonempty and begin with different letters. Then $\mathcal{X} = E_i\mathcal{Z}$ for some $1 \leq i \leq n$ and stack $\mathcal{Z}$, and $x = wx'$, $y = w'y'$ with $w, w' \in L^{(i)}$, $\mathcal{Z} \xrightarrow{x'} \mathcal{Y}$ and $\mathcal{Z} \xrightarrow{y'} \mathcal{Y}$.*

Proof. Only a top E_i with $i \geq 1$ admits two different letters, namely $\mathtt{b}$ and $\mathtt{c}$. Say $x = \mathtt{c}x_0$ and $y = \mathtt{b}y_0$, so $E_{i-1}\hat{\mathtt{c}}_i\mathcal{Z} \xrightarrow{x_0} \mathcal{Y}$ and $E_{i-1}E_{i-1}\hat{\mathtt{b}}_i\mathcal{Z} \xrightarrow{y_0} \mathcal{Y}$. By (D3) both runs are in the same case. Case (b) is impossible: it would give $\mathcal{Y} = \mathcal{Y}_1\mathcal{Z} = \mathcal{Y}_2\mathcal{Z}$ with $\mathcal{Y}_1$ ending in $\hat{\mathtt{c}}_i$ and $\mathcal{Y}_2$ in $\hat{\mathtt{b}}_i$. In case (a), $x_0 = x_1x'$ with $E_{i-1}\hat{\mathtt{c}}_i \xrightarrow{x_1} \varepsilon$ and $\mathcal{Z} \xrightarrow{x'} \mathcal{Y}$, so $w := \mathtt{c}x_1$ satisfies $E_i \xrightarrow{w} \varepsilon$ and $w \in L^{(i)}$ by Lemma G.1(i). The same argument gives $y = w'y'$. □

Proposition G.4. *If $\mathcal{X} \xrightarrow{x} \mathcal{Y}$ and $\mathcal{X} \xrightarrow{y} \mathcal{Y}$, then $x \equiv y$. Consequently L_n is Clark–Eyraud substitutable.*

Proof. Induct on $|x| + |y|$. If x or y is empty, then $\mathcal{X} = \mathcal{Y}$, and the other is empty by (D2). If $x = cx'$ and $y = cy'$ for a letter c, (D1) gives $\mathcal{X} \xrightarrow{c} \mathcal{X}'$ with $\mathcal{X}' \xrightarrow{x'} \mathcal{Y}$ and $\mathcal{X}' \xrightarrow{y'} \mathcal{Y}$, and induction

applies. Otherwise Lemma G.3 gives $x = wx'$, $y = w'y'$; induction gives $x' \equiv y'$, Lemma G.2 gives $w \equiv w'$, and $\equiv$ is a congruence.

For the consequence, let $x, y \in \Sigma^+$ with $uxv, uyv \in L_n$. By (D1) and Lemma G.1(i) there are runs $E_n \xrightarrow{u} \mathcal{X} \xrightarrow{x} \mathcal{Y} \xrightarrow{v} \varepsilon$ and $E_n \xrightarrow{u} \mathcal{X} \xrightarrow{y} \mathcal{Y}' \xrightarrow{v} \varepsilon$; $\mathcal{Y}$ and $\mathcal{Y}'$ are reachable, so Lemma G.1(ii) gives $\mathcal{Y} = \mathcal{Y}'$, and $x \equiv y$. Letters outside $\{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$ only have empty distributions. □

Proof of Proposition 7.1. (i) Every template is nonempty and uses each variable once; at fan-out one, nonpermutation and nonmerging are automatic. The grammar is reduced since $E_i \Rightarrow^* \mathtt{c}^i\mathtt{a}\mathtt{c}^i$. It has $n+2$ nonterminals and rules of sizes 3 (start), 2 (base), 7 and 5 (the two rules of each level) (Definition 2.5), whence $\|G_n\| = 13n+7$. A shortest yield of E_i has length $2i+1$, and the levels above E_i add at least $2(n-i)$ letters, attained by the rules $E_j \to \mathtt{c}E_{j-1}\mathtt{c}$. So the start, base and unary rules have rule thickness $2n+1$, and the binary rule of level i has $2(n-i)+2+2(2i-1) = 2n+2i$; hence $\bar{\theta}_{G_n} = 4n$.

(ii) The first claim is Proposition G.4. It is the condition of Definition 2.8 at $f = 1$ for the trivial typing, and the finite language L_n is a 1-MCFL; item (a) after Definition 2.8 gives $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$.

(iii) A derivation yields a word without $\mathtt{c}$ iff it uses no rule $E_i \to \mathtt{c}E_{i-1}\mathtt{c}$. Such a derivation from E_i is the complete binary tree of height i and yields W_i, and $|W_i| = 2|W_{i-1}| + 2$ with $|W_0| = 1$.

(iv) By (ii), $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$, and by (iii), $1 \in \mathsf{S}_{L_n}$ with $\mu_1(L_n) = |W_n|$. Proposition 4.6 with $m = 1$ forces a word of type 1, which can only be W_n. □

H Isolated Letters at Fan-Out One

This appendix proves Theorem 7.10 and Corollary 7.11. Throughout, $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ is a typing, $\mathtt{a} \in \Sigma$ is isolated for s (Definition 7.9), and $L \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$; we may assume $\varepsilon \notin L$. A word of Σ^+ is *pure* if it lies in $\mathtt{a}^+$, $\mathtt{a}$-*free* if it does not contain $\mathtt{a}$, and $\mathtt{a}$-*mixed* otherwise, that is, if it contains $\mathtt{a}$ and another letter. Isolation is used only through two facts:

(I1) all pure words have the type $s(\mathtt{a})$, and the words of this type are the pure words;

(I2) all $\mathtt{a}$-mixed words have one type.

For $x \in \Sigma^+$ write $\mathcal{D}(x) := \mathcal{D}^{(1)}_L(x)$ and $x \sim y$ if $\mathcal{D}(x) = \mathcal{D}(y)$; this is a congruence on Σ^+, and words with empty distribution are $\sim$-equivalent. Thus $x \equiv_{L,s} y$ if and only if $x \sim y$ and $s(x) = s(y)$.

H.1 Preliminaries

Lemma H.1 (Cancellation)**.** *Let $x, y \in \Sigma^+$ have the same type, $z \in \Sigma^+$, $\mathcal{D}(zx) \neq \emptyset$ and $zx \sim zy$. Then $x \sim y$. The same holds on the right.*

Proof. For $(l, r) \in \mathcal{D}(zx)$ we get $lzyr \in L$, so x and y share the context (lz, r). □

We apply the lemma only to pure x, y or to $\mathtt{a}$-mixed x, y, which have the same type by (I1), respectively (I2). If a nonterminal X has a derivational context (l, r), two yields of X of the same type share (l, r) and are $\sim$-equivalent (Lemma 3.8). For a type t realized by X we write $\tau(X, t)$ for the representative length of the typed nonterminal X_t, the least length of a word of type t that is $\sim$-equivalent to the yields of X of type t.

Binarization. Let G be a reduced good presentation of L of fan-out one, and let σ_G be the number of letters occurring in its rules, so $\sigma_G \le \min\{|\Sigma|, \|G\|\}$. Replace every rule $X \to Y_1 \cdots Y_t$ with $t \ge 3$ by a right-linear chain through fresh nonterminals, replace terminals occurring in right sides of length at least two by preterminals $T_{\mathtt{x}} \to \mathtt{x}$ (only for the letters $\mathtt{x}$ that are used), and trim. The resulting grammar G' is reduced and good, has rules of the forms $X \to YZ$, $X \to Y$ and $X \to \mathtt{x}$ with $\mathtt{x} \in \Sigma$, and generates L. A rule $X \to \gamma$ of G gives at most $\max\{0, |\gamma| - 2\}$ fresh nonterminals and at most $\max\{1, |\gamma| - 1\}$ rules, both at most the size $1 + r + |\gamma|$ of the rule in $\|G\|$, so

$$k_V := |V'| \le \|G\| + \sigma_G \le 2\|G\|, \qquad |P'| \le \|G\| + \sigma_G,$$

and $k_V \le \|G\| + 2$, $|P'| \le \|G\| + 2$ if $|\Sigma| = 2$. Moreover $\theta := \bar{\theta}_{G'} \le \bar{\theta}_G$, since a derivation using a new rule corresponds to one using the rule of G it comes from, and a derivation using a preterminal rule $T_{\mathtt{x}} \to \mathtt{x}$ to one using a rule of G that writes $\mathtt{x}$. Every nonterminal of G' has a yield of length $\le \theta$ and a derivational context (l, r) with $|l| + |r| \le \theta$: take a shortest sentence through a rule with that nonterminal on the left, respectively on the right. The nonterminals of G are nonterminals of G' with the same yields.

H.2 The Tree and the Base

Fix a nonterminal A of G' with a pure yield; let $\mathtt{a}^n$ be the shortest one, T a derivation tree of $\mathtt{a}^n$ from A with the least number of nodes, and (l, r) a derivational context of A with $|l| + |r| \le \theta$. Replacing subtrees of T by derivations from the same labels gives yields w of A, hence $lwr \in L$. A node is *heavy* if its yield is longer than θ, and a heavy node without heavy children is a *seed*.

Lemma H.2. (a) *No label repeats on a root-to-leaf path of T; such a path has at most k_V nonterminal nodes.*

(b) *All nodes of T with the same label Y have the same yield $\mathtt{a}^{n_Y}$.*

(c) *Every label Y of a heavy node has a yield $\omega_Y \notin \mathtt{a}^+$ of length $\le \theta$.*

(d) *Seeds have yields of length $\le 2\theta$; seeds are pairwise disjoint; a subtree containing $k \ge 1$ seeds has a yield of length $\le k(k_V + 2)\theta$; and the left ends of two consecutive seeds differ by at most $2k_V\theta$.*

Proof. (a) If an ancestor and a descendant share a label, replacing the ancestor's subtree by the descendant's gives a shorter pure yield of A or a smaller tree. (b) Otherwise replace the longer subtree by the shorter one. (c) Let ρ be the rule at a heavy node labelled Y. A sentence of length $\le \theta$ using ρ contains a yield y of Y with $|y| \le \theta < n_Y$; if $y \in \mathtt{a}^+$, inserting it into T gives a shorter pure yield of A. (d) A seed is not a unit node and has two light children. If a seed v lay below a seed u, the child of u above v would be heavy. The heavy nodes of a subtree with k seeds lie on paths to seeds, so there are at most kk_V of them, each non-seed one with at most one light child. Between consecutive seeds u, v lie the yields of the subtrees hanging off the two paths from their least common ancestor, at most $2(k_V - 1)$ of them; each is light, since a heavy one would contain a seed between u and v. □

Let X_0 be the root of T and X_{i+1} the child of X_i with the longer yield, so that $n_{X_i} \le 2n_{X_{i+1}}$. If the root subtree has no seed, the root is light and $n \le \theta$; if it has at least one and at most k_V seeds, then $n \le k_V(k_V + 2)\theta$. Otherwise let i^* be the largest index whose subtree contains at least $k_V + 1$ seeds; then X_{i^*+1} exists, $i^* \le k_V - 2$ by Lemma H.2(a), and $B_0 := n_{X_{i^*}} \le 2k_V(k_V + 2)\theta$; we may assume $n_{X_{i^*}} < n$, and $k_V \ge 2$ since T has a binary node. The seeds below X_{i^*} are

consecutive in T; among k_V+1 consecutive ones two have the same label Y, at positions $x_1<x_2$ with $d:=x_2-x_1\le 2k_V^2\theta$. Put $g_0:=d-n_Y\ge 0$, $\omega:=\omega_Y$, $x:=x_1$, $y:=n-x_2-n_Y$, and $y_1:=n-x_1-n_Y=d+y$. Replacing both seeds, the first, or the second by ω gives the yields

$$W_1=\mathtt{a}^x\omega\mathtt{a}^{g_0}\omega\mathtt{a}^y,\qquad W_2=\mathtt{a}^x\omega\mathtt{a}^{d+y},\qquad W_3=\mathtt{a}^{x+d}\omega\mathtt{a}^y$$

of A. Here $x+g_0+y\ge 1$: otherwise the two seeds would cover the yield of the root, which contains at least $k_V+1\ge 3$ seeds. Since $d\ge n_Y\ge 1$ and $\omega\notin\mathtt{a}^+$, the words W_1,W_2,W_3 are $\mathtt{a}$-mixed.

Lemma H.3 (Base)**.** *Put $z_0:=\omega\mathtt{a}^{g_0}$ and $c_0:=d$ if $g_0\ge 1$ or ω is $\mathtt{a}$-mixed, and $z_0:=\omega\mathtt{a}^d$ and $c_0:=2d$ if $g_0=0$ and ω is $\mathtt{a}$-free. Then z_0 is $\mathtt{a}$-mixed, $\mathcal{D}(z_0)\ne\emptyset$, $d\mid c_0$,*

$$\mathtt{a}^dz_0\sim z_0\mathtt{a}^d\qquad\text{and}\qquad z_0z_0\sim\mathtt{a}^{c_0}z_0.$$

Proof. By (I2), W_1,W_2,W_3 have one type; they share (l,r), hence are $\sim$-equivalent. Cancelling $\mathtt{a}^x$ on the left and $\mathtt{a}^y$ on the right of $W_2\sim W_3$ (Lemma H.1; the remaining words contain $\mathtt{a}^d$ and ω, so they are $\mathtt{a}$-mixed, and the words cancelled from are factors of W_2, so their distributions are nonempty) gives $\omega\mathtt{a}^d\sim\mathtt{a}^d\omega$. If $g_0\ge 1$ or ω is $\mathtt{a}$-mixed, the same cancellation in $W_1\sim W_2$ (now $\omega\mathtt{a}^{g_0}\omega$ is $\mathtt{a}$-mixed) gives $\omega\mathtt{a}^{g_0}\omega\sim\omega\mathtt{a}^d$, whence $z_0z_0\sim\omega\mathtt{a}^{d+g_0}=z_0\mathtt{a}^d$ and $\mathtt{a}^dz_0\sim z_0\mathtt{a}^d$. If $g_0=0$ and ω is $\mathtt{a}$-free, then $n_Y=d$ and $x+y\ge 1$. For $y\ge 1$, cancelling $\mathtt{a}^x$ and $\mathtt{a}^{y-1}$ gives $\omega\omega\mathtt{a}\sim\omega\mathtt{a}^{d+1}$, so $z_0z_0\sim\omega\omega\mathtt{a}^{2d}\sim\omega\mathtt{a}^{3d}=z_0\mathtt{a}^{2d}$; for $y=0$, cancelling $\mathtt{a}^{x-1}$ gives $\mathtt{a}\omega\omega\sim\mathtt{a}\omega\mathtt{a}^d$, so, keeping one letter $\mathtt{a}$ in front of $\omega\omega$, $z_0z_0\sim\mathtt{a}^{2d}\omega\omega=\mathtt{a}^{2d-1}(\mathtt{a}\omega\omega)\sim\mathtt{a}^{2d-1}(\mathtt{a}\omega\mathtt{a}^d)=\mathtt{a}^{2d}z_0$; the $\mathtt{a}$-free word $\omega\omega$ is never compared with anything. The words compared here contain $\mathtt{a}$ and ω, so they are $\mathtt{a}$-mixed, and those cancelled from are factors of W_1. In both cases $z_0\mathtt{a}^{2d}\sim\mathtt{a}^{2d}z_0$ and $\mathtt{a}^dz_0\sim z_0\mathtt{a}^d$. Finally z_0 is a factor of W_1 or of W_2. □

For a typing onto U_1 in which $\mathtt{a}$ is the only letter of type 1, the $\mathtt{a}$-free words have the type of the $\mathtt{a}$-mixed ones, and the second case could be avoided; we do not use this.

An *annotated code* is a pair (z,e) with $e\ge 0$, $d\mid e$ and $z\sim\mathtt{a}^ez_0$; its *value* is $e+c_0$. A pure block $\mathtt{a}^R$ with $d\mid R$ has value R. For a specified product u of annotated codes and such blocks, $\mathrm{val}(u)$ is the sum of the values of its factors. (The exponent e is not determined by z in general, so values are attached to annotations.)

Lemma H.4. *If u contains at least one code, then $u\sim\mathtt{a}^{\mathrm{val}(u)-c_0}z_0$.*

Proof. Replace each code by $\mathtt{a}^ez_0$, move all blocks to the left with $\mathtt{a}^dz_0\sim z_0\mathtt{a}^d$, and use $z_0^t\sim\mathtt{a}^{(t-1)c_0}z_0$, which follows by induction from $z_0z_0\sim\mathtt{a}^{c_0}z_0$ and $d\mid c_0$. □

H.3 Codes along the Heavy Path

Let i_0 be the least index with $n_{X_i}<n$; then $1\le i_0\le i^*$. For $i_0\le i\le i^*$ let $(\mathtt{a}^{p_i},\mathtt{a}^{q_i})$ be the context of X_i inside T, so $p_i+q_i\ge 1$; the subtree of X_i contains both seeds; let L_i,R_i be the numbers of letters to the left and right of the first seed inside it, so $R_i\ge g_0+n_Y\ge 1$, and put $\Omega_i:=\mathtt{a}^{L_i}\omega\mathtt{a}^{R_i}$, an $\mathtt{a}$-mixed yield of X_i.

Lemma H.5. *If ω_{X_i} is $\mathtt{a}$-mixed, then $\omega_{X_i}\sim\Omega_i$. If ω_{X_i} is $\mathtt{a}$-free and $p_i\ge 1$, then $\mathtt{a}\omega_{X_i}\sim\mathtt{a}\Omega_i$; if ω_{X_i} is $\mathtt{a}$-free and $p_i=0$, then $\omega_{X_i}\mathtt{a}\sim\Omega_i\mathtt{a}$.*

Proof. Both $l\mathtt{a}^{p_i}\omega_{X_i}\mathtt{a}^{q_i}r$ and $l\mathtt{a}^{p_i}\Omega_i\mathtt{a}^{q_i}r$ lie in L. In the first case ω_{X_i} and Ω_i are $\mathtt{a}$-mixed and share the context $(l\mathtt{a}^{p_i},\mathtt{a}^{q_i}r)$. In the second case $\mathtt{a}\omega_{X_i}$ and $\mathtt{a}\Omega_i$ are $\mathtt{a}$-mixed and share the context $(l\mathtt{a}^{p_i-1},\mathtt{a}^{q_i}r)$; the third case is symmetric, with $q_i\ge 1$. Conclude by (I2). □

Choose $s_i', s_i \in [0,d]$ with $d \mid L_i + s_i'$, with $R_i + s_i \equiv g_0 \pmod d$ in the first case of Lemma H.3 and $R_i + s_i \equiv 0 \pmod d$ in the second, and with $s_i' \geq 1$, respectively $s_i \geq 1$, when a letter is borrowed in Lemma H.5. Put $\hat{z}_i := \mathtt{a}^{s_i'}\omega_{X_i}\mathtt{a}^{s_i}$. By Lemma H.5, $\hat{z}_i \sim \mathtt{a}^{s_i'+L_i}\omega\mathtt{a}^{R_i+s_i}$. In the first case $R_i + s_i = g_0 + kd$ with $k \geq 0$, and in the second $R_i + s_i = d + kd$ with $k \geq 0$, since $R_i \geq g_0 + n_Y$; so $\hat{z}_i \sim \mathtt{a}^{e_i} z_0$ with $e_i := s_i' + L_i + kd$, and $(\hat{z}_i, e_i)$ is an annotated code, of value

$$v_i = n_{X_i} + s_i + s_i', \qquad |\hat{z}_i| \leq \theta + 2d,$$

in both cases (using $n_{X_i} = L_i + n_Y + R_i$ and $d - g_0 = n_Y$, respectively $n_Y = d$). For $i < i_0$ put $v_i := n$ (no code). Since X_i is an ancestor of X_{i^*}, whose yield contains both seeds, $n_{X_i} \geq d + n_Y$ for $i \leq i^*$. Hence, for all $i \leq i^*$,

$$v_i \geq n_{X_i} > d, \qquad v_i \leq 2v_{i+1} + 2d \ \ (i < i^*), \tag{H.1}$$

and $v_{i^*} \leq B_0 + 2d$; moreover $n \leq 2n_{X_{i_0}} \leq 2v_{i_0}$.

The yield $\mathtt{a}^{x_1}\omega\mathtt{a}^{y_1}$ of A equals $\mathtt{a}^{x_1} z_0 \mathtt{a}^h$ with $h := y_1 - g_0$, respectively $h := y_1 - d$. Writing $h = qd + \rho$ and $x_1 + qd = e^* + \rho'$ with $0 \leq \rho, \rho' < d$ and $d \mid e^*$, we obtain $l_\circ \mathtt{a}^{e^*} z_0 r_\circ \in L$ for $l_\circ := l\mathtt{a}^{\rho'}$ and $r_\circ := \mathtt{a}^\rho r$, and $V^* := e^* + c_0 = n - \rho - \rho'$.

H.4 Periodicity and Membership

Lemma H.6. *Let $P_0 := \{p \geq 0 : \mathtt{a}^p z_0 \sim z_0\}$.*

(i) *$P_0 = \{0\}$ or $P_0 = p_0\mathbb{Z}_{\geq 0}$ for some $p_0 \geq 1$.*

(ii) *If $l'\mathtt{a}^f z_0 r' \in L$ for some f, then $\{f \geq 0 : l'\mathtt{a}^f z_0 r' \in L\}$ is a single point if $P_0 = \{0\}$ and a full residue class modulo p_0 otherwise.*

(iii) *If $P_0 \neq \{0\}$, then $\mathtt{a} \sim \mathtt{a}^{1+p_0}$, so every $\mathtt{a}^k$ is $\equiv_{L,s}$-equivalent to some $\mathtt{a}^{k'}$ with $1 \leq k' \leq p_0$.*

Proof. All words $\mathtt{a}^f z_0$ are $\mathtt{a}$-mixed, so they have one type by (I2). (i) P_0 is closed under sums. It is closed under differences $q - p$ with $q > p > 0$ in P_0 by Lemma H.1, applied to $\mathtt{a}^p(\mathtt{a}^{q-p} z_0) \sim \mathtt{a}^p z_0$, since $\mathcal{D}(\mathtt{a}^p z_0) = \mathcal{D}(z_0) \neq \emptyset$; the case $p = 0$ is trivial. (ii) Two members $f < f'$ give $\mathtt{a}^f z_0 \sim \mathtt{a}^{f'} z_0$, hence $f' - f \in P_0$ by Lemma H.1; conversely the set is closed under adding and, by $\mathtt{a}^{p_0} z_0 \sim z_0$, under subtracting p_0. (iii) For $(l', r') \in \mathcal{D}(z_0)$ both $l'\mathtt{a}^{p_0} z_0 r'$ and $l'\mathtt{a}^{2p_0} z_0 r'$ lie in L, so the pure words $\mathtt{a}$ and $\mathtt{a}^{1+p_0}$, which have one type by (I1), share the context $(l'\mathtt{a}^{p_0-1}, z_0 r')$. □

Let $D := \operatorname{lcm}(d, p_0)$, and $D := \infty$ if $P_0 = \{0\}$. By Lemma H.4 and Lemma H.6(ii), for every specified product u as above that contains a code,

$$l_\circ u r_\circ \in L \iff \operatorname{val}(u) \equiv V^* \pmod D, \tag{H.2}$$

with equality when $D = \infty$; values and V^* are multiples of d. Two annotations of the same word differ by a multiple of D, so membership does not depend on the annotation.

H.5 Counting

Lemma H.7. *Let $(z_1, e_1), \ldots, (z_M, e_M)$ be annotated codes, $M \geq 6$, with $|z_j| \leq \ell_z$, values w_j whose subset sums are pairwise distinct, and $\Sigma_w := \sum_j w_j < D$, and let Q be a fixed specified product of codes and blocks with $\Sigma_w + \operatorname{val}(Q) = V^*$. Put $m := |l_\circ| + |r_\circ| + |Q| + \sum_j |z_j|$. Then*

$$(27/4)^{\lfloor M/3 \rfloor} \leq (M+1)^2 k_V (m+1)^2 (M\ell_z + 1)^2.$$

Proof. For a permutation π, the word $w_\pi := l_\circ z_{\pi(1)} \cdots z_{\pi(M)} Q r_\circ$ lies in L by (H.2). Take a derivation tree of w_π in G', contract unit nodes, and give each node the weight "number of the codes $z_{\pi(1)}, \ldots, z_{\pi(M)}$ whose first letter lies in its yield". Leaves have weight ≤ 1 and the root has weight M; descending to the heavier child we reach a node u of weight $\kappa \in (M/3, 2M/3]$, with label X and yield interval $[p, q)$. Let $I(\pi)$ be the set of indices j such that z_j lies inside $[p, q)$; then $\kappa - 1 \le |I(\pi)| \le \kappa$. The *record* of π consists of X, p, q, and, for each end of $[p, q)$ lying strictly inside one of the M codes, the index and the offset of that code; there are at most $k_V(m+1)^2(M\ell_z+1)^2$ records. The three blocks $l_\circ$, $z_{\pi(1)} \cdots z_{\pi(M)}$ and $Q r_\circ$ occupy the same positions for all π. The codes whose first letter lies in $[p, q)$ are those of $I(\pi)$ and at most one code that starts inside and ends after q; a code through p starts before p. This gives $\kappa - 1 \le |I(\pi)|$. Each end of $[p, q)$ lies in $l_\circ$, in $Q r_\circ$, on a boundary between two codes, or strictly inside a code; only in the last case is a code cut there, and then its index and offset are part of the record.

Let π, π' have the same record. Replacing the subtree of u in the tree of w_π by the corresponding subtree of $w_{\pi'}$ gives a derivation of $w_\pi[0, p)\, w_{\pi'}[p, q)\, w_\pi[q, |w_\pi|) \in L$. Since an end inside a code has the same index and offset on both sides, the fragments join into complete copies of the same code, and the word is $l_\circ u'' Q r_\circ$ where u'' is a product of complete codes in which z_j occurs $1 - \mathbf{1}[j \in I(\pi)] + \mathbf{1}[j \in I(\pi')]$ times; this also covers ends in $l_\circ$ or $Q r_\circ$, and both ends cannot lie in $Q r_\circ$ because $\kappa \ge 2$. By (H.2), $\sum_{I(\pi')} w_j \equiv \sum_{I(\pi)} w_j \pmod D$; both sums lie in $[0, \Sigma_w) \subseteq [0, D)$, so they are equal, and $I(\pi') = I(\pi)$.

Hence a record and $I = I(\pi)$, $|I| = j$, determine π up to the order inside ($\le j!$) and the arrangement of the remaining codes around this block ($\le (M - j + 1)!$). Since $j \in \{\kappa - 1, \kappa\}$ and $M/3 < \kappa \le 2M/3$, j lies in $J := [\lfloor M/3 \rfloor, \lfloor 2M/3 \rfloor]$, and $\binom{M+1}{j} \le (M+1)k_V(m+1)^2(M\ell_z+1)^2$ for some $j \in J$. Put $k := \lfloor M/3 \rfloor$. As $\binom{M+1}{\cdot}$ is symmetric and unimodal and $\lfloor M/3 \rfloor + \lfloor 2M/3 \rfloor \le M$, every $j \in J$ satisfies $\binom{M+1}{j} \ge \binom{M+1}{k} \ge \binom{3k}{k}$. Finally, the numbers $t_i := \binom{3k}{i} 3^{-i} (2/3)^{3k-i}$, $0 \le i \le 3k$, sum to 1, and $t_{i+1}/t_i = (3k - i)/(2i + 2)$ is at least 1 if and only if $i \le k - 1$; so t_k is the largest of them and $t_k \ge 1/(3k+1) \ge 1/(M+1)$. Since $t_k = \binom{3k}{k} 4^k / 27^k$, this gives $\binom{3k}{k} \ge (27/4)^k/(M+1)$, and the lemma follows. □

H.6 Selection and Filler

Put $\Lambda := \min(V^*, D)$, so $\Lambda \le V^* \le n$.

Lemma H.8 (Selection)**.** *There are $M \ge 0$ indices $i^* = j_1 > j_2 > \cdots > j_M \ge i_0$ whose values $w_t := v_{j_t}$ are superincreasing ($w_t > w_1 + \cdots + w_{t-1}$), with $w_1 + \cdots + w_M < \Lambda$ and $\Lambda \le 3^M(v_{i^*} + d)$.*

Proof. If $v_{i^*} \ge \Lambda$, take $M := 0$. Otherwise take $j_1 := i^*$. Given j_t and $S_t := w_1 + \cdots + w_t < \Lambda$, let i be the largest index below j_t with $v_i > S_t$; it exists, since $v_0 = n \ge \Lambda$ (recall $i_0 \ge 1$). If $S_t + v_i \ge \Lambda$, stop with $M := t$; otherwise put $j_{t+1} := i$. The values are superincreasing and their sum is below Λ by construction. A taken value is less than $\Lambda \le n$, whereas $v_i = n$ for $i < i_0$; so all taken indices are at least i_0. By the choice of i, $v_{i+1} \le S_t$: for $i + 1 < j_t$ by maximality, and for $i + 1 = j_t$ because $v_{j_t} = w_t$. By (H.1), $v_i \le 2S_t + 2d$. Hence $S_{t+1} + d \le 3(S_t + d)$ whenever i is taken, and $\Lambda \le S_M + v_i < 3(S_M + d)$ when the process stops. With $S_1 = v_{i^*}$ this gives $\Lambda < 3^M(v_{i^*} + d)$. □

Lemma H.9 (Filler)**.** *Let $0 < X < n$ be a multiple of d. There is a specified product Q of at most $3k_V$ codes among $\hat{z}_{i_0}, \ldots, \hat{z}_{i^*}$ and at most one pure block $\mathtt{a}^{X'}$ with $d \mid X'$ and $X' < v_{i^*}$, such that $\mathrm{val}(Q) = X$ and $|Q| \le (3k_V + 2)\ell_z$, where $\ell_z := 5k_V^2\theta$.*

Proof. For $i = i_0, \ldots, i^*$ in turn, append $q_i := \lfloor X/v_i \rfloor$ copies of $\hat{z}_i$ and replace X by $X - q_i v_i$. Then $q_{i_0} \le 1$, since $X < n \le 2v_{i_0}$. For $i > i_0$ the current X is less than v_{i-1}, and $v_{i-1} \le 2v_i + 2d < 4v_i$

by (H.1), so $q_i \le 3$. The final remainder $X' < v_{i^*}$ is a multiple of d, since all values are, and is appended as a block. As $i^* - i_0 \le k_V - 3$, there are at most $1 + 3(k_V - 3) < 3k_V$ codes, each of length $\le \theta + 2d \le \ell_z$, and $X' < v_{i^*} \le B_0 + 2d \le (6k_V^2 + 4k_V)\theta \le 2\ell_z$. □

H.7 Proof of Theorem 7.10

Let A have a pure yield. By (I1) the yields of A of type $s(\mathtt{a})$ are its pure yields, so $\tau(A, s(\mathtt{a})) \le n$, and it suffices to show $\tau(A, s(\mathtt{a})) \le 11 \cdot 3^{M^*} k_V^2 \theta$, where

$$M^* := \max(\{5\} \cup \{M \ge 6 : (27/4)^{\lfloor M/3 \rfloor} \le (M+1)^4 k_V (3k_V + M + 4)^2 \ell_z^4\}),$$

a finite number since the left side grows exponentially in M. If the root subtree has at most k_V seeds, or $n_{X_{i^*}} = n$, then $n \le 2k_V(k_V + 2)\theta$. Otherwise take the indices of Lemma H.8.

Counting. Suppose $M \ge 6$. Let $X := V^* - \sum_t w_t$; then $0 < X < n$, since $\sum_t w_t < \Lambda \le V^* \le n$ and $w_1 > 0$, and X is a multiple of d. Let Q be given by Lemma H.9. The codes $\hat{z}_{j_1}, \dots, \hat{z}_{j_M}$ have length $\le \ell_z$ and superincreasing values, so their subset sums are distinct, and $\sum_t w_t < \Lambda \le D$. With $|l_\circ| + |r_\circ| \le \theta + 2d \le \ell_z$ we get $m \le (3k_V + M + 3)\ell_z$, and Lemma H.7, with $(m+1)^2(M\ell_z + 1)^2 \le (3k_V + M + 4)^2 (M+1)^2 \ell_z^4$, gives $M \le M^*$. So in every case $M \le M^*$, and by Lemma H.8

$$\Lambda \le 3^{M^*}(v_{i^*} + d) \le 3^{M^*}(B_0 + 3d) \le 10 \cdot 3^{M^*} k_V^2 \theta.$$

Conclusion. If $\Lambda = V^*$, then $n = V^* + \rho + \rho' < \Lambda + 2d$ and $\tau(A, s(\mathtt{a})) \le n$. If $\Lambda = D < V^*$, then $P_0 \ne \{0\}$, and by Lemma H.6(iii) $\mathtt{a}^n \equiv_{L,s} \mathtt{a}^{n'}$ with $n' \le p_0 \le D$, so $\tau(A, s(\mathtt{a})) \le n'$. Together with the small cases, $\tau(A, s(\mathtt{a})) \le \max\{2k_V(k_V+2)\theta,\ 10 \cdot 3^{M^*} k_V^2 \theta + 4k_V^2\theta\} \le 11 \cdot 3^{M^*} k_V^2 \theta$.

The exponents. Put $\mu := (27/4)^{1/3}$, $\gamma := \ln 3 / \ln \mu < 1.7260$, $\eta := 1/80$ and $C_\eta := \sup_{M \ge 0}(M+1)^6 \mu^{-\eta M} < e^{34}$ (the maximum of $6 \ln x - (x-1)\eta \ln \mu$ over $x > 0$ is attained at $x = 6/(\eta \ln \mu)$). For $M \ge 6$ in the set defining M^*, $\mu^{M-2} \le (27/4)^{\lfloor M/3 \rfloor}$ and $3k_V + M + 4 \le 5k_V(M+1)$ give

$$\mu^M \le 15625\, \mu^2 (M+1)^6 k_V^{11} \theta^4, \qquad \text{hence} \qquad \mu^{(1-\eta)M} \le 15625\, \mu^2 C_\eta k_V^{11} \theta^4,$$

and $3^M = \mu^{\gamma M} \le (15625\, \mu^2 C_\eta k_V^{11} \theta^4)^{\gamma/(1-\eta)}$; this also holds for $M = 5$. Since $\gamma/(1-\eta) = 80\gamma/79 < 1.7479$, we obtain

$$\tau(A, s(\mathtt{a})) \le 11 \cdot 3^{M^*} k_V^2 \theta \le C_0 k_V^{2+19.23} \theta^{1+6.992} \le C_0 k_V^{22} \theta^8, \qquad C_0 := 11(15625\, \mu^2 C_\eta)^{80\gamma/79} < 10^{36}.$$

As $k_V \le 2\|G\|$, $\theta \le \bar{\theta}_G$, and the nonterminals of G are nonterminals of G' with the same yields, the theorem follows with $C := 2^{22} C_0 < 10^{43}$. Letting $\eta \to 0$ shows that the exponents can be taken arbitrarily close to $2 + 11\gamma < 20.99$ and $1 + 4\gamma < 7.91$. □

H.8 Other Types and Proof of Corollary 7.11

Lemma H.10. *Let G' be as above and let A be a nonterminal of G'.*

(i) *If $s : \Sigma^+ \twoheadrightarrow U_1$ and $\mathtt{a}$ is the only letter of type 1, and A has a yield of type ζ, then $\tau(A, \zeta) \le (k_V - 1)\theta + 1$.*

(ii) *If $\Sigma = \{\mathtt{a}, \mathtt{b}\}$, $s = c$, and A has a yield of content Σ, then $\tau(A, \Sigma) \le 3k_V\theta + 2$.*

Proof. (i) Take a derivation of a yield of A of type ζ with the least number of nodes and mark a leaf whose letter is not $\mathtt{a}$. Replacing the subtree of a node on the path from the root to the marked leaf by the subtree of a lower node on the same path with the same label keeps the marked leaf, hence

the type; so no label repeats on this path, which has at most k_V nonterminal nodes. Replacing the at most $k_V - 1$ subtrees hanging off it by yields of length $\le \theta$ gives a yield of A of type ζ and length $\le (k_V - 1)\theta + 1$. (ii) Take a derivation of a yield of content Σ with the least number of nodes and mark an $\mathtt{a}$-leaf and a $\mathtt{b}$-leaf. By the same contraction within each of the three path segments between the root, the branching node and the two leaves (never between different segments), the skeleton has at most $3k_V$ nodes; replacing the at most $3k_V$ subtrees hanging off it by yields of length $\le \theta$ gives a yield of content Σ of length $\le 3k_V\theta + 2$. In both cases the new word is a yield of A of the same type, hence in the same class (Lemma 3.8). □

Proof of Corollary 7.11. In both cases G' is a reduced good presentation of $L \cap \Sigma^+$ of fan-out one and rank $2 \le r_{\max}$, with $\bar\theta_{G'} \le \bar\theta_G$, and Corollary 4.8 applied to G' gives $\|K\|_+ \le N_{\mathrm{typ}}(G', s)(1 + 2\bar\theta_G + r_{\max}\tau(G', s)) + \mathbf{1}_{\{\varepsilon\in L\}}$. It remains to bound $\tau(G', s)$ and $N_{\mathrm{typ}}(G', s)$.

(i) The positive image of s is generated by $s(\mathtt{a})$ and $s(\mathtt{b})$ in a semilattice, so $s(w)$ depends only on $c(w)$; thus $s \preceq c$, $\mathcal{C}^{\mathrm{mcf}}_{1,s} \subseteq \mathcal{C}^{\mathrm{mcf}}_{1,c}$, and $\equiv_{L,c}$ refines $\equiv_{L,s}$. For a typed nonterminal A_t of G' and the c-type t' of one of its yields, $\tau_{A_t}(G', s) \le \tau(A, t')$. Both letters are isolated for c, so Theorem 7.10, applied to c with $\mathtt{a}$ and with $\mathtt{b}$, and Lemma H.10(ii) give $\tau(G', s) \le C(\|G\| + 2)^{22}\bar\theta_G^8$. The typed refinement of G' has at most $3k_V$ typed nonterminals and at most nine typed rules per rule, so $N_{\mathrm{typ}}(G', s) \le 3k_V + 9|P'| \le 12(\|G\| + 2)$.

(ii) The typed nonterminals of G' are of the form A_1, with A having a yield in $\mathtt{a}^+$, and A_ζ. By Theorem 7.10 and Lemma H.10(i), $\tau(G', s) \le C(\|G\| + 2)^{22}\bar\theta_G^8$. There are at most $2k_V$ typed nonterminals and at most four typed rules per rule, so $N_{\mathrm{typ}}(G', s) \le 2k_V + 4|P'| \le 6(\|G\| + \sigma_G) \le 12\|G\|$.

In both cases $1 + 2\bar\theta_G \le 3\bar\theta_G$ gives the bound with $C' := 12(C + 2) < 10^{45}$. □

I Typings onto U_1 at Fan-Out One

I.1 Letter-Disjoint Images of Finite Substitutable Languages

Lemma I.1. *Let $L \subseteq \Delta^+$ be finite and Clark–Eyraud substitutable. There are no $P, Q \in \Delta^*$ and $T \in \Delta^+$ with $PQ \in L$ and $PTQ \in L$.*

Proof. Suppose otherwise. If $P \ne \varepsilon$, the words P and PT share the context (ε, Q), so $\mathcal{D}_L(P) = \mathcal{D}_L(PT)$; then $PT^jQ \in L$ implies $PT^{j+1}Q \in L$, and L contains the words PT^jQ of every length, contrary to finiteness. If $P = \varepsilon$, argue in the same way with Q and TQ, which share the context $(\varepsilon, \varepsilon)$. □

Lemma I.2. *Let $h : \Delta^* \to \Sigma^*$ be a morphism such that every $h(\delta)$, $\delta \in \Delta$, is nonempty and no letter occurs twice in the concatenation of all $h(\delta)$. If $L \subseteq \Delta^+$ is finite and Clark–Eyraud substitutable, so is $h(L)$.*

Proof. Each letter occurring in some $h(\delta)$ determines δ and its position in $h(\delta)$. Hence h is injective, and in a word $h(D)$ every letter occurrence belongs to a determined occurrence of a codeword $h(\delta)$; whether two adjacent letters lie in one codeword occurrence, and whether a letter begins or ends one, is read off the letters themselves.

Let $x, y \in \Sigma^+$ with $pxq = h(D_1)$ and $pyq = h(D_2)$ in $h(L)$, $x \ne y$. The codeword occurrences lying inside p, and those inside q, are the same in both words; so $D_i = PM_iQ$, where M_i consists of the codeword occurrences meeting x, respectively y. If p ends inside a codeword occurrence $h(\delta)$, then M_1 and M_2 both begin with δ, and if q begins inside one, $h(\delta')$, both end with δ'.

Suppose that x lies inside one codeword occurrence $h(\delta)$ and is not all of it; say p ends inside it (the other case is symmetric). Then $M_1 = \delta$ and M_2 begins with δ; $M_2 \neq M_1$, since otherwise $y = x$, so $M_2 = \delta T$ with $T \in \Delta^+$, and $D_1 = P\delta Q$, $D_2 = P\delta TQ$ contradict Lemma I.1. The same holds with x and y exchanged. So neither lies inside one codeword occurrence unless it is a whole codeword. Then $M_1 = [\delta]X[\delta']$ and $M_2 = [\delta]Y[\delta']$, where $X, Y \in \Delta^*$ are the codewords lying inside x, respectively y, and the bracketed letters are present iff the corresponding cut lies inside a codeword. If $X = \varepsilon \neq Y$ or $Y = \varepsilon \neq X$, Lemma I.1 is contradicted again, and $X = Y = \varepsilon$ gives $x = y$. Hence $X, Y \in \Delta^+$, and

$$x = \alpha\, h(X)\, \beta, \qquad y = \alpha\, h(Y)\, \beta,$$

where α is the part of $h(\delta)$ after p (or ε) and β the part of $h(\delta')$ before q (or ε). The words X and Y share a context in L, so $\mathcal{D}_L(X) = \mathcal{D}_L(Y)$.

Now let $p'xq' \in h(L)$. The cut between α and $h(X)$, and the one between $h(X)$ and β, are boundaries of codeword occurrences, which is read off the letters of x; so $p'\alpha = h(P')$ and $\beta q' = h(Q')$ for some P', Q', and $P'XQ' \in L$. Then $P'YQ' \in L$, and $h(P'YQ') = p'yq'$. By symmetry $\mathcal{D}_{h(L)}(x) = \mathcal{D}_{h(L)}(y)$. □

I.2 Quadratic Data for the Binary Typing

We prove Proposition 7.8. Put $k := 2^m$, $M := 2^n$ and $\psi(w) := |w|_{\mathtt{b}} + |w|_{\mathtt{a}}/k$, which is additive. For a word w let $\lambda(w)$ and $\varrho(w)$ be the lengths of its longest suffix and longest prefix in $\mathtt{a}^*$.

Proof of Proposition 7.8. (i) A word of Σ^+ lies in $L_{m,n}$ iff all its maximal runs of $\mathtt{a}$ have length divisible by k and $\psi(w) = M$: such a word factors uniquely into blocks $\mathtt{a}^k$ and $\mathtt{b}$, and ψ counts the blocks. Let $x, y \in \Sigma^+$ have the same type and let $uxv, uyv \in L_{m,n}$. By additivity $\psi(x) = \psi(y)$. If x, y are pure, this gives $x = y$. Otherwise write $x = \mathtt{a}^{\alpha}x'\mathtt{a}^{\beta}$ with x' beginning and ending with $\mathtt{b}$, and similarly $y = \mathtt{a}^{\alpha'}y'\mathtt{a}^{\beta'}$. The maximal $\mathtt{a}$-runs inside x' and y' are maximal runs of uxv and uyv, so their lengths are divisible by k; the runs through the two ends of x have lengths $\lambda(u) + \alpha$ and $\beta + \varrho(v)$, so $\alpha \equiv \alpha'$ and $\beta \equiv \beta'$ (mod k). For any context (p, q) the maximal $\mathtt{a}$-runs of pxq are those of p not at its end, a run of length $\lambda(p) + \alpha$, the runs inside x', a run of length $\beta + \varrho(q)$, and those of q not at its start; the same holds for pyq with α', β', y'. So the divisibility conditions agree, $\psi(pxq) = \psi(pyq)$, and $pxq \in L_{m,n} \iff pyq \in L_{m,n}$. Finite languages are context-free.

(ii) Take nonterminals $C_0, \dots, C_m$, $A_0, \dots, A_n$, start symbol A_n, and the rules $C_0 \to \mathtt{a}$, $C_j \to C_{j-1}C_{j-1}$ $(1 \le j \le m)$, $A_0 \to C_m \mid \mathtt{b}$ and $A_i \to A_{i-1}A_{i-1}$ $(1 \le i \le n)$. Then $L(C_j) = \{\mathtt{a}^{2^j}\}$, $L(A_0) = \{\mathtt{a}^k, \mathtt{b}\}$ and $L(A_i)$ is the set of concatenations of 2^i words of $L(A_0)$, so $L(A_n) = L_{m,n}$; the grammar is good and reduced. It has $m + n + 2$ nonterminals and rules of sizes 2, 5 (m rules), 3, 2 and 5 (n rules) (Definition 2.5), which gives $6m + 6n + 9$. At fan-out one, the shortest sentence using a rule $R : B \to \rho(B_1, \dots, B_q)$ has length $\theta_G(R) = o_B + c_R + \sum_j d_{B_j}$, where d_C is the length of a shortest yield of C, o_C the least number of letters around an occurrence of C in a sentence, and c_R the number of terminals written by R; here $d_{A_0} = 1$, $d_{A_i} = 2^i$, $d_{C_j} = 2^j$, $o_{A_i} = M - 2^i$, $o_{A_0} = o_{C_m} = M - 1$ and $o_{C_j} = M - 1 + k - 2^j$. This gives $\theta_G(R) = M$ for the rules of A_i and for $A_0 \to \mathtt{b}$, and $\theta_G(R) = M + k - 1$ for $A_0 \to C_m$ and the rules of the C_j; so $\bar{\theta}_{G_{m,n}} = 2^m + 2^n - 1$.

(iii) The only pure word of $L_{m,n}$ is $\mathtt{a}^{kM}$, so by Proposition 4.6 every characteristic sample contains it and $\|F\|_+ \ge kM + 1 > 2^{m+n}$. For $m = n$, $\bar{\theta}_{G_{n,n}} < 2^{n+1}$ gives $\bar{\theta}^2_{G_{n,n}}/4 < 4^n < \|F\|_+$. A bound $p(\|G\|)\bar{\theta}_G$ would give $4^n \le p(12n + 9)\, 2^{n+1}$ for all n, which fails for every polynomial p. □

J Content Typings on Two Letters at Fan-Out at Least Two

This appendix proves Theorem 8.6. Throughout, $\Sigma = \{\mathtt{a}, \mathtt{b}\}$, $c := c_\Sigma$, $f \geq 2$ is fixed, $L \in \mathcal{C}^{\mathrm{mcf}}_{f,c}$, and $G = (V, \Sigma, P, S, \mathrm{fan})$ is a reduced good grammar of fan-out $\leq f$ and rank $\leq r := r_{\max}$ with $L(G) = L \cap \Sigma^+$; put $v := |V|$ and $\theta := \bar{\theta}_G$; we also put $D_f := (6f+1)3^f$. Apart from Lemma J.1(i), which uses the condition of the arity of the typed nonterminal at hand, the arguments use only $(1,c)$- and $(2,c)$-tuple-substitutability, which hold since $f \geq 2$. For each $A \in V$ the context E_A is that of Lemma 4.1(iv), so $|E_A|_{\mathrm{fix}} \leq \theta$ and $E_A[L_A(G)] \subseteq L(G)$; for each $B \in V$ fix $\omega_B \in L_B(G)$ with $\|\omega_B\|_1 \leq \theta$ (item (ii)), and recall that every rule writes at most θ terminals (item (i)). In a derivation tree, the *yield* of a node u is the tuple it derives, ℓ_u is its total length, B_u its nonterminal, and u is *heavy* if $\ell_u > \theta$; a node *writes* the terminals of its template. The letters x, y always denote the two letters of Σ in some order. A *run* of a word is a maximal factor x^k ($k \geq 1$); a run is *internal* if it is neither the first nor the last run, so that an internal y-run is preceded and followed by x. The *x-skeleton* $\sigma_x(w)$ replaces every x-run of w by one x, and w has a *spare* x if $|w|_x$ exceeds the number of x-runs of w. We say that x^k *occurs* if some word of L has the factor x^k, and that L is *Parikh-saturated* if for $u, w \in \Sigma^+$ with the same Parikh vector $\Psi(u) = \Psi(w)$ we have $u \in L$ iff $w \in L$.

Map of the proof. The table lists, for each step, what it uses and where it is used.

Step	Uses	Used in
Lemma J.1(i) typed closure	Lemma 3.8, arity-d substitutability	Lemma J.6
Lemma J.1(ii) cutting identity paths	goodness of G	(P2), the mixed case of Lemma J.6
Lemma J.2 run exchanges	$(1,c)$, $(2,c)$-substitutability, x^4 occurs	Lemma J.3, Proposition J.5
Lemma J.3	Lemma J.2(i),(iii)	Proposition J.5
Lemma J.4 runs created by replacement	Lemma 4.2(i), node-minimal shortest T	Proposition J.5
Proposition J.5 (not saturated)	the three lemmas above; typed labels on paths	Theorem 8.6
Lemma J.6, union Σ	Lemma J.1; floors along paths	Theorem 8.6
(P1) row rigidity	$(1,c)$-substitutability, minimality of K	(P3), (P4)
(P2) height, mixed nodes	Lemma J.1(ii), minimality of K	(P3), (P6)
(P3) heavy pure nodes	(P1), (P2)	(P4), (P5)
(P4) common ratio α	(P3)	(P5), (P6)
(P5) bound on α	(P3), (P4)	(P6)
(P6) counting	(P2), (P4), (P5)	Theorem 8.6

Lemma J.1. (i) *For every typed nonterminal $X = A_{\mathbf{t}}$ of arity d, $\mathrm{Cl}_{L,c}(X) = \{\vec{z} \in (\Sigma^+)^d : c^{(d)}(\vec{z}) = \mathbf{t},\ E_A[\vec{z}] \in L\}$; hence τ_X is the least total length of such a tuple.*

(ii) *In a derivation tree of G, let u be a proper ancestor of u' with the same nonterminal, and suppose their yields have the same total Parikh vector. Then every rule on the path from u to u' has rank one, writes no terminal and is a componentwise identity, so replacing the subtree at u by the subtree at u' changes no derived tuple and removes at least one node.*

Proof. (i) Fix $\vec{u} \in L_X$. By Lemma 3.8, $\mathrm{Cl}_{L,c}(X)$ is the set of tuples with the type and the distribution of $\vec{u}$; they are accepted by $E_A \in \mathcal{D}_L(\vec{u})$. A tuple $\vec{z}$ of type $\mathbf{t}$ accepted by E_A shares this context with $\vec{u}$, so $\mathcal{D}_L(\vec{z}) = \mathcal{D}_L(\vec{u})$ by substitutability.

(ii) By linearity and nondeletion every letter of the yield of u' occurs in the yield of u, so equal Parikh vectors exclude terminals written on the path and siblings off the path, since siblings

have nonempty yields (ε-freeness). So every rule on the path has one child of some fan-out d', each of its d' variables occurs once (nondeletion) and in order (nonpermutation), and each parent component is nonempty (ε-freeness). Two variables in one component would be consecutive, which nonmerging forbids; hence every component is a single variable, the fan-outs agree, and the rule is the identity. □

Lemma J.2 (Run exchanges)**.** *Suppose that x^4 occurs.*

(i) *If $w, w' \in \Sigma^+$ satisfy $\sigma_x(w) = \sigma_x(w')$ and $|w|_x = |w'|_x$, then $w \in L$ iff $w' \in L$.*

(ii) *If some word of L has a factor $xxyx$ or $xyxx$, then L is Parikh-saturated.*

(iii) *If L is not Parikh-saturated, then in every word of L with a factor xx every internal y-run has length at least 2.*

(iv) *If L is not Parikh-saturated and y^4 does not occur, then every word of L with a factor xx has at most one internal y-run.*

Proof. (i) Since $c(x)$ is idempotent, Lemma 8.1 applies with $u = x$, $i_0 = p = 1$ and $i = \delta = j = 1$: as x^4 occurs, $w_{\mathrm{L}}xw_{\mathrm{M}}xxw_{\mathrm{R}} \in L$ iff $w_{\mathrm{L}}xxw_{\mathrm{M}}xw_{\mathrm{R}} \in L$ whenever $w_{\mathrm{M}} \neq \varepsilon$. Let R and R' be x-runs of a word w with $|R| \geq 2$, say R' to the left of R (the other case is symmetric). Write $w = U\,x\,Z\,xx\,V$, where the first displayed x is the last letter of R' and xx are the first two letters of R; the factor Z separates two runs, so $Z \neq \varepsilon$. Hence $U\,xx\,Z\,x\,V \in L$ iff $w \in L$. This moves one x from R to R' and keeps the x-skeleton and $|w|_x$. Words with the same x-skeleton have the same number ρ of x-runs, and their run lengths are compositions of $|w|_x$ into ρ positive parts; any two such compositions are connected by moves of one unit from a part of size at least 2 to another part.

(ii) Say $UxxyxV \in L$ (the other case is symmetric). By (i), $UxyxxV \in L$. The context (Ux, xV) accepts xy and yx, both of content Σ, so $\mathcal{D}_L(xy) = \mathcal{D}_L(yx)$: in every context xy may be replaced by yx and conversely. Adjacent transpositions connect all words with one Parikh vector.

(iii) Let $w \in L$ have a factor xx and an internal y-run of length 1. Then w has a spare x, and by (i) the spare letters can be moved into the x-run preceding that y-run; the resulting word of L has a factor $xxyx$, so L is Parikh-saturated by (ii).

(iv) Suppose that $w \in L$ has a factor xx and two internal y-runs. Then it has two consecutive internal y-runs y^e and $y^{e'}$, separated by an x-run R; let A_0 be the x-run preceding y^e. By (iii), $e, e' \geq 2$. Since w has a spare x, (i) gives two words of L with the x-skeleton and the number of letters x of w: a word $W_2 = U\,x\,y^e\,xx\,V'$ in which R has length 2 (the displayed x before y^e is the last letter of A_0, and V' begins with $y^{e'}$), and a word $W_1 = U_1\,x\,y^e\,x\,y^{e'}V_1$ in which R has length 1. The context (U, xV') accepts xy^ex, giving W_2, and xxy^e, giving $U\,xx\,y^e\,x\,V'$, which has the x-skeleton and the x-count of W_2 and lies in L by (i). Both tuples have content Σ, so $\mathcal{D}_L(xy^ex) = \mathcal{D}_L(xxy^e)$. The context $(U_1, y^{e'}V_1)$ accepts xy^ex, giving W_1, hence also xxy^e, so $U_1\,xx\,y^{e+e'}V_1 \in L$. As $e + e' \geq 4$, this word has the factor y^4, a contradiction. □

Lemma J.3. *If L is not Parikh-saturated and both $\mathtt{a}^4$ and $\mathtt{b}^4$ occur, then every run of every word of L with at least four runs has length 1.*

Proof. Let $w \in L$ have at least four runs. Its second and third runs are internal and consist of different letters, so for each letter y the word w has an internal y-run β and at least two y-runs. By Lemma J.2(i) for y, moving letters y from β to another y-run gives a word of L in which β has length 1 and the x-runs are those of w. If an x-run of w had length at least 2, this word would have a factor xx and an internal y-run of length 1, contrary to Lemma J.2(iii). So all x-runs of w have length 1, for both choices of x. □

Lemma J.4 (Runs created by short replacements)**.** *Let $X = A_{\mathbf{t}}$ be a typed nonterminal of arity $d \le f$, and let T be a derivation of $\widetilde{G}_0$ from X whose yield $\vec{u}$ has the least total length and, among those, the fewest nodes. If T has an antichain Q of heavy nodes with $|Q| \ge 6f + 2$, then for some letter x:* (a) *some component of $\vec{u}$ has content $\{x\}$ and length at least 5, so x^5 occurs; and* (b) *some word $W \in L$ has a factor xx and at least four runs, and, unless both $\mathtt{a}^5$ and $\mathtt{b}^5$ occur, W has two internal y-runs.*

Proof. In each component u_i of content $\{x\}$ (a *pure* component) mark the first and the last $\min\{2, |u_i|\}$ letters, and in each component of content Σ mark one $\mathtt{a}$ and one $\mathtt{b}$; there are at most $4f$ marks. Each mark is written by one node, and nodes of Q have disjoint subtrees, so the set Q_0 of nodes of Q whose subtrees write no mark has $|Q_0| \ge |Q| - 4f \ge 2f + 2$.

Let $h \in Q_0$ with nonterminal C_h. Erase the type indices and replace the subtree at h by a derivation of ω_{C_h}. The new root tuple lies in $L_A(G)$ and is shorter, since by nondeletion every letter of the yield of h occurs in $\vec{u}$ and $\|\omega_{C_h}\|_1 \le \theta < \ell_h$. The marks keep the content of every component of content Σ. If every pure component stayed pure, the new tuple would have type $\mathbf{t}$ and would lie in L_X by Lemma 4.2(i) with $\lambda := \widehat{c}$, contrary to the choice of T. Hence ω_{C_h} puts the other letter into a block of h (an interval of a root component occupied by a component of the yield of h) that lies in a pure component. Blocks contain no mark, so that pure component has length at least 5, and the block lies after its first two and before its last two letters. This proves (a), since $E_A[\vec{u}] \in L$.

Each $h \in Q_0$ owns at most f blocks. Join two owners when blocks of theirs in one root component are adjacent with no letter between them. Each block has at most two adjacent blocks, so this graph has maximum degree at most $2f$ and an independent set I with $|I| \ge |Q_0|/(2f + 1) > 1$, hence $|I| \ge 2$. Replace the subtrees at all $h \in I$ at once, giving $\vec{u}' \in L_A(G)$ and $W := E_A[\vec{u}'] \in L$. Fix a pure component of letter x and write y for the other letter. Its letters outside the blocks of I are unchanged, hence equal to x; this covers the blocks of owners outside I, which are kept, and the first two and the last two letters, which are marked. So every maximal y-run of the new component is preceded and followed by x, and it lies in a union of blocks of I that are consecutive with no letter between them. Such blocks belong to one owner: two blocks of distinct owners $h \ne h'$ of I with no block of I between them are separated by an unchanged letter, since their gap is nonempty (otherwise h and h' would be joined) or contains a block of an owner outside I. Hence every y-run of the new component has a unique *creating* owner in I, and by the previous paragraph every $h \in I$ creates at least one run, in a pure component whose letter we call x_h. For $h \ne h'$ in I the runs they create are distinct: if they lie in the same pure component, they have different creating owners; if they lie in different pure components, they are in different components of $\vec{u}'$.

Take $h_1 \ne h_2$ in I. If $x_{h_1} = x_{h_2} =: x$, then W has the factor xx (the first two letters of a pure component of length at least 5) and two internal y-runs, hence at least five runs. Otherwise both $\mathtt{a}^5$ and $\mathtt{b}^5$ occur, by the argument for (a) applied to h_1 and to h_2, and with $x := x_{h_1}$ and $y := x_{h_2}$ the two pure components contain the runs x, y, x and y, x, y respectively; these are six distinct runs of W, and W has the factor xx. □

Proposition J.5. *If L is not Parikh-saturated, then $t_{\mathrm{typ}}(G, c) \le D_f(r + 1)v\theta$.*

Proof. Let X and T be as in Lemma J.4. We first show that every antichain of heavy nodes of T has at most $6f + 1$ nodes. Otherwise Lemma J.4 applies. If neither $\mathtt{a}^4$ nor $\mathtt{b}^4$ occurs, (a) is impossible. If both occur, the word W of (b) has at least four runs and a run of length at least 2, contrary to Lemma J.3. If exactly one occurs, say x_0^4, then by (a) $x = x_0$ and y^5 does not occur, so by (b) W has a factor xx and two internal y-runs, contrary to Lemma J.2(iv).

A typed label is a nonterminal together with the contents of its components, so there are at most $3^f v$ of them. No typed label repeats on a path of T, since cutting between two equal labels

gives a derivation from X whose yield is not longer and which has fewer nodes; so paths have at most $3^f v$ nodes. Heavy nodes are closed under ancestors, and those at one depth form an antichain, so there are at most $(6f+1)3^f v = D_f v$ of them. If the root is not heavy, $\|\vec{u}\|_1 \le \theta$. Otherwise every letter of $\vec{u}$ is written by a heavy node or lies in the yield of a non-heavy child of one, and each heavy node writes at most θ letters and has at most r children. Hence $\|\vec{u}\|_1 \le D_f(r+1)v\theta$. □

Lemma J.6. *If L is Parikh-saturated, then $\tau(G,c) \le C_{f,r} v^2 \theta^4 H_\theta$, where $C_{f,r} := (r+1)(4r(3f+1)+D_f)+3$.*

Proof. Fix $X = A_{\mathbf{t}}$ of arity d, put $(p,q) := \Psi(E_A)$, so $p+q \le \theta$, and $\mathcal{P} := \Psi(L \cap \Sigma^+)$. By Lemma J.1(i) and Parikh saturation, τ_X is the least $\|\vec{z}\|_1$ over tuples $\vec{z}$ of type $\mathbf{t}$ with $(p,q) + \sum_i \Psi(z_i) \in \mathcal{P}$. Write $(x_B, y_B) := \Psi(\omega_B)$, and for a node u of a derivation tree let (a_u, b_u) be the Parikh vector of its yield. If u lies in a successful derivation of a word w, replacing its subtree by a derivation of ω_{B_u} gives a word of L with Parikh vector $\Psi(w) - (a_u, b_u) + (x_{B_u}, y_{B_u})$; for a proper descendant u' with $B_{u'} = B_u$, replacing the subtree at u by that at u' gives one with $\Psi(w) - (a_u - a_{u'}, b_u - b_{u'})$. Every comparison below is of this kind: it concerns Parikh vectors of words of L yielded by complete successful derivations obtained from a fixed one by such replacements, and by Parikh saturation membership in L of a word depends only on its Parikh vector. The union of the contents $t_1, \dots, t_d$ is Σ, or all of them equal one singleton.

The union is Σ. Let $t_{\mathtt{a}}$ and $t_{\mathtt{b}}$ be the numbers of components whose content contains $\mathtt{a}$, resp. $\mathtt{b}$; both lie in $\{1, \dots, f\}$. A tuple of type $\mathbf{t}$ with $n_{\mathtt{a}}$ letters $\mathtt{a}$ and $n_{\mathtt{b}}$ letters $\mathtt{b}$ exists iff $n_{\mathtt{a}} \ge t_{\mathtt{a}}$ and $n_{\mathtt{b}} \ge t_{\mathtt{b}}$, so τ_X is the least $n_{\mathtt{a}} + n_{\mathtt{b}}$ with these bounds and $(p + n_{\mathtt{a}}, q + n_{\mathtt{b}}) \in \mathcal{P}$. Fix a minimizing pair, put $c_{\mathtt{a}} := n_{\mathtt{a}} - t_{\mathtt{a}} + 1 \ge 1$ and $c_{\mathtt{b}} := n_{\mathtt{b}} - t_{\mathtt{b}} + 1 \ge 1$, and fix a successful derivation T of $W := \mathtt{a}^{p+n_{\mathtt{a}}}\mathtt{b}^{q+n_{\mathtt{b}}}$ with the fewest nodes. If a heavy node u had $a_u < c_{\mathtt{a}}$ and $b_u < c_{\mathtt{b}}$, its replacement would give a Parikh vector $(p + n'_{\mathtt{a}}, q + n'_{\mathtt{b}}) \in \mathcal{P}$ with $n'_{\mathtt{a}} \ge t_{\mathtt{a}}$, $n'_{\mathtt{b}} \ge t_{\mathtt{b}}$ and $n'_{\mathtt{a}} + n'_{\mathtt{b}} < n_{\mathtt{a}} + n_{\mathtt{b}}$; so every heavy node has $a_u \ge c_{\mathtt{a}}$ or $b_u \ge c_{\mathtt{b}}$. If u' is a proper descendant of u with the same nonterminal, then $a_u - a_{u'} \ge c_{\mathtt{a}}$ or $b_u - b_{u'} \ge c_{\mathtt{b}}$: a zero difference contradicts node-minimality (Lemma J.1(ii)), and another small difference contradicts the choice of the pair. Along a path, $(\lfloor a_u/c_{\mathtt{a}} \rfloor, \lfloor b_u/c_{\mathtt{b}} \rfloor)$ is componentwise nonincreasing, and no nonterminal repeats while it is constant. Since $\lfloor (p+n)/(n-t+1) \rfloor \le p + t$ for $n \ge t \ge 1$ and $p \ge 0$ (as $(p+t)(n-t+1) - (p+n) = (n-t)(p+t-1) \ge 0$), it takes at most $1 + (p + t_{\mathtt{a}}) + (q + t_{\mathtt{b}}) \le \theta + 2f + 1$ values, so paths have at most $v(\theta + 2f + 1)$ nodes. Heavy nodes at one depth have disjoint yields, so there are at most $(p + n_{\mathtt{a}})/c_{\mathtt{a}} + (q + n_{\mathtt{b}})/c_{\mathtt{b}} \le \theta + 2f$ of them. Heavy nodes are closed under ancestors, and every letter is written by a heavy node or lies in one of its at most r non-heavy children; hence $\tau_X \le |W| \le \max\{\theta, (r+1)\theta(\theta+2f)(\theta+2f+1)v\} \le (2f+1)(2f+2)(r+1)v\theta^3$.

All contents equal $\{\mathtt{a}\}$ (the case $\{\mathtt{b}\}$ is symmetric), and $h := d \in \{1, \dots, f\}$. Let $K := \min\{k \ge h : (p+k, q) \in \mathcal{P}\}$; then $\tau_X = K$, and we may assume $K > \theta + 2(f-1) \ge p + 2(h-1)$, since otherwise $\tau_X \le (2f-1)\theta$. Fix a successful derivation T of $W := \mathtt{b}^q\mathtt{a}^{p+K}$ with the fewest nodes.

(P1) *Row rigidity*: there are no $(c_0, y), (c_0 + \delta, y) \in \mathcal{P}$ with $c_0 \ge 1$ and $1 \le \delta \le K - h$. Otherwise the context $(\mathtt{b}^y\mathtt{a}^{c_0 - 1}, \varepsilon)$ accepts $\mathtt{a}$ and $\mathtt{a}^{1+\delta}$, so $\mathcal{D}_L(\mathtt{a}) = \mathcal{D}_L(\mathtt{a}^{1+\delta})$ by $(1,c)$-substitutability (both have content $\{\mathtt{a}\}$); the context $(\mathtt{b}^q\mathtt{a}^{p+K-\delta-1}, \varepsilon)$ accepts $\mathtt{a}^{1+\delta}$, hence $\mathtt{a}$, and $(p + K - \delta, q) \in \mathcal{P}$ with $h \le K - \delta < K$, contradicting the choice of K.

(P2) *Height.* If u' is a proper descendant of u with the same nonterminal and $b_u = b_{u'}$, then $a_u - a_{u'} \ge K - h + 1$: the value 0 contradicts node-minimality (Lemma J.1(ii)), and a value in $[1, K-h]$ contradicts the choice of K. Going down a path, b_u is nonincreasing (by nondeletion the yield of a child occurs in that of its parent), so the nodes of the path with one value of b_u form a segment; b_u takes at most $q+1$ values, and three nodes with one nonterminal and one value of b_u would give $p + K \ge a_u \ge 2(K - h + 1)$, false as $K > p + 2h - 2$. So paths have at most $2v(q+1)$

nodes. A node is *mixed* if its yield contains $\mathtt{b}$. For $1 \le k \le q$, the mixed nodes with $b_u = k$ that have no such proper ancestor have pairwise disjoint yields, so there are at most $\lfloor q/k \rfloor$ of them; below one of them, two such nodes are comparable (otherwise its yield would contain $2k$ letters $\mathtt{b}$), so they form a chain, which has at most $2v$ nodes by the argument above. Hence there are $M \le 2v \sum_{k=1}^{q} \lfloor q/k \rfloor \le 2vqH_q \le 2v\theta H_\theta$ mixed nodes.

(P3) *Heavy pure nodes.* Call a node u that is not mixed a *heavy pure* node if $\ell_u = a_u > \theta + f - 1$, and put $g_u := \ell_u - x_{B_u} \ge f$. Two comparable heavy pure nodes have different nonterminals, since by (P2) the upper one would have $\ell_u \ge K - h + 1 + \theta + f > p + K = |W|_{\mathtt{a}}$. In particular heavy pure nodes with one nonterminal have disjoint yields. For an antichain J of heavy pure nodes, replacing all of them gives a word with Parikh vector $(p + K - \sigma(J), q + Y(J))$, where $\sigma(J) := \sum_{u \in J} g_u$ and $Y(J) := \sum_{u \in J} y_{B_u}$; this word is yielded by a complete successful derivation, so its Parikh vector lies in $\mathcal{P}$. Hence two antichains J, J' with $\sigma(J), \sigma(J') \le K - h$ and $Y(J) = Y(J') =: Y$ give $(p + K - \sigma(J), q + Y), (p + K - \sigma(J'), q + Y) \in \mathcal{P}$, whose first coordinates are at least $p + h \ge 1$ and differ by $|\sigma(J) - \sigma(J')| \le K - h$; by (P1), $\sigma(J) = \sigma(J')$.

(P4) *Repeated nonterminals.* If B labels $m_B \ge 2$ heavy pure nodes, their yields are disjoint by (P3), so these nodes are pairwise incomparable and every set of them is an antichain, so each has $\ell_u \le p + K - (\theta + f) \le K - f$ and $g_u \le K - f \le K - h$; comparing two singletons by (P3) gives a common value g_B, and then a common ℓ_B. If moreover $y_B = 0$, one replacement gives $(p + K - g_B, q) \in \mathcal{P}$ with $h \le K - g_B < K$, which is impossible; so $y_B \ge 1$. Call B *frequent* if $m_B > \theta$. For frequent B, C, choose y_C nodes labelled B and, separately, y_B nodes labelled C, which is possible since $m_B > \theta \ge y_C$ and $m_C > \theta \ge y_B$; both sets are antichains with $Y = y_B y_C$, and $y_C g_B \le (m_B - 1)\ell_B \le p + K - \ell_B \le K - f \le K - h$, and likewise for C. By (P3), $y_C g_B = y_B g_C$, so all frequent B share a ratio $\alpha := g_B / y_B$, and $\ell_B = x_B + \alpha y_B \le (1 + \alpha)\theta$.

(P5) *The ratio.* Put $\alpha_* := (r + 1)(\theta + f - 1)(2\theta + 1)v \le 3f(r + 1)v\theta^2$; we show $\alpha \le \alpha_*$ when frequent nonterminals exist. This is clear if $\alpha \le \theta$. Otherwise fix a heavy pure node u labelled by a frequent nonterminal B, and let H_u be the number of heavy pure nodes in its subtree. These nodes are closed under ancestors within that subtree. Every letter in the yield of u is written by a heavy pure node or belongs to the yield of a non-heavy child of such a node. Each heavy pure node writes at most θ letters and has at most r such children, each of total yield length at most $\theta + f - 1$; hence $\ell_B \le (r + 1)(\theta + f - 1)H_u$. Group these nodes by their number of heavy pure proper ancestors in the subtree. A chain of heavy pure nodes has pairwise different nonterminals by (P3), hence at most v nodes, so this number is less than v and there are at most v groups; two comparable nodes have different numbers, so each group is an antichain. In one group the nodes with frequent labels have length at least α and disjoint yields inside that of u, so there are at most $(1 + \alpha)\theta/\alpha < \theta + 1$ of them; nodes with other labels number at most $v\theta$ in all of T. So $H_u \le v(2\theta + 1)$, and $\alpha \le g_B \le \ell_B \le \alpha_*$.

(P6) *Counting.* Call a node that is not mixed *large* if $\ell_u > T_* := (\alpha_* + 1)\theta \le (3f + 1)(r + 1)v\theta^3$. By (P4)–(P5) no frequent nonterminal labels a large node, so there are at most $v\theta$ large nodes, and they are closed under ancestors among nodes that are not mixed. Mixed nodes are closed under ancestors, since by nondeletion the yield of a node occurs in that of its parent. Let a letter $\mathtt{a}$ of W be written by a node w that is neither mixed nor large. By the two closure properties, the ancestors of w (including w) that are neither mixed nor large form a final segment of the path from the root to w; let z be its top node. Then z is the root of T or a child of a mixed or a large node, and the subtree at z contains no mixed or large node, so $\ell_z \le T_*$. Thus every letter $\mathtt{a}$ of W is written by a mixed node, is written by a large node, or lies in the yield of such a z; there are at most $1 + rM + rv\theta$ nodes z. Hence

$$\begin{aligned} p + K &\le \theta M + (rM + rv\theta + 1)T_* + \theta \cdot v\theta \\ &\le 3v\theta^2 H_\theta + 4rv\theta H_\theta \cdot (3f + 1)(r + 1)v\theta^3 \le \big(4r(3f + 1)(r + 1) + 3\big)v^2\theta^4 H_\theta, \end{aligned}$$

using $M \le 2v\theta H_\theta$ and $rM + rv\theta + 1 \le 4rv\theta H_\theta$. As $(2f+1)(2f+2) \le D_f$, in both cases, and when $K \le \theta + 2(f-1)$, $\tau_X \le C_{f,r} v^2 \theta^4 H_\theta$. □

Proof of Theorem 8.6. If L is Parikh-saturated, Lemma J.6 applies. Otherwise Proposition J.5 gives $\tau(G,c) \le t_{\mathrm{typ}}(G,c) \le D_f(r+1)v\theta$. In both cases $\tau(G,c) \le C_{f,r} v^2 \theta^4 H_\theta$. The image of c has three elements, so $|\mathsf{S}^\sharp| = 4$ and $N_{\mathrm{typ}} \le C_c(G) \le 4^{fr}\|G\|$. By Corollary 4.8 there is a common characteristic sample with $\|K\|_+ \le N_{\mathrm{typ}}(1 + 2\theta + r\tau(G,c)) + \mathbf{1}_{\{\varepsilon \in L\}}$. With $v \le \|G\|$, this gives the first bound using $1 + 2\theta \le 3\|G\|^2\theta^4 H_\theta$ and $\tau \le C_{f,r}\|G\|^2\theta^4 H_\theta$, and the second using $1 + 2\theta \le 3\|G\|\theta$ and $\tau \le D_f(r+1)\|G\|\theta$. □

K The Linear Factor at Every Fan-Out

This appendix proves Proposition 6.1. Ψ denotes the Parikh map.

Proof of Proposition 6.1. Put $t := 2f$ and $L := L^{(f)}_{q,n}$.

(i) The language $R := \mathtt{a}_1^*\mathtt{d}_1\mathtt{a}_2^*\mathtt{d}_2 \cdots \mathtt{a}_t^*\mathtt{d}_t\mathtt{z}^*$ is the path language from 0 to t of the graph on $\{0,\dots,t\}$ with edges $\mathtt{a}_h : h-1 \to h-1$, $\mathtt{d}_h : h-1 \to h$ and $\mathtt{z} : t \to t$, and a word of R is determined by its Parikh vector. In w_i the marker $\mathtt{d}_h$ occurs once, at position $h(i+1)$, so the position of one marker, and the distance $(h'-h)(i+1)$ of two, determine i; all words have length W.

Let $J \subseteq L$ and $1 \le d \le f$, let $\vec{x}, \vec{y}$ be nonempty d-tuples and $E = (v_0,\dots,v_d)$ a d-context with $E[\vec{x}] = w_i \in J$ and $E[\vec{y}] = w_j \in J$, and let F be a d-context with $F[\vec{x}] \in J$. We show $F[\vec{y}] \in J$; the converse is symmetric. By Lemma 3.17, $F[\vec{y}] \in R$. If $i = j$, then $\sum_h \Psi(x_h) = \sum_h \Psi(y_h)$, so $F[\vec{y}]$ has the Parikh vector of $F[\vec{x}]$ and equals it. Let $i \ne j$. A marker in v_0 would have the same position in w_i and w_j, a marker in v_d the same distance from the end, and two markers in one v_h the same distance; each forces $i = j$. So the fixed part of E carries at most $d-1$ markers, and $\vec{x}$ carries at least $t - d + 1 \ge d + 1$ of them. Some component x_h carries two, and every word of L with factor x_h is w_i. Hence $F[\vec{x}] = w_i = E[\vec{x}]$, the fixed parts of F and E have equal Parikh vectors, $\Psi(F[\vec{y}]) = \Psi(E[\vec{y}])$, and $F[\vec{y}] = w_j \in J$. Finite languages are context-free, and the inclusion is item (a) after Definition 2.8.

(ii) For each i, $L \setminus \{w_i\} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ by (i), so every characteristic sample contains w_i by Lemma 4.5. Hence $F = L$. Conversely, $K = L$ is the only finite K with $L \subseteq K \subseteq L$, and the canonical operators return L on it by sample consistency and soundness.

(iii) Take nonterminals $P_0,\dots,P_q$ of fan-out f and $S_0,\dots,S_q$, $C_0,\dots,C_n$ of fan-out one, start symbol S_q, and the rules

$$
\begin{aligned}
&P_0 \to (\mathtt{d}_1, \mathtt{d}_3, \dots, \mathtt{d}_{2f-1}), \qquad C_0 \to (\mathtt{z}), \qquad C_h \to (\xi_1^1\xi_2^1)(C_{h-1}, C_{h-1})\ (1 \le h \le n),\\
&P_i \to (\mathtt{a}_1\xi_1^1\mathtt{a}_2,\ \mathtt{a}_3\xi_1^2\mathtt{a}_4,\ \dots,\ \mathtt{a}_{2f-1}\xi_1^f\mathtt{a}_{2f})(P_{i-1})\ (1 \le i \le q),\\
&S_j \to (\xi_1^1\mathtt{d}_2\xi_1^2\mathtt{d}_4 \cdots \xi_1^f\mathtt{d}_{2f}\xi_2^1)(P_j, C_n)\ (0 \le j \le q), \qquad S_j \to (\xi_1^1\mathtt{z}^{2f})(S_{j-1})\ (1 \le j \le q).
\end{aligned}
$$

By induction $L(C_h) = \{\mathtt{z}^{2^h}\}$,

$$
L(P_i) = \{(\mathtt{a}_1^i\mathtt{d}_1\mathtt{a}_2^i, \dots, \mathtt{a}_{2f-1}^i\mathtt{d}_{2f-1}\mathtt{a}_{2f}^i)\}, \qquad L(S_j) = \{\mathtt{a}_1^i\mathtt{d}_1 \cdots \mathtt{a}_{2f}^i\mathtt{d}_{2f}\mathtt{z}^{2f(j-i)+2^n} : 0 \le i \le j\},
$$

so $L(S_q) = L$. Every variable occurs once and in order, consecutive components of P_j are separated by $\mathtt{d}_2, \mathtt{d}_4, \dots$, and no component is empty; so the grammar is good, and it is reduced. It has $2q + n + 3$ nonterminals and $1, q, q+1, q, 1, n$ rules of sizes $f+1$, $3f+2$, $2f+4$, $2f+3$, 2, 5 for P_0, P_i, the first and second rules of S_j, C_0 and C_h (Definition 2.5); this gives the stated size, and $n \le q$ gives $(7f+17)(q+1)$. In a reduced presentation every rule occurs in a successful derivation and all

words have length W, so $\bar{\theta}_G = W$. Finally let G be any reduced good fan-out-$\leq f$ presentation of L. For the trivial typing, Theorem 3.13 makes $\mathsf{W}(\widetilde{G}_0)$ characteristic for $\mathcal{B}_{f,s_{\mathrm{triv}}}$, and it has at most $|V| + |P| \leq \|G\|$ words (Definition 3.9 and the remark after Theorem 4.3); by (ii) for s_{triv}, it has $q + 1$ words. The final estimate follows from (ii), (iii) and $|w_i| = W$.

(iv) The identification changes no length and no template shape, so the grammar stays reduced and good with the same size and thickness, and the arguments for (ii) and (iii) apply unchanged once (i) is shown. For (i), let $J \subseteq L$, and let $u\,x\,v = w_i \in J$ and $u\,y\,v = w_j \in J$ with $x, y \in \Sigma^+$; if $i = j$ then $x = y$. Let $i \neq j$. All words have length W. If u contained $\mathtt{d}$, the first $\mathtt{d}$ would have position $i + 1$ in w_i and $j + 1$ in w_j; if v contained $\mathtt{d}$, the last $\mathtt{d}$ would be followed by $2(q - i) + 2^n$ and by $2(q - j) + 2^n$ letters. So x contains both letters $\mathtt{d}$ of w_i and has the factor $\mathtt{da}^i\mathtt{d}$, which occurs in no other word of L and only once in w_i. Hence $p\,x\,q' \in J$ forces $(p, q') = (u, v)$, and then $p\,y\,q' = w_j \in J$; the converse is symmetric. □

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the author used ChatGPT (OpenAI) and Claude (Anthropic) to assist with organization of the manuscript, expression and translation of text, cross-checks of references and notation, and editing of LaTeX source, and also to assist with exploring and checking proof approaches and with pre-submission reviewing. The author checked all mathematical content. After using these tools, the author reviewed and edited the material as needed and takes full responsibility for the content of the publication.

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